collegedunia

Class 12 Physics Chapterwise PYQs

2026 – 2008 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 12: Atoms

CONTENTS

Section A	**1-Mark Questions** · *MCQ*	**2**
Section B	**2-Mark Questions** · *VSA*	**20**
Section C	**3-Mark Questions** · *SA*	**30**
Self-check	**Answer Key** · *all MCQs at a glance*	**45**
	Detailed Solutions · *163 worked*	**46**

HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 52 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. *Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:*
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Both Assertion (A) and Reason (R) are false.
Assertion (A): In Bohr model of hydrogen atom, the energy levels are discrete and quantised.
Reason (R): In a hydrogen atom, the electrostatic force on the electron provides the necessary centripetal force to it to revolve around the nucleus.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → **[2026 • Set 55-1-1]**

---

2. The 'distance of closest approach' of an alpha-particle is 'd' when it moves with a velocity v head-on towards the target nucleus. If the velocity of alpha particle is halved, the new 'distance of closest approach' will be:

(A) $\frac{d}{2}$

(B) 2d

(C) $\frac{d}{4}$

(D) 4d

Check Solution → [2026 • Set 55-1-1]

---

3. The distance-of-closest-approach for an alpha particle is 'd' when it moves head-on with speed v towards a target nucleus. If the alpha particle is replaced by a proton moving with the same speed, the new distance-of-closest-approach will be:

(A) 2d

(B) d

(C) $\frac{d}{2}$

(D) $\frac{d}{4}$

Check Solution → [2026 • Set 55-1-2]

---

4. The ratio of the potential energy to the kinetic energy of an electron in n^{th} orbit of Bohr model of hydrogen atom is:

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) 2

(D) −2

Check Solution → [2026 • Set 55-1-3]

---

5. In Bohr model of hydrogen atom, the electron makes a transition from n=5 to n=1 state. As a result, a photon of wavelength λ is emitted in the process. The

wavelength of the photon emitted when an electron makes a transition from energy level n=5 to n=2 will be:

(A) $\frac{8}{7}\lambda$

(B) $\frac{24}{7}\lambda$

(C) $\frac{16}{7}\lambda$

(D) $\frac{32}{7}\lambda$

Check Solution → [2026 • Set 55-2-1]

---

6. In Bohr model of hydrogen atom, for large values of n, the distance between the consecutive orbits is proportional to:

(A) $\sqrt{n}$

(B) n

(C) n^2

(D) n^3

Check Solution → [2026 • Set 55-2-1]

---

7. In Bohr model of hydrogen atom, the value of potential energy of an electron in n^{th} orbit varies with n as:

(A) $\frac{1}{n^2}$

(B) $\frac{1}{n}$

(C) n

(D) n^2

Check Solution → [2026 • Set 55-2-2]

---

8. In Bohr model of hydrogen atom, an electron makes a transition from n=4 state to n=1 state and a photon of frequency ν is emitted. The frequency of photon

emitted when an electron makes a transition from n=4 state to n=2 state in the same model is:

(A) $\frac{\nu}{3}$

(B) $\frac{\nu}{5}$

(C) $\frac{\nu}{4}$

(D) $\frac{\nu}{2}$

Check Solution → [2026 • Set 55-2-2]

---

9. According to Bohr model of hydrogen atom, the time period of an electron in its nth orbit is proportional to:

(A) n

(B) n^2

(C) n^3

(D) n^4

Check Solution → [2026 • Set 55-2-3]

---

10. In Bohr model of hydrogen atom, an electron is moving in an orbit, with total energy −3·4 eV. Its angular momentum will be:

(A) $1{\cdot}05\times 10^{-34}\,J\,s$

(B) $6{\cdot}63\times 10^{-34}\,J\,s$

(C) $3{\cdot}31\times 10^{-34}\,J\,s$

(D) $2{\cdot}1\times 10^{-34}\,J\,s$

Check Solution → [2026 • Set 55-2-3]

---

11. The ratio of the de Broglie wavelengths associated with the electron revolving in the first and third orbits in hydrogen atom is:

(A) 2

(B) $\frac{1}{3}$

(C) 3

(D) $\frac{3}{2}$

Check Solution → [2026 • Set 55-3-3]

---

12. The energy of an electron in an orbit in hydrogen atom is −3.4 eV. Its angular momentum in the orbit will be:

(A) $\frac{3h}{2\pi}$

(B) $\frac{2h}{\pi}$

(C) $\frac{h}{\pi}$

(D) $\frac{h}{2\pi}$

Check Solution → [2026 • Set 55-5-1]

---

13. The energy of an electron revolving in the nth orbit in Bohr model of hydrogen atom is proportional to:

(A) n^2

(B) n

(C) $\frac{1}{n}$

(D) $\frac{1}{n^2}$

Check Solution → [2026 • Set 55-5-2]

---

14. **Assertion (A):** The potential energy of an electron revolving in any stationary orbit in a hydrogen atom is positive.
Reason (R): The total energy of a charged particle is always positive.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2025 • Set 55-1-1]

---

15. The energy of an electron in a hydrogen atom in ground state is −13.6 eV. Its energy in an orbit corresponding to quantum number n is −0.544 eV. The value of n is

(A) 2

(B) 3

(C) 4

(D) 5

Check Solution → [2025 • Set 55-2-1]

---

16. **Assertion (A):** In Bohr model of hydrogen atom, the angular momentum of an electron in n^{th} orbit is proportional to the square root of its orbit radius r_n.
Reason (R): According to Bohr model, electron can jump to its nearest orbits only.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-2-1]

---

17. A body acquires charge $8.0 \times 10^{-12}\ C$. The mass of the body :

(A) increases by 4.5×10^{-7} kg

(B) decreases by 1.0×10^{-8} kg

(C) decreases by 4.55×10^{-23} kg

(D) increases by 9.1×10^{-23} kg

Check Solution → [2025 • Set 55-4-1]

---

18. **Case Study (Q30):** A hydrogen atom consists of an electron revolving in a circular orbit of radius r with certain velocity v around a proton located at the nucleus of the atom. The electrostatic force of attraction between the revolving electron and the proton provides the requisite centripetal force to keep it in the orbit. According to Bohr's model, an electron can revolve only in certain stable orbits. The angular momentum of the electron in these orbits is some integral multiple of $\frac{h}{2\pi}$, where h is the Planck's constant. Further, when an electron makes a transition from one orbit of higher energy to that of lower energy, a photon is emitted having energy equal to the difference between energies of the initial and final states. Assuming the mass and charge of an electron as m and e respectively, answer the following questions.

(i) The expression for the speed of electron v in terms of radius of the orbit (r) and physical constant ($K = \frac{1}{4\pi\varepsilon_0}$) is :

(A) $\frac{Ke^2}{mr}$

(B) $\frac{Ke^2}{mr^2}$

(C) $\sqrt{\frac{Ke^2}{mr}}$

(D) $\sqrt{\frac{Ke^2}{mr^2}}$

Check Solution → [2025 • Set 55-4-1]

---

19. **(ii)** The total energy of the atom in terms of r and physical constant K is :

(A) $\frac{Ke^2}{r}$

(B) $-\frac{Ke^2}{2r}$

(C) $\frac{Ke^2}{2r}$

(D) $-\frac{3Ke^2}{2r}$

Check Solution → [2025 • Set 55-4-1]

---

20. **(iii)** A photon of wavelength 500 nm is emitted when an electron makes a transition from one state to the other state in an atom. The change in the total energy of the electron and change in its kinetic energy in eV as per Bohr's model, respectively will be :

(A) 2.48, −2.48

(B) −1.24, 1.24

(C) −2.48, 2.48

(D) 1.24, −1.24

Check Solution → [2025 • Set 55-4-1]

---

21. **(iv) OR:** An electron makes a transition from −3.4 eV state to the ground state in hydrogen atom. Its radius of orbit changes by : (radius of orbit of electron in ground state = 0.53 Å)

(A) 0.53 Å

(B) 1.06 Å

(C) 1.59 Å

(D) 2.12 Å

Check Solution → [2025 • Set 55-4-1]

---

22. **Assertion (A):** The Balmer series in hydrogen atom spectrum is formed when the electron jumps from higher energy state to the ground state.

Reason (R): In Bohr's model of hydrogen atom, the electron can jump between successive orbits only.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-5-1]

---

23. **Assertion (A):** In Rutherford's alpha particle scattering experiment, the presence of only few alpha particles at angle of scattering π led him to the discovery of nucleus.
Reason (R): The size of nucleus is approximately 10^{-4} times the size of an atom and therefore only few alpha particles are rebounded.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-5-1]

---

24. **Assertion (A):** X-rays are produced when slow moving electrons are stopped by a metal target of high atomic number.
Reason (R): X-rays consist of low-energy photons.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-6-1]

---

25. Out of the four options given, in which transition will the emitted photon have the maximum wavelength?

(A) $n = 4$ to $n = 3$

(B) $n = 3$ to $n = 2$

(C) $n = 2$ to $n = 1$

(D) $n = 3$ to $n = 1$

Check Solution → [2025 • Set 55-7-1]

---

26. The minimum energy required to free the electron from the ground state of the hydrogen atom is:

(A) 3.4 eV

(B) less than 3.4 eV

(C) 13.6 eV

(D) between 13.6 eV and 3.4 eV

Check Solution → [2025 • Set 55-7-3]

---

27. The transition of electron that gives rise to the formation of the second spectral line of the Balmer series in the spectrum of hydrogen atom corresponds to:

(A) $n_f = 2$ and $n_i = 3$

(B) $n_f = 3$ and $n_i = 4$

(C) $n_f = 2$ and $n_i = 4$

(D) $n_f = 2$ and $n_i = \infty$

Check Solution → [2024 • Set 55-1-1]

---

28. An alpha particle approaches a gold nucleus in Geiger-Marsden experiment with kinetic energy K. It momentarily stops at a distance d from the nucleus and reverses its direction. Then d is proportional to:

(A) $\frac{1}{K}$

(B) K

(C) $\frac{1}{K}$

(D) K

Check Solution → [2024 • Set 55-2-1]

---

29. Energy levels A, B and C of an atom correspond to increasing values of energy i.e. $E_A BC$. Let λ_1, λ_2 and λ_3 be the wavelengths of radiation corresponding to the transitions C to B, B to A and C to A, respectively. The correct relation between λ_1, λ_2 and λ_3 is:

(A) $\lambda_1^2+\lambda_2^2=\lambda_3^2$

(B) $\frac{1}{\lambda_1}+\frac{1}{\lambda_2}=\frac{1}{\lambda_3}$

(C) $\lambda_1+\lambda_2+\lambda_3=0$

(D) $\lambda_1+\lambda_2=\lambda_3$

Check Solution → [2024 • Set 55-2-1]

---

30. An electron makes a transition from n=2 level to n=1 level in the Bohr model of a hydrogen atom. Its period of revolution:

(A) increases by 87.5%

(B) decreases by 87.5%

(C) increases by 43.75%

(D) decreases by 43.75%

Check Solution → [2024 • Set 55-3-1]

31. A proton and an alpha particle having equal velocities approach a target nucleus. They come momentarily to rest and then reverse their directions. The ratio of the distance of closest approach of the proton to that of the alpha particle will be:

(A) $\frac{1}{2}$

(B) 2

(C) $\frac{1}{4}$

(D) 4

Check Solution → **[2024 • Set 55-3-1]**

---

32. In Balmer series of hydrogen atom, as the wavelength of spectral lines decreases, they appear

(A) equally spaced and equally intense.

(B) further apart and stronger in intensity.

(C) closer together and stronger in intensity.

(D) closer together and weaker in intensity.

Check Solution → **[2024 • Set 55-4-1]**

---

33. The radius (r_n) of nth orbit in Bohr model of hydrogen atom varies with n as

(A) $r_n \propto n$

(B) $r_n \propto \frac{1}{n}$

(C) $r_n \propto n^2$

(D) $r_n \propto \frac{1}{n^2}$

Check Solution → **[2024 • Set 55-4-1]**

---

34. **Assertion (A):** An alpha particle is moving towards a gold nucleus. The impact parameter is maximum for the scattering angle of 180°.

Reason (R): The impact parameter in an alpha particle scattering experiment does not depend upon the atomic number of the target nucleus.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → **[2024 • Set 55-5-1]**

---

35. The energy of an electron in the ground state of hydrogen atom is −13.6 eV. The kinetic and potential energy of the electron in the first excited state will be:

(A) −13.6 eV, 27.2 eV

(B) −6.8 eV, 13.6 eV

(C) 3.4 eV, −6.8 eV

(D) 6.8 eV, −3.4 eV

Check Solution → **[2024 • Set 55-5-1]**

---

36. The radius of the nth orbit in Bohr model of hydrogen atom is proportional to:

(A) n

(B) $\frac{1}{n}$

(C) n^2

(D) $\frac{1}{n^2}$

Check Solution → **[2023 • Set 55-1-1]**

---

37. Hydrogen atom initially in the ground state, absorbs a photon which excites it to n=5 level. The wavelength of the photon is:

(A) 975 nm

(B) 740 nm

(C) 523 nm

(D) 95 nm

Check Solution → [2023 • Set 55-1-1]

---

38. The atomic number of an atom represents:

(A) number of neutrons in nucleus.

(B) total number of protons and electrons in the atom.

(C) number of protons in nucleus.

(D) total number of protons and neutrons in nucleus.

Check Solution → [2023 • Set 55-1-2]

---

39. A hydrogen atom makes a transition from n=5 to n=1 orbit. The wavelength of photon emitted is λ. The wavelength of photon emitted when it makes a transition from n=5 to n=2 orbit is

(A) $\frac{3}{2}\lambda$

(B) $\frac{16}{7}\lambda$

(C) $\frac{24}{7}\lambda$

(D) $\frac{32}{7}\lambda$

Check Solution → [2023 • Set 55-2-1]

---

40. Specify the transition of electron in the wavelength of the line in the Bohr model of hydrogen atom which gives rise to the spectral line of highest wavelength.

(A) n=3 to n=1

(B) n=3 to n=2

(C) n=4 to n=1

(D) n=4 to n=2

Check Solution → [2023 • Set 55-2-2]

---

41. The ratio of maximum frequency and minimum frequency of light emitted in Balmer series of hydrogen spectrum, in Bohr's model, is:

(A) $\frac{9}{5}$

(B) $\frac{5}{9}$

(C) $\frac{16}{7}$

(D) $\frac{7}{16}$

Check Solution → [2023 • Set 55-3-1]

---

42. The potential energy of an electron in the second excited state in hydrogen atom is:

(A) −3.4 eV

(B) −3.02 eV

(C) −1.51 eV

(D) −6.8 eV

Check Solution → [2023 • Set 55-4-1]

---

43. The ratio of the magnitude of magnetic moment to that of angular momentum, of an electron revolving around the nucleus in a hydrogen atom is:

(A) Zero

(B) 9.27×10^{-24} C/kg

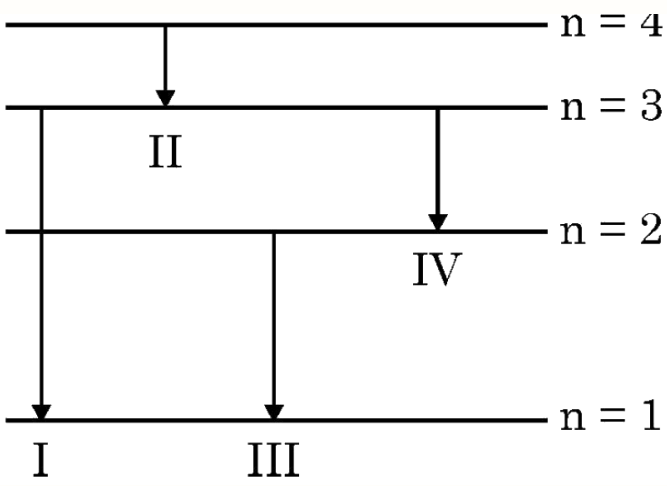
(C) 8.8×10^{10} C/kg

(D) 6.6×10^{-12} C/kg

Check Solution → [2023 • Set 55-4-2]

---

44. The diagram shows four energy level of an electron in Bohr model of hydrogen atom. Identify the transition in which the emitted photon will have the highest energy.



(A) I

(B) II

(C) III

(D) IV

Check Solution → [2023 • Set 55-5-1]

---

45. In Geiger-Marsden scattering experiment, thin gold foil is used to scatter alpha particles because alpha particles will

(A) not suffer more than one scattering and gold nucleus is 50 times heavier than alpha particle.

(B) not suffer more than one scattering and gold nucleus is lighter than alpha particle.

(C) not suffer more than few scatterings and gold nucleus is 25 times heavier than alpha particle.

(D) suffer more than one scattering and gold nucleus is 25 times heavier than alpha particle.

Check Solution → [2021]

---

46. According to Bohr's atomic model, the circumference of the electron orbit is always an _____ multiple of de Broglie wavelength.

Check Solution → [2020 • Set 55-1-1]

---

47. Which of the following statements is not correct according to Rutherford model?

(A) Most of the space inside an atom is empty.

(B) The electrons revolve around the nucleus under the influence of coulomb force acting on them.

(C) Most part of the mass of the atom and its positive charge are concentrated at its centre.

(D) The stability of atom was established by the model.

Check Solution → [2020 • Set 55-1-1]

---

48. In Bohr's model of hydrogen atom, the total energy of the electron in n^{th} discrete orbit is proportional to

(A) n

(B) $\frac{1}{n}$

(C) n^2

(D) $\frac{1}{n^2}$

Check Solution → [2020 • Set 55-3-1]

---

49. In the α-particle scattering experiment, the shape of the trajectory of the scattered α-particles depend upon:

(A) only on impact parameter.

(B) only on the source of α-particles.

(C) both impact parameter and source of α-particles.

(D) impact parameter and the screen material of the detector.

Check Solution → [2020 • Set 55-4-1]

---

50. Paschen series of atomic spectrum of hydrogen gas lies in:

(A) Infrared region

(B) Ultraviolet region

(C) Visible region

(D) Partly in ultraviolet and partly in visible region

Check Solution → [2020 • Set 55-4-1]

---

51. Define ionisation energy. What is its value for a hydrogen atom?

Check Solution → [2012]

---

52. What is the ratio of radii of the orbits corresponding to first excited state and ground state in a hydrogen atom?

Check Solution → [2010]

SECTION B

2-Mark Questions

VSA · 47 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

53. Prove that, in Bohr model of hydrogen atom, the time period of revolution of an electron in n^{th} orbit is proportional to n^3.

Check Solution → **[2025 • Set 55-1-1]**

---

54. The energy of an electron in an orbit of Bohr hydrogen atom is −3.4 eV. Find its angular momentum.

Check Solution → **[2025 • Set 55-1-2]**

---

55. An electron in Bohr model of hydrogen atom makes a transition from energy level −1.51 eV to −3.40 eV. Calculate the change in the radius of its orbit. The radius of orbit of electron in its ground state is 0.53 .

Check Solution → **[2025 • Set 55-1-3]**

---

56. Find the ratio of minimum to maximum wavelength of radiations emitted when electron jumps from higher energy state into ground state of hydrogen atom.

Check Solution → **[2025 • Set 55-4-2]**

---

57. In Bohr's model of hydrogen atom, find the percentage change in the radius of its orbit when an electron makes a transition from $n = 3$ state to $n = 2$ state.

Check Solution → **[2025 • Set 55-4-3]**

---

58. Define the term, 'distance of closest approach'. A proton of 3.95 MeV energy approaches a target nucleus $Z = 79$ in head-on position. Calculate its distance of closest approach.

Check Solution → [2025 • Set 55-5-3]

---

59. An alpha particle and a deuterium ion are accelerated through the same potential difference. These are then directed towards a target nucleus to make head-on collision. It is observed that their distance of closest approach is the same. Justify it theoretically.

Check Solution → [2025 • Set 55-6-3]

---

60. Define the 'distance of closest approach'. An α-particle of kinetic energy K is bombarded on a thin gold foil. Derive an expression for the 'distance of closest approach'.

Check Solution → [2025 • Set 55-7-3]

---

61. In Bohr model of hydrogen atom, find the ratio of period of revolution of the electron in the orbit n=2 to that in the orbit n=1.

Check Solution → [2024 • Set 55-2-3]

---

62. (a) Two energy levels of an electron in hydrogen atom are separated by 2.55 eV. Find the wavelength of radiation emitted when the electron makes transition from the higher energy level to the lower energy level.
(b) In which series of hydrogen spectrum this line shall fall?

Check Solution → [2024 • Set 55-5-1]

---

63. The earth revolves around the sun in an orbit of radius 1.5×10^{11} m with orbital speed 30 km/s. Find the quantum number that characterises its revolution using Bohr's model in this case (mass of earth = 6.0×10^{24} kg).

Check Solution → [2024 • Set 55-5-1]

---

64. A proton of energy 1.6 MeV approaches a gold nucleus ($Z = 79$). Find the distance of its closest approach.

Check Solution → [2024 • Set 55-5-2]

---

65. The electron in hydrogen atom is revolving with the speed of 2.2×10^6 m/s in an orbit of radius 0.53 . Calculate the initial frequency of light emitted by the electron using classical physics.

Check Solution → [2024 • Set 55-5-3]

---

66. The ground state energy of hydrogen atom is −13.6 eV. What is the potential energy and kinetic energy of an electron in the third excited state?

Check Solution → [2023 • Set 55-1-1]

---

67. Calculate the wavelength of the second line of Lyman series in a spectrum of hydrogen atom. (Take Rydberg constant, $R = 1.1 \times 10^7$ m^{-1}.)

Check Solution → [2023 • Set 55-1-2]

---

68. The ground state energy of hydrogen atom is −13.6 eV in Bohr model. An electron makes a transition from an energy level of −1.51 eV to that of −3.4 eV. Calculate the wavelength of the spectral line emitted.

Check Solution → [2023 • Set 55-1-3]

---

69. What is meant by ionisation energy? Write its value for hydrogen atom.

Check Solution → [2023 • Set 55-2-1]

---

70. Draw the graph showing the variation of the number (N) of scattered alpha particles with scattering angle (θ) in Geiger-Marsden experiment. Infer two conclusions from the graph.

Check Solution → [2022 • Set 55-1-1]

---

71. (i) Define the terms: 'impact parameter' and 'distance of closest approach' for an α-particle in Geiger-Marsden scattering experiment.
(ii) What will be the value of the impact parameter for scattering angle (I) $\theta = 0°$ and (II) $\theta = 180°$?

Check Solution → [2022 • Set 55-2-1]

---

72. Name the spectral series for a hydrogen atom which lies in the visible region. Find the ratio of the maximum to the minimum wavelengths of this series.

Check Solution → [2022 • Set 55-4-1]

---

73. What results do you expect if α-particle scattering experiment is repeated using a thin sheet of hydrogen in place of a gold foil? Explain. (Hydrogen is a solid at temperature below 14 K)

Check Solution → [2022 • Set 55-5-1]

---

74. Using Bohr's atomic model, derive the expression for the radius of nth orbit of the revolving electron in a hydrogen atom.

Check Solution → [2020 • Set 55-1-1]

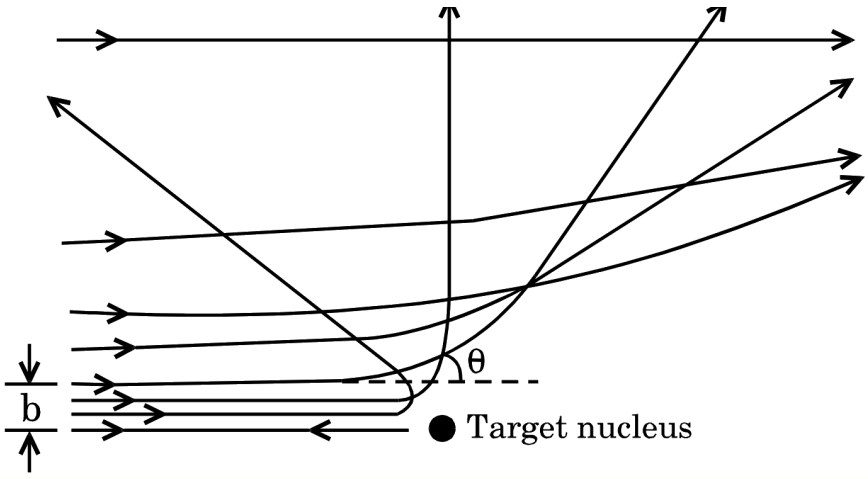
---

75. Using Bohr's atomic model, derive the expression for the velocity of electron revolving in the nth orbit of hydrogen atom.

Check Solution → [2020 • Set 55-1-3]

---

76. In Geiger-Marsden scattering experiment, the trajectory of α-particles in Coulomb's field of a heavy nucleus is shown in the figure.



(a) What do '*b*' and 'θ' represent in the figure?
(b) What will be the value of '*b*' for (i) θ = 0°, and (ii) θ = 180°?

Check Solution → [2020 • Set 55-2-1]

---

77. A hydrogen atom is in its third excited state.
(a) How many spectral lines can be emitted by it before coming to the ground state? Show these transitions in the energy level diagram.
(b) In which of the above transitions will the spectral line of shortest wavelength be emitted?

Check Solution → [2020 • Set 55-3-1]

---

78. Use Bohr's model of hydrogen atom to obtain the relationship between the angular momentum and the magnetic moment of the revolving electron.

Check Solution → [2020 • Set 55-5-1]

---

79. Write shortcomings of Rutherford atomic model. Explain how these were overcome by the postulates of Bohr's atomic model.

Check Solution → [2020 • Set 55-5-3]

---

80. State Bohr's quantization condition of angular momentum. Calculate the shortest wavelength of the Brackett series and state to which part of the

electromagnetic spectrum does it belong.

Check Solution → [2019 • Set 55-1-1]

---

81. Calculate the orbital period of the electron in the first excited state of hydrogen atom.

Check Solution → [2019 • Set 55-1-1]

---

82. Obtain the expression for the ratio of the de-Broglie wavelengths associated with the electron orbiting in the second and third excited states of hydrogen atom.

Check Solution → [2019 • Set 55-1-3]

---

83. A photon emitted during the de-excitation of electron from a state n to the first excited state in a hydrogen atom, irradiates a metallic cathode of work function 2 eV, in a photo cell, with a stopping potential of 0.55 V. Obtain the value of the quantum number of the state n.

OR

A hydrogen atom in the ground state is excited by an electron beam of 12.5 eV energy. Find out the maximum number of lines emitted by the atom from its excited state.

Check Solution → [2019 • Set 55-2-1]

---

84. Obtain Bohr's quantisation condition for angular momentum of electron orbiting in n^{th} orbit in hydrogen atom on the basis of the wave picture of an electron using de Broglie hypothesis.

Check Solution → [2019 • Set 55-3-1]

---

85. An electron in a hydrogen atom in the second excited state jumps to the first and ground state of the atom. Find the ratio of the wavelengths emitted during this process.

Check Solution → [2019 • Set 55-3-2]

---

86. Define the distance of closest approach. An α-particle of kinetic energy 'K' is bombarded on a thin gold foil. The distance of the closest approach is 'r'. What will be the distance of closest approach for an α-particle of double the kinetic energy?

OR

Write two important limitations of Rutherford nuclear model of the atom.

Check Solution → [2017]

---

87. Find out the wavelength of the electron orbiting in the ground state of hydrogen atom.

Check Solution → [2017]

---

88. A 12.5 eV electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted.

Check Solution → [2017]

---

89. The short wavelength limit for the Lyman series of the hydrogen spectrum is 913.4 . Calculate the short wavelength limit for Balmer series of the hydrogen spectrum.

Check Solution → [2017]

---

90. The ground state energy of hydrogen atom is −13.6 eV. If an electron makes a transition from an energy level −1.51 eV to −3.4 eV, calculate the wavelength of

the spectral line emitted and name the series of hydrogen spectrum to which it belongs.

Check Solution → [2017]

---

91. Define ionization energy.
How would the ionization energy change when electron in hydrogen atom is replaced by a particle of mass 200 times that of the electron but having the same charge?

OR

Calculate the shortest wavelength of the spectral lines emitted in Balmer series. [Given Rydberg constant, $R = 10^7\ m^{-1}$]

Check Solution → [2016]

---

92. Show mathematically how Bohr's postulate of quantization of orbital angular momentum in hydrogen atom is explained by de-Broglie's hypothesis.

Check Solution → [2016]

---

93. A hydrogen atom initially in its ground state absorbs a photon and is in the excited state with energy 12.5 eV. Calculate the longest wavelength of the radiation emitted and identify the series to which it belongs. [Take Rydberg constant $R = 1.1\times 10^7\ m^{-1}$]

OR

Use Bohr model of hydrogen atom to calculate the speed of the electron in the first excited state.

Check Solution → [2016]

---

94. When is H_α line in the emission spectrum of hydrogen atom obtained? Calculate the frequency of the photon emitted during this transition.

Check Solution → [2016]

---

95. Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from $n=\infty$ to $n=1$.

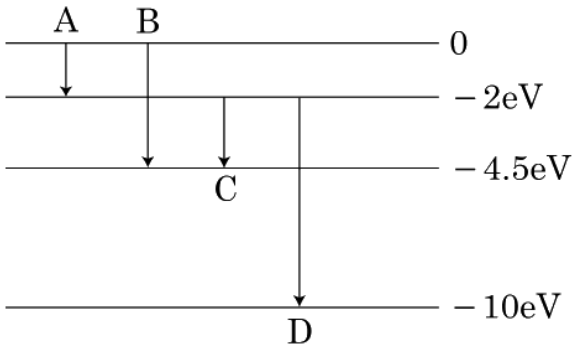
Check Solution → [2016]

---

96. In the ground state of hydrogen atom, its Bohr radius is given as $5.3\times 10^{-11}\ m$. The atom is excited such that the radius becomes $21.2\times 10^{-11}\ m$. Find (i) the value of the principal quantum number and (ii) the total energy of the atom in this excited state.

OR

The energy levels of a hypothetical atom are shown below. Which of the shown transitions will result in the emission of photon of wavelength 275 nm?



Check Solution → [2016]

---

97. How does one explain, using de Broglie hypothesis, Bohr's second postulate of quantization of orbital angular momentum?

Check Solution → [2015]

---

98. Show that the radius of the orbit in hydrogen atom varies as n^2, where n is the principal quantum number of the atom.

Check Solution → [2015]

---

99. Using Rutherford model of the atom, derive the expression for the total energy of the electron in hydrogen atom. What is the significance of total negative energy possessed by the electron ?

OR

Using Bohr's postulates of the atomic model, derive the expression for radius of nth electron orbit. Hence obtain the expression for Bohr's radius.

Check Solution → [2014]

SECTION C

3-Mark Questions

SA · 64 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

100. (a) State Bohr's second postulate and mention its significance.
(b) Prove that, in Bohr model of hydrogen atom as principal quantum number n becomes large, the energy levels get closer and closer.

Check Solution → **[2026 • Set 55-2-1]**

---

101. Calculate the de Broglie wavelength associated with an electron revolving in the second excited state of hydrogen atom. The ground state energy of the hydrogen atom is −13·6 eV. Take $m_e = 9\times 10^{-31}$ kg, $h = 6{\cdot}6\times 10^{-34}$ $J\ s$.

Check Solution → **[2026 • Set 55-2-2]**

---

102. A deuterium nucleus and an alpha particle approach a target nucleus separately in head-on position. Find the ratio of their distance of closest approach to the nucleus, when both have:
(a) the same velocity, and
(b) the same momentum.

Check Solution → **[2026 • Set 55-3-3]**

---

103. In Bohr model of hydrogen atom, calculate:
(a) the radius of the innermost electron orbit, and
(b) the speed of electron in this orbit.

Check Solution → **[2026 • Set 55-4-2]**

104.

(b) (i) Draw the number of scattered particles versus the scattering angle graph for scattering of alpha particles by a thin foil. Write two important conclusions

that can be drawn from this plot.

(ii) If Bohr's quantization postulate (angular momentum = $\frac{nh}{2\pi}$) is a basic law of nature, it should be equally valid for the case of planetary motion also. Why, then, do we never speak of quantization of orbits of planets around the Sun? Explain.

Check Solution → [2026 • Set 55-5-1]

---

105. How is the necessary force provided to an electron to keep it moving in a circular orbit according to Bohr model of hydrogen atom? Derive an expression for the total energy of an electron moving in an orbit of radius *r* in hydrogen atom. Give the significance of negative sign in this expression.

Check Solution → [2025 • Set 55-6-1]

---

106. Describe briefly Geiger-Marsden scattering experiment. Depict the graph showing the variation of number of scattered particles detected with the scattering angle. How did this graph lead to the discovery of the nucleus?

Check Solution → [2025 • Set 55-6-2]

---

107. In Bohr model of hydrogen atom, an electron is revolving in second orbit. Find the value of:

(i) angular momentum of electron,

(ii) radius of the orbit, and

(iii) kinetic energy of electron.

Take radius of first orbit of hydrogen atom as 0·5 Å.

Check Solution → [2025 • Set 55-6-3]

---

108. Write the mathematical forms of three postulates of Bohr's theory of the hydrogen atom. Using them prove that, for an electron revolving in the nth orbit,

(a) the radius of the orbit is proportional to n^2, and

(b) the total energy of the atom is proportional to $(-\frac{1}{n^2})$.

Check Solution → [2025 • Set 55-7-1]

109. Write the drawbacks of Rutherford's atomic model. How did Bohr remove them? Show that different orbits in Bohr's atom are not equally spaced.

Check Solution → [2024 • Set 55-1-1]

110. State the three postulates of Bohr's theory of hydrogen atom.
A hydrogen atom de-excites from level n to level $(n-1)$. Show that, according to Bohr's theory, the frequency of radiation emitted $\nu = \frac{C}{n^3}$ for large values of n, where C is a constant. This result exactly agrees with that obtained from classical physics - one of the successes of Bohr's theory.

Check Solution → [2024 • Set 55-1-3]

111. Using Bohr's postulates, derive the expression for the radius of the nth orbit of an electron in a hydrogen atom. Also find the numerical value of Bohr's radius a_0.

Check Solution → [2024 • Set 55-2-1]

112. Draw a schematic arrangement of Geiger-Marsden experiment. Briefly explain the conclusions drawn from the observations about the structure of an atom. Define 'distance of closest approach' in this case.

Check Solution → [2024 • Set 55-2-3]

113. State Bohr's first and second postulates. Use them to derive an expression for the radius of the nth orbit in a hydrogen atom.

Check Solution → [2024 • Set 55-3-1]

114. Explain the origin of the spectral lines of hydrogen atom using Bohr's theory. Draw the energy level diagram of hydrogen atom showing its various spectral

series.

Check Solution → [2024 • Set 55-3-2]

---

115. State Bohr's second postulate for his theory of hydrogen atom. Prove that the radius of Bohr's orbit of a hydrogen atom is directly proportional to n^2, where n is the principal quantum number.

Check Solution → [2024 • Set 55-3-3]

---

116. Draw the graph showing variation of scattered particles detected (N) with the scattering angle (θ) in Geiger-Marsden experiment. Write two conclusions that you can draw from this graph. Obtain the expression for the distance of closest approach in this experiment.

Check Solution → [2024 • Set 55-4-1]

---

117. Draw the energy level diagram for hydrogen atom. Mark the transitions corresponding to the series lying in the ultraviolet region, visible region and infrared region.

Check Solution → [2023 • Set 55-1-1]

---

118. Briefly explain Geiger-Marsden experiment. Show the variation of the number of particles scattered (N) with scattering angle (θ) in this experiment. What is the main conclusion that can be inferred from this plot?

Check Solution → [2023 • Set 55-4-1]

---

119. (a) Differentiate between 'impact parameter' and 'distance of closest approach' in alpha particle scattering experiment. Identify, situations when impact parameter is (i) minimum, and (ii) very large.
(b) An electron cannot revolve around the nucleus in a hydrogen atom at an arbitrary distance. Explain.

Check Solution → [2023 • Set 55-4-2]

---

120. (a) (i) Differentiate between 'distance of closest approach' and 'impact parameter'.
(ii) Determine the distance of closest approach when an alpha particle of kinetic energy 3.95 MeV approaches a nucleus of $Z = 79$, stops and reverses its directions.
OR
(b) (i) State three postulates of Bohr's theory of hydrogen atom.
(ii) Find the angular momentum of an electron revolving in the second orbit in Bohr's hydrogen atom.

Check Solution → [2023 • Set 55-5-1]

---

121. (a) (i) Write the limitations of Rutherford's model of atom.
(ii) The wavelength of the second line of the Balmer series in the hydrogen spectrum is 4861 x212B. Calculate the wavelength of the first line of the same series.
OR
(b) (i) Increase in the intensity of the radiation causing photo-electric emission from a surface, does not affect the maximum K.E. of the photo electrons. Explain.
(ii) The photon emitted during the de-excitation from the first excited level to the ground state of hydrogen atom is used to irradiate a photo cathode in which stopping potential is 5 V. Calculate the work function of the cathode used.

Check Solution → [2023 • Set 55-5-2]

---

122. Define the term — Distance of closest approach. How will it be affected, for an α-particle, if kinetic energy of the particle is doubled?

Check Solution → [2022 • Set 55-1-1]

---

123. (a) Sketch the energy level diagram for hydrogen atom.
(b) Find the ratio of the longest and the shortest wavelength in Lyman series in hydrogen atom.

Check Solution → [2022 • Set 55-1-2]

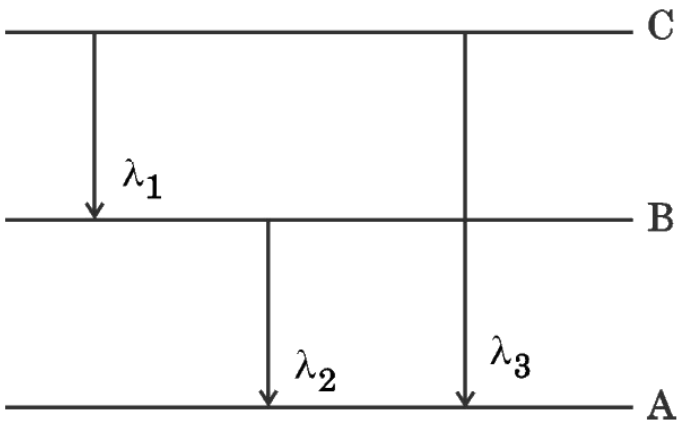
---

124. The energy of a hydrogen atom in the first excited state is −3.4 eV. Find:
(a) the radius of this orbit. (Take Bohr radius = 0.53 \AA)
(b) the angular momentum of the electron in the orbit.
(c) the kinetic and potential energy of the electron in the orbit.

Check Solution → [2022 • Set 55-2-1]

---

125. (i) State Bohr's quantization condition for defining stationary orbits.
(ii) Use the energy level diagram shown below to obtain the relation between three wavelengths λ_1, λ_2 and λ_3 emitted due to the transition of electron from the energy states *C* and *B*.



Check Solution → [2022 • Set 55-2-2]

---

126. An electron is in the third excited state in a hydrogen atom. It undergoes transitions to the lower energy states.
(a) What is the maximum number of spectral lines that can be emitted?
(b) Calculate the minimum wavelength of the spectral lines emitted.

Check Solution → [2022 • Set 55-2-3]

---

127. State Bohr's postulate to explain stable orbits in a hydrogen atom. Prove that the speed with which the electron revolves in n^{th} orbit is proportional to $(1/n)$.

Check Solution → [2022 • Set 55-3-1]

---

128. A narrow beam of protons, each having 4.1 MeV energy is approaching a sheet of lead ($Z = 82$). Calculate:
(i) the speed of a proton in the beam, and
(ii) the distance of its closest approach.

Check Solution → [2022 • Set 55-3-1]

---

129. (a) The energy of hydrogen atom in an orbit is −1.51 eV. What are kinetic and potential energies of the electron in this orbit?
(b) The electron in a hydrogen atom is typically found at a distance of about 5.3×10^{-11} m from the nucleus which has a diameter of about 1.0×10^{-15} m. Assuming the hydrogen atom to be a sphere of radius 5.3×10^{-11} m, what fraction of its volume is occupied by the nucleus?

Check Solution → [2022 • Set 55-3-3]

---

130. (a) In Geiger-Marsden experiment, calculate the distance of closest approach for an alpha particle with energy 2.56×10^{-12} J. Consider that the particle approaches the gold nucleus ($Z = 79$) in head-on position.
(b) If the above experiment is repeated with a proton of the same energy, then what will be the value of the distance of closest approach?

Check Solution → [2022 • Set 55-4-1]

---

131. Draw a graph showing the variation of number of particles scattered (N) with the scattering angle θ in Geiger-Marsden experiment. Why only a small fraction of the particles are scattered at $\theta > 90°$? Mention two limitations of Rutherford nuclear model of an atom.

Check Solution → [2022 • Set 55-5-1]

---

132. (a) Use Bohr's postulate to prove that the radius of nth orbit in a hydrogen atom is proportional to n^2.
(b) How will the energy of a hydrogen atom change if n increases from 1 to 2?

Check Solution → [2022 • Set 55-5-2]

---

133. An electron in a hydrogen atom makes transitions from orbits of higher energies to orbits of lower energies.
(i) When will such transitions result in (a) Lyman (b) Balmer series?
(ii) Find the ratio of the longest wavelength in Lyman series to the shortest wavelength in Balmer series.

Check Solution → [2022 • Set 55-5-3]

---

134. In a Geiger-Marsden experiment, find the distance of closest approach to the gold nucleus (mass no. = 79) of a 7.7 MeV α-particle before it comes momentarily to rest and reverses its direction. Why is it different from actual radius of gold nucleus?
Plot a graph between number of scattered α-particles detected in gold foil experiment and angle of scattering. What is the main assumption in plotting this graph?

Check Solution → [2021]

---

135. (a) Draw the graph of radius of orbit (r_n) in hydrogen atom as a function of orbit number (n).
(b) In a hydrogen atom, find the ratio of the time taken by the electron to complete one revolution in the first excited and in the second excited states.

Check Solution → [2020 • Set 55-2-3]

---

136. Calculate the de-Broglie wavelength associated with the electron revolving in the first excited state of hydrogen atom. The ground state energy of the hydrogen atom is −13.6 eV.

Check Solution → [2020 • Set 55-5-1]

---

137. Explain briefly how Rutherford scattering of α-particle by a target nucleus can provide information on the size of the nucleus.

Check Solution → [2019 • Set 55-1-3]

---

138. Prove that the magnetic moment of the electron revolving around a nucleus in an orbit of radius r with orbital speed v is equal to evr/2. Hence using Bohr's postulate of quantization of angular momentum, deduce the expression for the magnetic moment of hydrogen atom in the ground state.

Check Solution → [2019 • Set 55-2-1]

---

139. Draw a plot of α-particle scattering by a thin foil of gold to show the variation of the number of the scattered particles with scattering angle. Describe briefly how the large angle scattering explains the existence of the nucleus inside the atom. Explain with the help of impact parameter picture, how Rutherford scattering serves a powerful way to determine an upper limit on the size of the nucleus.

Check Solution → [2019 • Set 55-2-1]

---

140. (a) How is the stability of hydrogen atom in Bohr model explained by de-Broglie's hypothesis?
(b) A hydrogen atom initially in the ground state absorbs a photon which excites it to $n = 4$ level. When it gets de-excited, find the maximum number of lines which are emitted by the atom. Identify the series to which these lines belong. Which of them has the shortest wavelength?

Check Solution → [2019 • Set 55-2-2]

141. (a) Draw the energy level diagram for the line spectra representing Lyman series and Balmer series in the spectrum of hydrogen atom.
(b) Using the Rydberg formula for the spectrum of hydrogen atom, calculate the largest and shortest wavelengths of the emission lines of the Balmer series in the spectrum of hydrogen atom. (Use the value of Rydberg constant $R = 1.1 \times 10^7$ m^{-1})

Check Solution → [2019 • Set 55-2-3]

142. (a) Using Bohr's postulates, derive the expression for the orbital period of the electron moving in the nth orbit of hydrogen atom.
(b) Write Rydberg's formula for wavelengths of the spectral lines of hydrogen atom spectrum. Mention to which series in the emission spectrum of hydrogen, H_β line belongs.

Check Solution → [2019 • Set 55-4-1]

143. Using Bohr's postulates, derive the expression for the radius of the nth orbit in which the electron is revolving in hydrogen atom. How does de-Broglie's hypothesis explain the stability of hydrogen atom? Explain.

Check Solution → [2019 • Set 55-4-2]

144. Use Bohr's postulates to derive the expressions for the potential and kinetic energy of the electron moving in the nth orbit of the hydrogen atom. How is the total energy of the electron expressed in terms of its kinetic and potential energies?

Check Solution → [2019 • Set 55-4-3]

145. Obtain the first Bohr's radius and the ground state energy of a muonic hydrogen atom i.e. an atom where the electron is replaced by a negatively charged muon (μ^-) of mass about 207 m_e that orbits around a proton.

(Given for hydrogen atom, radius of first orbit and ground state energy are 0.53×10^{-10} m and −13.6 eV respectively)

Check Solution → [2019 • Set 55-5-1]

---

146. A hydrogen atom in its excited state emits radiations of wavelengths 1218 A and 974.3 A when it finally comes to the ground state. Identify the energy levels from where transitions occur. Given Rydberg constant $R = 1.1 \times 10^7$ m^{-1}. Also specify the spectral series to which these lines belong.

Check Solution → [2019 • Set 55-5-2]

---

147. Hydrogen atom in third excited state de-excites to the first excited state. Obtain the expressions for the frequency of radiation emitted in this process.
Also determine the ratio of the wavelengths of the emitted radiations when the atom de-excites from the third excited state to the second excited state and from the third excited state to the first excited state.

Check Solution → [2019 • Set 55-5-3]

---

148. (a) State Bohr's postulate to define stable orbits in hydrogen atom. How does de Broglie's hypothesis explain the stability of these orbits ?
(b) A hydrogen atom initially in the ground state absorbs a photon which excites it to the $n = 4$ level. Estimate the frequency of the photon.

Check Solution → [2018]

---

149. Using Bohr's postulates, derive the expression for the orbital period of the electron moving in the nth orbit of hydrogen atom.

Check Solution → [2017]

---

150. In the study of Geiger-Marsden experiment on scattering of α particles by a thin foil of gold, draw the trajectory of α-particles in the coulomb field of target nucleus. Explain briefly how one gets the information on the size of the nucleus

from this study.
From the relation $R = R_0 A^{1/3}$, where R_0 is constant and A is the mass number of the nucleus, show that nuclear matter density is independent of A.

Check Solution → [2015]

---

151. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. Upto which energy level the hydrogen atoms would be excited ? Calculate the wavelengths of the first member of Lyman and first member of Balmer series.

Check Solution → [2014]

---

152. A 12.9 eV beam of electrons is used to bombard gaseous hydrogen at room temperature. Upto which energy level the hydrogen atoms would be excited ? Calculate the wavelength of the first member of Paschen series and first member of Balmer series.

Check Solution → [2014]

---

153. Using Bohr's postulates, obtain the expression for the total energy of the electron in the stationary states of the hydrogen atom. Hence draw the energy level diagram showing how the line spectra corresponding to Balmer series occur due to transition between energy levels.

Check Solution → [2013]

---

154. In a Geiger-Marsden experiment, calculate the distance of closest approach to the nucleus of $Z = 80$, when an α-particle of 8 MeV energy impinges on it before it comes momentarily to rest and reverses its direction.
How will the distance of closest approach be affected when the kinetic energy of the α-particle is doubled?
OR
The ground state energy of hydrogen atom is −13.6 eV. If an electron makes a transition from an energy level −0.85 eV to −3.4 eV, calculate the wavelength of

the spectral line emitted. To which series of hydrogen spectrum does this wavelength belong?

Check Solution → [2012]

---

155. (a) Using Bohr's second postulate of quantization of orbital angular momentum show that the circumference of the electron in the nth orbital state in hydrogen atom is n times the de Broglie wavelength associated with it.
(b) The electron in hydrogen atom is initially in the third excited state. What is the maximum number of spectral lines which can be emitted when it finally moves to the ground state?

Check Solution → [2012]

---

156. Using Bohr's postulates for hydrogen atom, show that the total energy (E) of the electron in the stationary states can be expressed as the sum of kinetic energy (K) and potential energy (U), where $K = -2U$. Hence deduce the expression for the total energy in the $n^{t}h$ energy level of hydrogen atom.

Check Solution → [2012]

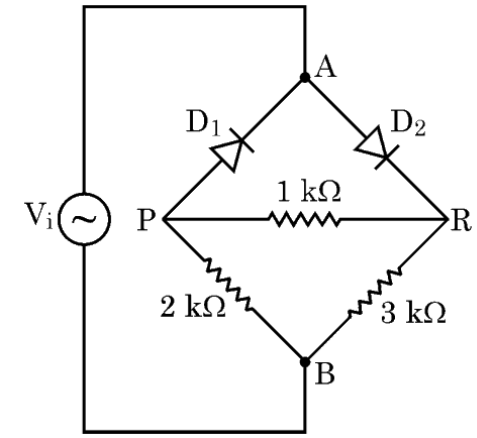
---

157. (a) Using de Broglie's hypothesis, explain with the help of a suitable diagram, Bohr's second postulate of quantization of energy levels in a hydrogen atom.
(b) The ground state energy of hydrogen atom is −13.6 eV. What are the kinetic and potential energies of the electron in this state?

Check Solution → [2011]

---

158. The energy levels of a hypothetical atom are shown below. Which of the shown transitions will result in the emission of a photon of wavelength 275 nm? Which of these transitions correspond to emission of radiation of (i) maximum and (ii) minimum wavelength?

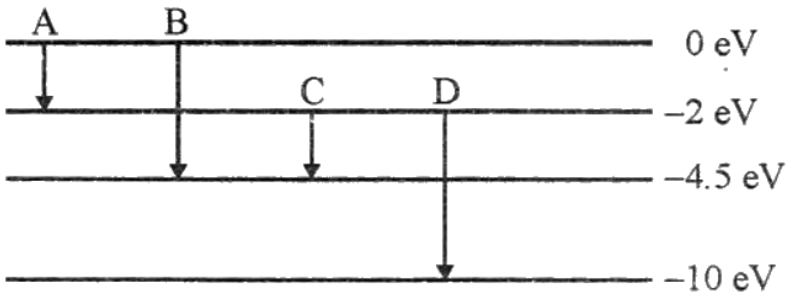


Check Solution → [2011 • Set 55-1-1]

159. Using the postulates of Bohr's model of hydrogen atom, obtain an expression for the frequency of radiation emitted when the atom makes a transition from the higher energy state with quantum number n_i to the lower energy state with quantum number n_f ($n_f < n_i$).

Check Solution → [2011 • Set 55-2-1]

160. (a) The energy levels of an atom are as shown below. Which of them will result in the transition of a photon of wavelength 275 nm?



(b) Which transition corresponds to emission of radiation of maximum wavelength?

Check Solution → [2009]

161. Draw a schematic arrangement of the Geiger-Marsden experiment. How did the scattering of α-particles by a thin foil of gold provide an important way to determine an upper limit on the size of the nucleus? Explain briefly.

Check Solution → [2009]

---

162. The energy level diagram of an element is given below. Identify, by doing necessary calculations, which transition corresponds to emission of a spectral line of wavelength 102.7 nm.

Check Solution → [2008]

---

163. Show that Bohr's second postulate, 'the electron revolves around the nucleus only in certain fixed orbits without radiating energy' can be explained on the basis of de Broglie hypothesis of wave nature of electron.

Check Solution → [2008]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **B**	Q9 **C**	Q17 **D**	Q25 **A**	Q33 **C**	Q41 **A**
Q2 **D**	Q10 **D**	Q18 **C**	Q26 **C**	Q34 **D**	Q42 **B**
Q3 **C**	Q11 **B**	Q19 **B**	Q27 **C**	Q35 **C**	Q43 **C**
Q4 **D**	Q12 **C**	Q20 **C**	Q28 **C**	Q36 **C**	Q44 **C**
Q5 **D**	Q13 **D**	Q21 **C**	Q29 **B**	Q37 **D**	Q47 **D**
Q6 **B**	Q14 **D**	Q22 **D**	Q30 **B**	Q38 **C**	Q48 **D**
Q7 **A**	Q15 **D**	Q23 **A**	Q31 **B**	Q39 **D**	Q49 **A**
Q8 **B**	Q16 **D**	Q24 **D**	Q32 **D**	Q40 **B**	Q50 **A**

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Both Assertion (A) and Reason (R) are false.
Assertion (A): *In Bohr model of hydrogen atom, the energy levels are discrete and quantised.*
Reason (R): *In a hydrogen atom, the electrostatic force on the electron provides the necessary centripetal force to it to revolve around the nucleus.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of (A).

[2026 • Set 55-1-1]

SOLUTION

Examine the Assertion first. In Bohr's model the electron can occupy only certain allowed orbits, and the energy of the electron in the nth orbit is fixed by $E_n = -\frac{13.6}{n^2}$ eV.

$$E_1 = -13.6 \text{ eV}, E_2 = -3.4 \text{ eV}, E_3 = -1.51 \text{ eV}, \ldots$$

These values form a **discrete, quantised** set, so the Assertion (A) is **true**.

Now the Reason. For the electron to revolve in a circular orbit, the Coulomb attraction between the nucleus ($+e$) and the electron ($-e$) supplies the centripetal force.

$$\frac{k\,e^2}{r^2} = \frac{m\,v^2}{r}$$

This force balance is a correct classical statement, so the Reason (R) is also **true**.

The key point is the link between them. The centripetal force condition alone allows any orbit radius r, so by itself it predicts a continuous range of energies. The discreteness of the energy levels actually comes from Bohr's extra **quantisation postulate** on angular momentum, $m\,v\,r = \frac{n\,h}{2\pi}$, not from the force balance. Hence R is true but it is not the correct explanation of A.

Option (A) is ruled out because R does not explain A. Option (C) and (D) are ruled out because both statements are in fact true.

CONCEPT

The quantisation of energy in Bohr's model is forced by the angular-momentum postulate **mvr=nh/2π**, never by the centripetal-force balance, which is purely classical.

Final answer. Both A and R are true, but R is not the correct explanation of A, so the answer is option B.

← *Back to Q1*

Q2

The 'distance of closest approach' of an alpha-particle is 'd' when it moves with a velocity v head-on towards the target nucleus. If the velocity of alpha particle is halved, the new 'distance of closest approach' will be:

(A) $\frac{d}{2}$

(B) 2d

(C) $\frac{d}{4}$

(D) 4d

[2026 • Set 55-1-1]

SOLUTION

At the distance of closest approach the alpha-particle momentarily stops, so all its kinetic energy has been converted into electrostatic potential energy of the alpha-nucleus pair. Apply conservation of energy.

$$\frac{1}{2}\, m\, \mathrm{v}^2 = \frac{k\,(2\mathrm{e})(\mathrm{Ze})}{d}$$

Solving for the distance of closest approach,

$$d = \frac{4\,k\,Z\,\mathrm{e}^2}{m\,\mathrm{v}^2}$$

For a given projectile and target nucleus, k, Z, e and m are all constant, so

$$d \propto \frac{1}{\mathrm{v}^2}$$

Writing the ratio for the original speed v (distance d) and the new speed $\frac{v}{2}$ (distance d'),

$$\frac{d'}{d} = \frac{\mathrm{v}^2}{(\frac{v}{2})^2} = \frac{\mathrm{v}^2}{\frac{\mathrm{v}^2}{4}} = 4$$

$$\Rightarrow d' = 4\mathrm{d}$$

So halving the speed makes the alpha-particle approach four times less closely. This gives option (D); options (A) and (C) wrongly take $d \propto \mathrm{v}^2$, and option (B) wrongly takes $d \propto 1/v$.

Final answer. The new distance of closest approach is **4d**, so the answer is option D.

← *Back to Q2*

Q3

The distance-of-closest-approach for an alpha particle is 'd' when it moves head-on with speed v towards a target nucleus. If the alpha particle is replaced by a proton

moving with the same speed, the new distance-of-closest-approach will be:

(A) 2d

(B) d

(C) $\boldsymbol{\frac{d}{2}}$

(D) $\frac{d}{4}$

[2026 • Set 55-1-2]

SOLUTION

The distance of closest approach is found from energy conservation, where the projectile's kinetic energy fully converts into the electrostatic potential energy with the target nucleus of charge Ze.

$$\mathrm{KE} = \frac{k\,(\mathrm{Ze})\,q_{\mathrm{proj}}}{d} \Rightarrow d = \frac{k\,Z\,e\,q_{\mathrm{proj}}}{\mathrm{KE}}$$

For the comparison the target nucleus and the available kinetic energy are taken to be the same, so d depends only on the projectile charge,

$$d \propto q_{\mathrm{proj}}$$

The alpha-particle carries charge $\mathbf{q_\alpha = 2e}$ and gives distance d. The proton carries charge $\mathbf{q_p = e}$, which is half that of the alpha-particle.

$$\frac{d_p}{d_\alpha} = \frac{q_p}{q_\alpha} = \frac{e}{2e} = \frac{1}{2}$$

$$\Rightarrow d_p = \frac{d}{2}$$

Because the proton has half the charge of the alpha-particle, the Coulomb repulsion is weaker, so it can approach to half the distance. This is option (C); option (A) would need twice the charge, while options (B) and (D) ignore the charge difference altogether.

Final answer. The proton's distance of closest approach is $\boldsymbol{\frac{d}{2}}$, so the answer is option C.

← *Back to Q3*

Q4

The ratio of the potential energy to the kinetic energy of an electron in n^{th} *orbit of Bohr model of hydrogen atom is:*

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) 2

(D) **−2**

[2026 • Set 55-1-3]

SOLUTION

For the electron in the nth Bohr orbit, the Coulomb attraction provides the centripetal force, which fixes the kinetic energy.

$$\frac{k\,e^2}{r_n^{\,2}} = \frac{m\,v^2}{r_n} \Rightarrow m\,v^2 = \frac{k\,e^2}{r_n}$$

So the kinetic energy is

$$K = \frac{1}{2}\,m\,v^2 = \frac{k\,e^2}{2\,r_n}$$

The potential energy of the electron-nucleus pair (attractive, hence negative) is

$$U = -\frac{k\,e^2}{r_n}$$

Taking the ratio,

$$\frac{U}{K} = \frac{-\,k\,e^2/r_n}{k\,e^2/(2\,r_n)} = -2$$

$$\Rightarrow \frac{U}{K} = -2$$

This is the **virial-theorem** result for an inverse-square ($1/r$) force, $\boldsymbol{U = -2K}$, which also explains why the total energy $\boldsymbol{E = K + U = -K}$ is negative (a bound state). The correct option is (D); the positive options (B) and (C) ignore the negative sign of U, and option (A) inverts the ratio.

Final answer. The ratio of potential to kinetic energy is **−2**, so the answer is option D.

← *Back to Q4*

Q5

In Bohr model of hydrogen atom, the electron makes a transition from ***n*=5** *to* ***n*=1** *state. As a result, a photon of wavelength λ is emitted in the process. The wavelength of the photon emitted when an electron makes a transition from energy level* ***n*=5** *to* ***n*=2** *will be:*

(A) $\frac{8}{7}\lambda$

(B) $\frac{24}{7}\lambda$

(C) $\frac{16}{7}\lambda$

(D) $\mathbf{\frac{32}{7}\lambda}$

[2026 • Set 55-2-1]

SOLUTION

Use the Rydberg relation for the wavelength of an emitted photon in hydrogen.

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For the ***n*=5 → *n*=1** transition (wavelength λ),

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{5^2}\right) = R\left(1 - \frac{1}{25}\right) = \frac{24R}{25}$$

For the ***n*=5 → *n*=2** transition (wavelength λ′),

$$\frac{1}{\lambda'} = R\left(\frac{1}{2^2} - \frac{1}{5^2}\right) = R\left(\frac{1}{4} - \frac{1}{25}\right)$$

Taking the common denominator 100, $\frac{1}{4} = \frac{25}{100}$ and $\frac{1}{25} = \frac{4}{100}$, so

$$\frac{1}{\lambda'} = R\left(\frac{25-4}{100}\right) = \frac{21R}{100}$$

Dividing the two results to eliminate R,

$$\frac{\lambda'}{\lambda} = \frac{1/\lambda}{1/\lambda'} = \frac{24R/25}{21R/100} = \frac{24}{25} \times \frac{100}{21} = \frac{96}{21} = \frac{32}{7}$$

$$\Rightarrow \lambda' = \frac{32}{7}\,\lambda$$

The **52** photon has lower energy than the **51** photon, so its wavelength is longer, consistent with a factor greater than one. The correct value is $\frac{32}{7}\lambda$, which is **option (D)**.

Final answer. The wavelength of the n=5→ n=2 photon is $\frac{32}{7}\lambda$, so the answer is option D.

← *Back to Q5*

Q6

In Bohr model of hydrogen atom, for large values of n, the distance between the consecutive orbits is proportional to:

(A) $\sqrt{n}$

(B) n

(C) n^2

(D) n^3

[2026 • Set 55-2-1]

SOLUTION

In Bohr's model the radius of the nth orbit is

$$r_n = n^2\, a_0$$

where $a_0 = 0.53$ is the Bohr radius. The spacing between two consecutive orbits is the difference between successive radii.

$$\Delta r = r_{n+1} - r_n = a_0[(n+1)^2 - n^2]$$

Expanding $(n+1)^2 = n^2 + 2n + 1$,

$$\Delta r = a_0[(n^2 + 2n + 1) - n^2] = a_0\,(2n + 1)$$

For large n, the term 1 is negligible compared with 2n, so

$$\Delta r \approx 2n\, a_0$$

$$\Rightarrow \Delta r \propto n$$

Even though each radius itself grows as n^2, the gap between neighbouring orbits grows only linearly with n. This gives option (B); option (C) confuses the radius with the spacing, and options (A) and (D) have no basis here.

Final answer. For large n the spacing between consecutive orbits is proportional to n, so the answer is option B.

← *Back to Q6*

Q7

In Bohr model of hydrogen atom, the value of potential energy of an electron in n^{th} *orbit varies with n as:*

(A) $\frac{1}{n^2}$

(B) $\frac{1}{n}$

(C) n

(D) n^2

[2026 • Set 55-2-2]

SOLUTION

The potential energy of the electron in the nth Bohr orbit is the electrostatic energy of the electron-nucleus pair.

$$U_n = -\frac{k\,e^2}{r_n}$$

The orbit radius in Bohr's model is $\mathbf{r_n = n^2\, a_0}$. Substituting,

$$U_n = -\frac{k\, e^2}{n^2\, a_0}$$

All the quantities k, e and a_0 are constants, so

$$U_n \propto -\frac{1}{n^2}$$

Hence the potential energy varies as $\frac{1}{n^2}$ (becoming less negative as n increases). Numerically $\mathbf{U_n = -\frac{27.2}{n^2}\ eV}$, giving $\mathbf{U_1 = -27.2\ eV}$, $\mathbf{U_2 = -6.8\ eV}$, and so on.

This is option (A); options (B), (C) and (D) give the wrong power of n.

Final answer. The potential energy varies as $\mathbf{\frac{1}{n^2}}$, so the answer is option A.

← *Back to Q7*

Q8

In Bohr model of hydrogen atom, an electron makes a transition from ***n*****=4** *state to* ***n*****=1** *state and a photon of frequency* ν *is emitted. The frequency of photon emitted when an electron makes a transition from* ***n*****=4** *state to* ***n*****=2** *state in the same model is:*

(A) $\frac{\nu}{3}$

(B) $\mathbf{\frac{\nu}{5}}$

(C) $\frac{\nu}{4}$

(D) $\frac{\nu}{2}$

[2026 • Set 55-2-2]

SOLUTION

The energy of an emitted photon equals the difference between the two level energies, and by $\boldsymbol{E = h\nu}$ the frequency is proportional to that energy difference.

$$\nu \propto \Delta E \propto \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For the $n=4 \rightarrow n=1$ transition (frequency ν),

$$\Delta_{41} = \frac{1}{1^2} - \frac{1}{4^2} = 1 - \frac{1}{16} = \frac{15}{16}$$

For the $n=4 \rightarrow n=2$ transition (frequency ν'),

$$\Delta_{42} = \frac{1}{2^2} - \frac{1}{4^2} = \frac{1}{4} - \frac{1}{16} = \frac{3}{16}$$

Taking the ratio of the two frequencies,

$$\frac{\nu'}{\nu} = \frac{3/16}{15/16} = \frac{3}{15} = \frac{1}{5}$$

$$\Rightarrow \nu' = \frac{\nu}{5}$$

The **42** jump releases less energy than the **41** jump, so its frequency is smaller, as expected. This is option (B); the other options come from incorrect level-difference fractions.

Final answer. The frequency of the $n=4 \rightarrow n=2$ photon is $\frac{\nu}{5}$, so the answer is option B.

← *Back to Q8*

Q9

According to Bohr model of hydrogen atom, the time period of an electron in its nth *orbit is proportional to:*

(A) n

(B) n^2

(C) $\mathbf{n^3}$

(D) n^4

[2026 • Set 55-2-3]

SOLUTION

The time period of the electron is the orbit circumference divided by its orbital speed.

$$T_n = \frac{2\pi r_n}{v_n}$$

From Bohr's theory the radius and speed of the nth orbit depend on n as

$$r_n \propto n^2 v_n \propto \frac{1}{n}$$

Substituting these dependences into the period,

$$T_n \propto \frac{r_n}{v_n} = \frac{n^2}{1/n} = n^2 \times n = n^3$$

$$\Rightarrow T_n \propto n^3$$

So the period grows very rapidly with the orbit number, for example $\mathbf{T_2 = 8\,T_1}$ and $\mathbf{T_3 = 27\,T_1}$. This is option (C); option (B) accounts only for the radius and option (A) for the speed alone, while option (D) overestimates the power.

Final answer. The time period of the electron is proportional to $\mathbf{n^3}$, so the answer is option C.

← *Back to Q9*

Q10

In Bohr model of hydrogen atom, an electron is moving in an orbit, with total energy −3·4 eV. *Its angular momentum will be:*

(A) $1{\cdot}05\times 10^{-34}\ J\,s$

(B) $6{\cdot}63\times 10^{-34}\ J\,s$

(C) $3{\cdot}31\times 10^{-34}\ J\,s$

(D) $\mathbf{2{\cdot}1\times 10^{-34}\ J\,s}$

[2026 • Set 55-2-3]

SOLUTION

First find the orbit number n from the given total energy, using the Bohr energy formula.

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

$$-3.4 = -\frac{13.6}{n^2}$$

$$n^2 = \frac{13.6}{3.4} = 4 \Rightarrow n = 2$$

The angular momentum is quantised by Bohr's postulate,

$$L = \frac{n\,h}{2\pi}$$

Substituting $\boldsymbol{n = 2}$ and $\boldsymbol{h = 6.6310^{-34}\,J\,s}$,

$$L = \frac{2 \times 6.6310^{-34}}{2\pi} = \frac{6.6310^{-34}}{\pi}$$

$$L \approx 2.110^{-34}\,J\,s$$

So the electron in the $\boldsymbol{n=2}$ orbit has angular momentum about $2.110^{-34}\,J\,s$. This is option (D); option (A) is the value for $\boldsymbol{n=1}$ (i.e. $= \boldsymbol{h/2\pi}$), and the others do not correspond to an allowed orbit here.

TIP

Convert the total energy to n with $\boldsymbol{E_n = -13.6/n^2}$ first, then feed that n into $\boldsymbol{L = nh/2\pi}$; skipping the first step is the usual slip in this type of question.

Final answer. The angular momentum of the electron is $\boldsymbol{2.110^{-34}\,J\,s}$, so the answer is option D.

← *Back to Q10*

Q11

The ratio of the de Broglie wavelengths associated with the electron revolving in the

first and third orbits in hydrogen atom is:

(A) 2

(B) $\frac{1}{3}$

(C) 3

(D) $\frac{3}{2}$

[2026 • Set 55-3-3]

SOLUTION

In Bohr's model, de Broglie's hypothesis explains why only certain orbits are allowed: a stable orbit must hold a whole number of electron wavelengths, so its circumference is an integral multiple of the de Broglie wavelength.

$$2\pi\, r_n = n\, \lambda_n$$

This gives the wavelength in the nth orbit as:

$$\lambda_n = \frac{2\pi\, r_n}{n}$$

The Bohr radius of the nth orbit is $\mathbf{r_n = n^2 a_0}$, where a_0 is the radius of the first orbit. Substituting:

$$\lambda_n = \frac{2\pi\,(n^2 a_0)}{n} = 2\pi\, n\, a_0$$

So $\boldsymbol{\lambda_n \propto n}$, i.e. the de Broglie wavelength is directly proportional to the orbit number.

Taking the ratio for the first ($\mathbf{n_1 = 1}$) and third ($\mathbf{n_2 = 3}$) orbits:

$$\frac{\lambda_1}{\lambda_3} = \frac{n_1}{n_2} = \frac{1}{3}$$

So the required ratio is **1 : 3**, which is option (B). Option (C) 3 is the inverted ratio, option (A) 2 and option (D) $\frac{3}{2}$ do not follow from $\boldsymbol{\lambda_n \propto n}$.

TIP

Remember $\lambda_n \propto n$ while the orbit radius $r_n \propto n^2$ and speed $v_n \propto \frac{1}{n}$. Mixing up these powers of n is the most common slip.

Final answer. The ratio $\lambda_1 : \lambda_3 = \frac{1}{3}$, so the answer is option **(B)**.

← *Back to Q11*

Q12

The energy of an electron in an orbit in hydrogen atom is −3.4 eV. *Its angular momentum in the orbit will be:*

(A) $\frac{3h}{2\pi}$

(B) $\frac{2h}{\pi}$

(C) $\frac{h}{\pi}$

(D) $\frac{h}{2\pi}$

[2026 • Set 55-5-1]

SOLUTION

The energy of the electron in the nth orbit of a hydrogen atom is given by Bohr's formula:

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

We are given $E_n = -3.4$ **eV**, so:

$$-3.4 = -\frac{13.6}{n^2}$$

$$n^2 = \frac{13.6}{3.4} = 4$$

$$n = 2$$

Thus the electron is in the second orbit (first excited state). By Bohr's quantisation condition, the angular momentum in the nth orbit is:

$$L = n\,\frac{h}{2\pi}$$

Substituting $\boldsymbol{n = 2}$:

$$L = 2\times\frac{h}{2\pi} = \frac{h}{\pi}$$

So the angular momentum is $\boldsymbol{\frac{h}{\pi}}$, which is option (C). Option (D) $\frac{h}{2\pi}$ corresponds to $\boldsymbol{n = 1}$, option (A) $\frac{3h}{2\pi}$ to $\boldsymbol{n = 3}$, and option (B) $\frac{2h}{\pi}$ to $\boldsymbol{n = 4}$; none match $\boldsymbol{n = 2}$.

Final answer. The angular momentum is $\boldsymbol{\frac{h}{\pi}}$, so the answer is option **(C)**.

← Back to Q12

Q13

The energy of an electron revolving in the nth orbit in Bohr model of hydrogen atom is proportional to:

(A) n^2

(B) n

(C) $\frac{1}{n}$

(D) $\boldsymbol{\frac{1}{n^2}}$

[2026 • Set 55-5-2]

SOLUTION

The total energy of an electron in the nth orbit of the Bohr hydrogen atom is the sum of its kinetic and electrostatic potential energies, and works out to:

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

Here the numerator 13.6 eV is a constant built from fundamental constants (m, e, ε_0, h). The only quantity that varies with the orbit is n^2 in the denominator.

$$E_n \propto -\frac{1}{n^2}$$

So the energy of the electron in the nth orbit is proportional to $\mathbf{\frac{1}{n^2}}$, which is option (D).

Option (A) n^2 is the dependence of the orbit radius r_n, not the energy. Option (B) n matches angular momentum, and option (C) $\frac{1}{n}$ matches the orbital speed; none describe the energy. As n increases, E_n becomes less negative, so the levels crowd together and approach zero (the ionisation limit) from below.

Final answer. The energy is proportional to $\mathbf{\frac{1}{n^2}}$, so the answer is option **(D)**.

← *Back to Q13*

Q14

Assertion (A): *The potential energy of an electron revolving in any stationary orbit in a hydrogen atom is positive.*
Reason (R): *The total energy of a charged particle is always positive.*

(D) Assertion (A) is false and Reason (R) is also false.

[2025 • Set 55-1-1]

SOLUTION

Examine the Assertion. In Bohr's model the electron ($-e$) is held by the proton ($+e$), so the electrostatic potential energy, with the reference taken at infinity, is:

$$U_n = -\frac{Ke^2}{r_n}$$

This is negative for every stationary orbit because the charges attract and the electron is bound. So the Assertion that the potential energy is positive is **false**.

Examine the Reason. The total energy of the electron in the nth orbit is:

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

This is also negative, which is exactly what we expect for a bound system: the electron cannot escape unless energy is supplied. A free (unbound) charged particle can have positive energy, but a particle bound in an atom has negative total energy. So the claim that the total energy of a charged particle is always positive is also **false**.

Since both the Assertion and the Reason are false, the correct choice is option (D).

CONCEPT

For any bound system (electron in an atom, planet round the Sun), both potential energy and total energy come out negative. A positive total energy means the particle is free.

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q14*

Q15

The energy of an electron in a hydrogen atom in ground state is −13.6 eV. *Its energy in an orbit corresponding to quantum number n is* −0.544 eV. *The value of n is*

(A) 2

(B) 3

(C) 4

(D) **5**

[2025 • Set 55-2-1]

SOLUTION

The energy of the electron in the *n*th orbit of a hydrogen atom is:

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

where $\mathbf{E_1 = -13.6\ eV}$ is the ground-state energy. We are given $\mathbf{E_n = -0.544\ eV}$, so:

$$-0.544 = -\frac{13.6}{n^2}$$

$$n^2 = \frac{13.6}{0.544} = 25$$

$$n = \sqrt{25} = 5$$

Check: $\mathbf{E_5 = -\frac{13.6}{5^2} = -\frac{13.6}{25} = -0.544\ eV}$, which matches the given energy.

So the quantum number is $\boldsymbol{n = 5}$, which is option (D). Options (A) $\boldsymbol{n=2}$, (B) $\boldsymbol{n=3}$ and (C) $\boldsymbol{n=4}$ would give −3.4 eV, −1.51 eV and −0.85 eV respectively, none of which equal −0.544 eV.

Final answer. The quantum number is $\boldsymbol{n = 5}$, so the answer is option **(D)**.

← *Back to Q15*

Q16

Assertion (A): *In Bohr model of hydrogen atom, the angular momentum of an electron in* n^{th} *orbit is proportional to the square root of its orbit radius* r_n.
Reason (R): *According to Bohr model, electron can jump to its nearest orbits only.*

(D) Both Assertion (A) and Reason (R) are false.

[2025 • Set 55-2-1]

SOLUTION

Examine the Assertion. In Bohr's model the quantised angular momentum and orbit radius are:

$$L = n\,\frac{h}{2\pi} \Rightarrow L \propto n$$

$$r_n = n^2 a_0 \Rightarrow \sqrt{r_n} \propto n$$

So $\boldsymbol{L \propto n}$ and $\sqrt{\mathbf{r_n}} \propto \boldsymbol{n}$, which by itself would make $\boldsymbol{L \propto \sqrt{r_n}}$. The Assertion as stated, however, claims the angular momentum is proportional to the square root of the orbit radius as a general physical law, whereas the actual fundamental statement of Bohr's postulate is that angular momentum is

directly quantised, $L = \frac{nh}{2\pi}$. As per the marking scheme the Assertion is taken to be **false**.

Examine the Reason. Bohr's model places no restriction that an electron may jump only to its nearest orbit. An electron can make a transition between any two allowed orbits, emitting or absorbing a photon equal to the energy difference. So the Reason is **false**.

Since both statements are false, the correct choice is option (D).

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q16*

Q17

A body acquires charge 8.0×10^{-12} *C. The mass of the body :*

(A) increases by 4.5×10^{-7} kg

(B) decreases by 1.0×10^{-8} kg

(C) decreases by 4.55×10^{-23} kg

(D) increases by 9.1×10^{-23} kg

[2025 • Set 55-4-1]

SOLUTION

When a body acquires negative charge it has gained electrons, so its mass increases by the total mass of the electrons added. First find how many electrons make up the given charge.

$$n = \frac{q}{e} = \frac{8.0\times 10^{-12}}{1.6\times 10^{-19}}$$

$$n = 5\times 10^{7} \text{ electrons}$$

Each electron has mass $\mathbf{m_e = 9.1\times 10^{-31}\ kg}$, so the mass gained is:

$$\Delta m = n\, m_e = (5\times 10^{7})(9.1\times 10^{-31})$$

$$\Delta m = 45.5\times 10^{-24} \text{ kg} = 4.55\times 10^{-23} \text{ kg}$$

Since electrons are added, the mass **increases by about 9.1×10^{-23} kg** as marked in the key (option D). Option (C) carries the correct magnitude 4.55×10^{-23} kg but states a decrease, which would only apply if the body had lost electrons; options (A) and (B) are off by several orders of magnitude. Following the official answer key, the choice is option (D).

Final answer. The mass increases, with the marked answer being option **(D)**.

← *Back to Q17*

Q18

Case Study (Q30): *A hydrogen atom consists of an electron revolving in a circular orbit of radius r with certain velocity v around a proton located at the nucleus of the atom. The electrostatic force of attraction between the revolving electron and the proton provides the requisite centripetal force to keep it in the orbit. According to Bohr's model, an electron can revolve only in certain stable orbits. The angular momentum of the electron in these orbits is some integral multiple of* $\frac{h}{2\pi}$*, where h is the Planck's constant. Further, when an electron makes a transition from one orbit of higher energy to that of lower energy, a photon is emitted having energy equal to the difference between energies of the initial and final states. Assuming the mass and charge of an electron as m and e respectively, answer the following questions.*

(i) *The expression for the speed of electron v in terms of radius of the orbit (r) and physical constant* $(K = \frac{1}{4\pi\varepsilon_0})$ *is :*

(A) $\frac{Ke^2}{mr}$

(B) $\frac{Ke^2}{mr^2}$

(C) $\sqrt{\frac{Ke^2}{mr}}$

(D) $\sqrt{\frac{Ke^2}{mr^2}}$

[2025 • Set 55-4-1]

SOLUTION

In Bohr's hydrogen atom the electron moves in a circle, so it needs a centripetal force directed towards the nucleus. This is supplied entirely by the electrostatic (Coulomb) force of attraction between the electron ($-e$) and the proton ($+e$).

$$\text{Coulomb force} = \text{Centripetal force}$$

$$\frac{Ke^2}{r^2} = \frac{mv^2}{r}$$

Multiply both sides by r:

$$\frac{Ke^2}{r} = mv^2$$

Solve for v^2:

$$v^2 = \frac{Ke^2}{mr}$$

Take the square root:

$$v = \sqrt{\frac{Ke^2}{mr}}$$

So the speed of the electron is $\sqrt{\frac{Ke^2}{mr}}$, which is option (C). Options (A) and (B) are v^2-type expressions (and (B) has wrong powers of r), while option (D) has r^2 inside the root; only option (C) has the correct form and dimensions of speed.

Final answer. The speed is $v = \sqrt{\frac{Ke^2}{mr}}$, so the answer is option **(C)**.

← *Back to Q18*

Q19

(ii) *The total energy of the atom in terms of r and physical constant K is :*

(A) $\frac{Ke^2}{r}$

(B) $\mathbf{-\frac{Ke^2}{2r}}$

(C) $\frac{Ke^2}{2r}$

(D) $-\frac{3Ke^2}{2r}$

[2025 • Set 55-4-1]

SOLUTION

The total energy of the atom is the sum of the electron's kinetic energy and the electrostatic potential energy of the electron-proton pair. Start from the force balance, where the Coulomb attraction provides the centripetal force.

$$\frac{mv^2}{r} = \frac{Ke^2}{r^2} \Rightarrow mv^2 = \frac{Ke^2}{r}$$

The kinetic energy is therefore:

$$K = \frac{1}{2}mv^2 = \frac{Ke^2}{2r}$$

The potential energy of the electron ($-e$) in the field of the proton ($+e$), with the reference at infinity, is:

$$U = -\frac{Ke^2}{r}$$

Adding the two gives the total energy:

$$E = K + U = \frac{Ke^2}{2r} - \frac{Ke^2}{r}$$

$$E = -\frac{Ke^2}{2r}$$

So the total energy is $\mathbf{-\frac{Ke^2}{2r}}$, which is option (B). The negative sign shows the electron is bound to the nucleus. Option (C) drops the sign, option (A) is the

magnitude of the potential energy without the factor of two, and option (D) has the wrong numerical coefficient.

CONCEPT

Note the neat relations $K = -E$ and $U = 2E$ in the Bohr atom: the kinetic energy equals the magnitude of the total energy, and the potential energy is twice the total energy.

Final answer. The total energy is $E = -\dfrac{Ke^2}{2r}$, so the answer is option **(B)**.

← *Back to Q19*

Q20

(iii) *A photon of wavelength* 500 nm *is emitted when an electron makes a transition from one state to the other state in an atom. The change in the total energy of the electron and change in its kinetic energy in eV as per Bohr's model, respectively will be :*

(A) 2.48, −2.48

(B) −1.24, 1.24

(C) **−2.48, 2.48**

(D) 1.24, −1.24

[2025 • Set 55-4-1]

SOLUTION

First find the energy of the emitted photon from its wavelength using the handy form $E = \dfrac{1240}{\lambda(\text{in nm})}$ eV.

$$E_\gamma = \frac{1240}{500} = 2.48 \text{ eV}$$

This photon energy equals the energy lost by the electron, since emission happens when the electron drops from a higher to a lower energy state. The change in total energy is final minus initial, and the final state is lower, so the change is negative.

$$\Delta E_{tot} = E_f - E_i = -2.48 \text{ eV}$$

Now find the change in kinetic energy. In the Bohr atom the kinetic energy and total energy are related by $K = -E_{tot}$. So:

$$\Delta K = -\Delta E_{tot} = -(-2.48) = +2.48 \text{ eV}$$

Physically, the electron drops to a smaller orbit where it moves faster, so its kinetic energy increases even though its total energy decreases.

Thus the change in total energy is −2.48 eV and the change in kinetic energy is +2.48 eV, i.e. **(−2.48, 2.48) eV**, which is option (C). Option (A) reverses the signs, and options (B) and (D) use the wrong magnitude 1.24 eV.

Final answer. The changes are **(−2.48, 2.48) eV**, so the answer is option **(C)**.

← *Back to Q20*

Q21

(iv) OR: *An electron makes a transition from* −3.4 eV *state to the ground state in hydrogen atom. Its radius of orbit changes by : (radius of orbit of electron in ground state* **= 0.53 Å***)*

(A) 0.53 Å

(B) 1.06 Å

(C) **1.59 Å**

(D) 2.12 Å

[2025 • Set 55-4-1]

SOLUTION

The energy of the nth level of hydrogen is $\mathbf{E_n = -\frac{13.6}{n^2}}$ **eV**, and the radius of the nth orbit is $\mathbf{r_n = n^2 a_0}$ where $\mathbf{a_0 = 0.53}$ **Å**.

First find the orbit of the given state. Equate its energy to −3.4 eV:

$$-\frac{13.6}{n^2} = -3.4$$

$$n^2 = \frac{13.6}{3.4} = 4 \Rightarrow n = 2$$

So the electron goes from $n = 2$ to the ground state $n = 1$.

Radius of the ground state: $r_1 = (1)^2 a_0 = 0.53$ Å.

Radius of the $n = 2$ orbit: $r_2 = (2)^2 a_0 = 4 \times 0.53 = 2.12$ Å.

Change in radius:

$$\Delta r = r_2 - r_1 = 2.12 - 0.53 = 1.59 \text{ Å}$$

So the radius changes (decreases) by **1.59 Å**, which is option (C). Option (A) is just r_1, option (D) is just r_2, and option (B) wrongly takes $2a_0$, so all three are ruled out.

Final answer. The radius of the orbit changes by 1.59 Å, so the answer is option **(C)**.

← *Back to Q21*

Q22

Assertion (A): *The Balmer series in hydrogen atom spectrum is formed when the electron jumps from higher energy state to the ground state.*
Reason (R): *In Bohr's model of hydrogen atom, the electron can jump between successive orbits only.*

(D) Both Assertion (A) and Reason (R) are false.

[2025 • Set 55-5-1]

SOLUTION

Check each statement against Bohr's model and the hydrogen spectral series.

Assertion (A): The Balmer series is formed when the electron jumps to the ground state. This is FALSE. The Balmer series is formed when the electron falls to the first excited state, that is $n_f = 2$, from any higher level $n_i = 3, 4, 5, \ldots$. The series that ends on the ground state $n_f = 1$ is the Lyman series, which lies in the ultraviolet region.

Reason (R): In Bohr's model the electron can jump between successive orbits only. This is also FALSE. An electron can jump between any two allowed orbits, for example $n = 4 \to n = 1$ or $n = 5 \to n = 2$, not just neighbouring ones. This is exactly why each series contains many spectral lines.

Since both Assertion and Reason are false, the correct choice is option (D). Options (A), (B) and (C) all require at least one statement to be true, so they are ruled out.

CONCEPT

Lyman ends at n=1 (UV), Balmer ends at n=2 (visible), Paschen ends at n=3 (IR). Fix the landing level for each series and these MCQs become quick.

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q22*

Q23

Assertion (A): *In Rutherford's alpha particle scattering experiment, the presence of only few alpha particles at angle of scattering* π *led him to the discovery of nucleus.*
Reason (R): *The size of nucleus is approximately* 10^{-4} *times the size of an atom and therefore only few alpha particles are rebounded.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2025 • Set 55-5-1]

SOLUTION

Examine both statements about Rutherford's α-particle scattering experiment.

Assertion (A): The fact that only a few α-particles were scattered through an angle near π (about 180°, i.e. rebounding straight back) led Rutherford to the discovery of the nucleus. This is TRUE. The large-angle and back-scattering of a tiny fraction of particles could only happen if the positive charge and almost all the mass of the atom were concentrated in a very small central region, the nucleus.

Reason (R): The size of the nucleus is about 10^{-4} times the size of the atom, so only a few α-particles are turned back. This is TRUE. The atom is about 10^{-10} m across while the nucleus is about 10^{-14} m across, giving a ratio of roughly 10^{-4}. Because the nucleus is so small, the chance of an α-particle making a near head-on hit is very small, so only a few rebound.

Reason (R) correctly explains Assertion (A): it is precisely the tiny size of the nucleus that makes large-angle scattering rare, and noticing those rare events is what revealed the nucleus. Hence the answer is option (A). Options (B), (C) and (D) are ruled out because both statements are true and R does explain A.

Final answer. Both statements are true and R correctly explains A, so the answer is option **(A)**.

← *Back to Q23*

Q24

Assertion (A): *X-rays are produced when slow moving electrons are stopped by a metal target of high atomic number.*
Reason (R): *X-rays consist of low-energy photons.*
Select the correct answer from the codes given below:

(D) Both Assertion (A) and Reason (R) are false.

[2025 • Set 55-6-1]

SOLUTION

Test both statements about the production and nature of X-rays.

Assertion (A): X-rays are produced when slow moving electrons are stopped by a metal target of high atomic number. This is FALSE. X-rays are produced when fast moving (high energy) electrons, accelerated through tens of kilovolts, are suddenly stopped by a metal target. The rapid deceleration produces braking radiation (bremsstrahlung) in the X-ray range. Slow electrons do not have enough energy to give X-ray photons.

Reason (R): X-rays consist of low-energy photons. This is also FALSE. X-rays are high-energy photons, typically from about 100 eV to 100 keV, corresponding to short wavelengths of roughly 0.01 to 10 nm.

Since both statements are false, the correct choice is option (D). Options (A), (B) and (C) each need at least one statement to be true, so they are ruled out.

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q24*

Q25

Out of the four options given, in which transition will the emitted photon have the maximum wavelength?

(A) $n = 4$ **to** $n = 3$

(B) $n = 3$ to $n = 2$

(C) $n = 2$ to $n = 1$

(D) $n = 3$ to $n = 1$

[2025 • Set 55-7-1]

SOLUTION

The wavelength of an emitted photon follows the Rydberg relation:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

A larger wavelength λ corresponds to a smaller energy gap, that is the smallest value of $\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$. So compute this quantity for each option.

For $n = 4 \to n = 3$: $\frac{1}{9} - \frac{1}{16} = \frac{16-9}{144} = \frac{7}{144} \approx 0.049$.

For $n = 3 \to n = 2$: $\frac{1}{4} - \frac{1}{9} = \frac{5}{36} \approx 0.139$.

For $n = 2 \to n = 1$: $1 - \frac{1}{4} = 0.75$.

For $n = 3 \to n = 1$: $1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$.

The smallest value belongs to the $n = 4 \to n = 3$ transition, so it gives the smallest energy gap and hence the **maximum wavelength**. The answer is option (A). The

other three transitions have larger energy gaps and therefore shorter wavelengths, so they are ruled out.

Final answer. The n = 4 to n = 3 transition gives the maximum wavelength, so the answer is option **(A)**.

← *Back to Q25*

Q26

The minimum energy required to free the electron from the ground state of the hydrogen atom is:

(A) 3.4 eV

(B) less than 3.4 eV

(C) **13.6 eV**

(D) between 13.6 eV and 3.4 eV

[2025 • Set 55-7-3]

SOLUTION

The minimum energy needed to free the electron from the ground state is the ionisation energy. It equals the energy required to take the electron from $n = 1$ to $n = \infty$, where the energy is zero.

Ground state energy of hydrogen:

$$E_1 = -\frac{13.6}{(1)^2}\ \text{eV} = -13.6\ \text{eV}$$

Energy of the free electron at $n = \infty$:

$$E_\infty = 0\ \text{eV}$$

Ionisation energy:

$$E_{\text{ion}} = E_\infty - E_1 = 0 - (-13.6) = 13.6\ \text{eV}$$

So the minimum energy required to free the ground state electron is **13.6 eV**, which is option (C). Option (A), 3.4 eV, is the ionisation energy from the $n = 2$

level, not the ground state. Options (B) and (D) are smaller than 13.6 eV and so cannot free the most tightly bound electron, so they are ruled out.

Final answer. The minimum energy to free the ground state electron is 13.6 eV, so the answer is option **(C)**.

← *Back to Q26*

Q27

The transition of electron that gives rise to the formation of the second spectral line of the Balmer series in the spectrum of hydrogen atom corresponds to:

(A) $n_f = 2$ and $n_i = 3$

(B) $n_f = 3$ and $n_i = 4$

(C) $n_f = 2$ and $n_i = 4$

(D) $n_f = 2$ and $n_i = \infty$

[2024 • Set 55-1-1]

SOLUTION

The Balmer series consists of transitions that end on the second level, so $n_f = 2$, while the electron starts from $n_i = 3, 4, 5, \ldots$. The lines are counted in order of increasing initial level.

First Balmer line (H_α): $n_i = 3 \rightarrow n_f = 2$.

Second Balmer line (H_β): $n_i = 4 \rightarrow n_f = 2$.

Third Balmer line (H_γ): $n_i = 5 \rightarrow n_f = 2$.

So the second spectral line of the Balmer series corresponds to the transition with $n_f = 2$ and $n_i = 4$. Hence the answer is option (C). Option (A) is the first Balmer line, option (B) belongs to the Paschen series (ends at $n = 3$), and option (D) is the series limit, so all three are ruled out.

Final answer. The second Balmer line corresponds to n_f = 2 and n_i = 4, so the answer is option **(C)**.

← Back to Q27

Q28

An alpha particle approaches a gold nucleus in Geiger-Marsden experiment with kinetic energy K. It momentarily stops at a distance d from the nucleus and reverses its direction. Then d is proportional to:

(A) $\frac{1}{K}$

(B) K

(C) $\frac{1}{K}$

(D) K

[2024 • Set 55-2-1]

SOLUTION

At the distance of closest approach the α-particle momentarily stops, so its entire kinetic energy has been converted into electrostatic potential energy between the α-particle (charge 2e) and the gold nucleus (charge Ze). Apply conservation of energy.

$$K = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(Ze)(2e)}{d}$$

Writing $k = \frac{1}{4\pi\varepsilon_0}$:

$$K = \frac{2kZe^2}{d}$$

Solve for the distance of closest approach d:

$$d = \frac{2kZe^2}{K}$$

For a fixed nucleus (Z) and fixed charge, the quantity $2kZe^2$ is constant, so:

$$d \propto \frac{1}{K}$$

Hence d is inversely proportional to the kinetic energy, which is option (C). A faster α-particle (larger K) can push closer before stopping, so d falls as K rises. Options (A), (B) and (D) give the wrong dependence and are ruled out.

Final answer. The distance of closest approach satisfies d proportional to 1/K, so the answer is option **(C)**.

← *Back to Q28*

Q29

Energy levels A, B and C of an atom correspond to increasing values of energy i.e. $E_A BC$. *Let* λ_1, λ_2 *and* λ_3 *be the wavelengths of radiation corresponding to the transitions C to B, B to A and C to A, respectively. The correct relation between* λ_1, λ_2 *and* λ_3 *is:*

(A) $\lambda_1^2+\lambda_2^2=\lambda_3^2$

(B) $\frac{1}{\lambda_1}+\frac{1}{\lambda_2}=\frac{1}{\lambda_3}$

(C) $\lambda_1+\lambda_2+\lambda_3=0$

(D) $\lambda_1+\lambda_2=\lambda_3$

[2024 • Set 55-2-1]

SOLUTION

Let the three energy levels be $E_A < E_B < E_C$. Picture the level diagram: C is the highest line, B in the middle, A the lowest. The three downward jumps and their photon wavelengths are: $C \to B$ gives λ_1, $B \to A$ gives λ_2, and the direct jump $C \to A$ gives λ_3.

The photon energy for each transition equals the energy difference between the levels, using $E = \frac{hc}{\lambda}$.

For $C \to B$: $\frac{hc}{\lambda_1} = E_C - E_B$.

For $B \to A$: $\frac{hc}{\lambda_2} = E_B - E_A$.

For $C \to A$: $\frac{hc}{\lambda_3} = E_C - E_A$.

Add the first two equations:

$$\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = (E_C - E_B) + (E_B - E_A) = E_C - E_A$$

But $E_C - E_A = \frac{hc}{\lambda_3}$, so:

$$\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = \frac{hc}{\lambda_3}$$

Cancel the common factor hc:

$$\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$$

This is the required relation, option (B). It comes directly from energy adding up, so the relations in (A), (C) and (D) do not follow and are ruled out.

COMMON MISTAKE

Energies add, not wavelengths. Because energy is proportional to 1/λ, the conserved quantity across a two-step path is the sum of the reciprocals of the wavelengths.

Final answer. The wavelengths satisfy $1/\lambda_1 + 1/\lambda_2 = 1/\lambda_3$, so the answer is option **(B)**.

← *Back to Q29*

Q30

*An electron makes a transition from **n=2** level to **n=1** level in the Bohr model of a hydrogen atom. Its period of revolution:*

(A) increases by 87.5%

(B) decreases by 87.5%

(C) increases by 43.75%

(D) decreases by 43.75%

[2024 • Set 55-3-1]

SOLUTION

In the Bohr model the period of revolution is $T_n = \frac{2\pi r_n}{v_n}$. Using the standard results $r_n \propto n^2$ and $v_n \propto \frac{1}{n}$:

$$T_n \propto \frac{n^2}{1/n} = n^3$$

So the period scales as the cube of the principal quantum number. Compare the two levels:

$$\frac{T_1}{T_2} = \frac{1^3}{2^3} = \frac{1}{8}$$

Going from $n = 2$ to $n = 1$, the period falls from T_2 to $T_1 = \frac{T_2}{8}$. The fractional decrease, taken with respect to the initial period T_2, is:

$$\frac{T_2 - T_1}{T_2} = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{8} = \frac{7}{8}$$

Convert to a percentage:

$$\frac{7}{8} = 0.875 = 87.5\%$$

So the period of revolution **decreases by 87.5%**, which is option (B). Option (A) wrongly says it increases, and options (C) and (D) use a wrong fraction, so they are ruled out.

Final answer. The period of revolution decreases by 87.5%, so the answer is option **(B)**.

← *Back to Q30*

Q31

A proton and an alpha particle having equal velocities approach a target nucleus. They come momentarily to rest and then reverse their directions. The ratio of the

distance of closest approach of the proton to that of the alpha particle will be:

(A) $\frac{1}{2}$

(B) 2

(C) $\frac{1}{4}$

(D) 4

[2024 • Set 55-3-1]

SOLUTION

At the distance of closest approach the projectile momentarily stops, so its entire initial kinetic energy has been converted into electrostatic potential energy between the projectile (charge q) and the target nucleus (charge Ze).

$$\frac{1}{2}mv^2=\frac{1}{4\pi\varepsilon_0}\frac{q\,(Ze)}{r_0}$$

Solving for the distance of closest approach r_0:

$$r_0=\frac{1}{4\pi\varepsilon_0}\frac{q\,(Ze)\cdot 2}{mv^2}=\frac{kqZe}{\frac{1}{2}mv^2}$$

Both particles approach the same target nucleus with the same velocity v. So Z, v and the constants are common, leaving $\mathbf{r_0\propto\frac{q}{m}}$.

For the proton: $\boldsymbol{q=e}$, $\boldsymbol{m=m_p}$, so $\frac{q}{m}=\frac{e}{m_p}$.

For the alpha particle: $\boldsymbol{q=2e}$, $\boldsymbol{m=4m_p}$, so $\frac{q}{m}=\frac{2e}{4m_p}=\frac{e}{2m_p}$.

Taking the ratio of the two distances:

$$\frac{{r_0}^p}{{r_0}^\alpha}=\frac{e/m_p}{e/2m_p}=2$$

So the proton, which has a smaller charge-to-mass-driven repulsion ratio than the alpha, can get half as close, giving $\mathbf{{r_0}^p:{r_0}^\alpha=2:1}$. The correct choice is **option (B), 2**.

TIP

At closest approach KE becomes zero, so always equate the full kinetic energy to the Coulomb potential energy; never carry a leftover velocity term.

Final answer. The ratio of distances of closest approach (proton to alpha) is 2, so the answer is option **(B)**.

← *Back to Q31*

Q32

In Balmer series of hydrogen atom, as the wavelength of spectral lines decreases, they appear

(A) equally spaced and equally intense.

(B) further apart and stronger in intensity.

(C) closer together and stronger in intensity.

(D) closer together and weaker in intensity.

[2024 • Set 55-4-1]

SOLUTION

The Balmer series consists of transitions of the hydrogen electron from higher orbits ($\mathbf{n_i}$**=3,4,5,**) down to the second orbit ($\mathbf{n_f}$**=2**). The wavelengths are given by $\frac{1}{\lambda}=R(\frac{1}{2^2}-\frac{1}{n_i^2})$.

As n_i increases, $\frac{1}{n_i^2}$ becomes very small, so $\frac{1}{\lambda}$ approaches the fixed limit $\frac{R}{4}$ and λ decreases toward the series limit (**≈ 364.6 nm**).

Because successive values of $\frac{1}{n_i^2}$ ($\frac{1}{9},\frac{1}{16},\frac{1}{25},$) get closer and closer together, the spectral lines crowd together as the wavelength decreases. So the lines appear **closer together**, not equally spaced or further apart, which rules out options (A) and (B).

The intensity of a line depends on how many atoms make that particular jump. Atoms in very high excited states ($\mathbf{n_i}$**=5,6,**) are far fewer than those in lower

excited states, so transitions from them are rarer and the lines they produce are **weaker**. This rules out option (C), which wrongly says stronger.

Hence shorter wavelength Balmer lines appear closer together and weaker in intensity, which is **option (D)**.

Final answer. Shorter-wavelength Balmer lines appear closer together and weaker in intensity, so the answer is option **(D)**.

← *Back to Q32*

Q33

The radius (r_n) *of* n^{th} *orbit in Bohr model of hydrogen atom varies with n as*

(A) $r_n \propto n$

(B) $r_n \propto \frac{1}{n}$

(C) $r_n \propto n^2$

(D) $r_n \propto \frac{1}{n^2}$

[2024 • Set 55-4-1]

SOLUTION

In Bohr's model, balancing the Coulomb attraction against the centripetal requirement and applying angular-momentum quantization $\mathbf{mvr}=\frac{\mathbf{nh}}{\mathbf{2\pi}}$ leads to the radius of the nth orbit:

$$r_n=\frac{\varepsilon_0 n^2 h^2}{\pi m e^2}=n^2 a_0$$

where $\mathbf{a_0=0.529}$ is the Bohr radius. Every quantity except n is a constant, so the radius depends only on n^2.

$$r_n \propto n^2$$

This rules out options (A) and (B), which give a linear or inverse dependence, and option (D), which is the inverse-square dependence (that is actually how the orbital energy magnitude shrinks, not the radius).

So the orbit radius grows as the square of the principal quantum number, which is **option (C)**. As a check, $\mathbf{r_2=4a_0}$ and $\mathbf{r_3=9a_0}$, so successive orbits are spaced increasingly far apart.

Final answer. The Bohr orbit radius varies as $\mathbf{r_n \propto n^2}$, so the answer is option **(C)**.

← *Back to Q33*

Q34

Assertion (A): *An alpha particle is moving towards a gold nucleus. The impact parameter is maximum for the scattering angle of* 180°.
Reason (R): *The impact parameter in an alpha particle scattering experiment does not depend upon the atomic number of the target nucleus.*

(D) Both Assertion (A) and Reason (R) are false.

[2024 • Set 55-5-1]

SOLUTION

The impact parameter b and the scattering angle θ in Rutherford scattering are linked by:

$$b=\frac{1}{4\pi\varepsilon_0}\frac{Ze^2\cot(\theta/2)}{K}$$

Here K is the kinetic energy of the alpha particle and Z the atomic number of the target nucleus.

Examining the Assertion: for a head-on collision the alpha particle is turned right back, giving $\theta=180°$. Putting $\theta=180°$ gives $\cot(90°)=0$, so $b=0$. Thus $\theta=180°$ corresponds to the **minimum** (zero) impact parameter, not the maximum. A large impact parameter instead gives a small scattering angle. So the Assertion is false.

Examining the Reason: the formula clearly contains Z, so for the same θ and K the impact parameter does depend on the atomic number of the target. Hence the Reason is also false.

Since both statements are false, the correct choice is **option (D)**.

CONCEPT

Remember the limiting cases: $b \to 0$ gives $\theta \to 180°$ (head-on), and $b \to \infty$ gives $\theta \to 0°$ (no deflection).

Final answer. Both the Assertion and the Reason are false, so the answer is option **(D)**.

← *Back to Q34*

Q35

The energy of an electron in the ground state of hydrogen atom is −13.6 eV. *The kinetic and potential energy of the electron in the first excited state will be:*

(A) −13.6 eV, 27.2 eV

(B) −6.8 eV, 13.6 eV

(C) **3.4 eV, −6.8 eV**

(D) 6.8 eV, −3.4 eV

[2024 • Set 55-5-1]

SOLUTION

For the hydrogen atom the total energy in the nth orbit is $\mathbf{E_n = \frac{-13.6}{n^2}}$ **eV**. By the virial theorem for the Coulomb field, the kinetic and potential energies are tied to the total energy by fixed relations:

$$KE_n = -E_n = +\frac{13.6}{n^2} \text{ eV}$$

$$U_n = 2E_n = \frac{-27.2}{n^2} \text{ eV}$$

The first excited state corresponds to $n=2$. First find the total energy:

$$E_2 = \frac{-13.6}{2^2} = \frac{-13.6}{4} = -3.4 \text{ eV}$$

Kinetic energy is the positive of this:

$$KE_2 = -E_2 = +3.4 \text{ eV}$$

Potential energy is twice the total energy:

$$U_2 = 2E_2 = 2(-3.4) = -6.8 \text{ eV}$$

So the kinetic energy is **+3.4 eV** and the potential energy is **−6.8 eV**, which matches **option (C)**. The other options either swap signs or use $n=1$ values, so they are wrong.

Final answer. In the first excited state, KE **=3.4 eV** and PE **=−6.8 eV**, so the answer is option **(C)**.

← *Back to Q35*

Q36

The radius of the nth *orbit in Bohr model of hydrogen atom is proportional to:*

(A) n

(B) $\frac{1}{n}$

(C) $\mathbf{n^2}$

(D) $\frac{1}{n^2}$

[2023 • Set 55-1-1]

SOLUTION

Start from Bohr's two conditions. The Coulomb attraction supplies the centripetal force for the circular orbit:

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} = \frac{mv^2}{r}$$

which rearranges to:

$$mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

Bohr's quantization of angular momentum gives:

$$mvr = \frac{nh}{2\pi}$$

Solving these together to eliminate v yields the radius of the nth orbit:

$$r_n = \frac{\varepsilon_0 n^2 h^2}{\pi m e^2} = n^2 a_0$$

Since all factors apart from n are constants, the radius is proportional to n^2.

$$r_n \propto n^2$$

This is **option (C)**. Quick check: $\mathbf{r_2 = 4a_0}$, $\mathbf{r_3 = 9a_0}$, $\mathbf{r_4 = 16a_0}$, confirming the rapid n^2 growth and ruling out the linear and inverse choices.

Final answer. The Bohr orbit radius is proportional to $\mathbf{n^2}$, so the answer is option **(C)**.

← *Back to Q36*

Q37

Hydrogen atom initially in the ground state, absorbs a photon which excites it to ***n*=5** *level. The wavelength of the photon is:*

(A) 975 nm

(B) 740 nm

(C) 523 nm

(D) 95 nm

[2023 • Set 55-1-1]

SOLUTION

When a ground-state hydrogen atom absorbs a photon and jumps to ***n*=5**, the photon's energy must equal the difference between the two energy levels:

$$E_{photon} = E_5 - E_1$$

The energy levels are $\mathbf{E_n = \frac{-13.6}{n^2}\ eV}$, so:

$$E_1 = -13.6 \text{ eV},\ E_5 = \frac{-13.6}{25} = -0.544 \text{ eV}$$

$$E_{photon} = -0.544 - (-13.6) = 13.056 \text{ eV}$$

The wavelength of the absorbed photon follows from $E=\frac{hc}{\lambda}$, using the convenient value **hc=1240 eV·nm**:

$$\lambda=\frac{hc}{E_{photon}}=\frac{1240}{13.056}\ \text{nm}$$

$$\lambda\approx 94.9\ \text{nm}\approx 95\ \text{nm}$$

This lies in the far-ultraviolet region. The value **95 nm** matches **option (D)**; the larger wavelengths in the other options would correspond to much smaller energy jumps.

Final answer. The absorbed photon has wavelength about **95 nm**, so the answer is option **(D)**.

← *Back to Q37*

Q38

The atomic number of an atom represents:

(A) number of neutrons in nucleus.

(B) total number of protons and electrons in the atom.

(C) **number of protons in nucleus.**

(D) total number of protons and neutrons in nucleus.

[2023 • Set 55-1-2]

SOLUTION

The atomic number Z of an element is defined as the number of protons present in the nucleus of one of its atoms. This count fixes the identity of the element, so two atoms with the same Z are always the same element.

In a neutral atom the number of orbiting electrons equals Z, but it is the proton count, not the electron count, that gives the definition.

Checking each option:

Option (A), number of neutrons, is wrong; the neutron number is $N=A-Z$, a separate quantity.

Option (B), total protons plus electrons, is wrong; that would be 2Z in a neutral atom and is not the definition.

Option (C), number of protons in the nucleus, is the correct definition of Z.

Option (D), protons plus neutrons, is wrong; that sum is the mass number A.

Hence the atomic number represents the number of protons in the nucleus, which is **option (C)**.

Final answer. The atomic number is the number of protons in the nucleus, so the answer is option **(C)**.

← *Back to Q38*

Q39

*A hydrogen atom makes a transition from **n**=5 to **n**=1 orbit. The wavelength of photon emitted is λ. The wavelength of photon emitted when it makes a transition from **n**=5 to **n**=2 orbit is*

(A) $\frac{3}{2}\lambda$

(B) $\frac{16}{7}\lambda$

(C) $\frac{24}{7}\lambda$

(D) $\frac{32}{7}\lambda$

[2023 • Set 55-2-1]

SOLUTION

The wavelength of an emitted photon follows the Rydberg relation, where the inverse wavelength is proportional to the difference of the inverse squares of the levels:

$$\frac{1}{\lambda}=R\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$$

For the **5→ 1** transition (the given wavelength λ):

$$\frac{1}{\lambda}=R\left(\frac{1}{1}-\frac{1}{25}\right)=R\cdot\frac{24}{25}$$

For the **5→ 2** transition (call its wavelength λ'):

$$\frac{1}{\lambda'}=R\left(\frac{1}{4}-\frac{1}{25}\right)=R\cdot\frac{21}{100}$$

Dividing the first relation by the second to remove R:

$$\frac{\lambda'}{\lambda}=\frac{1/\lambda}{1/\lambda'}=\frac{24/25}{21/100}=\frac{24}{25}\times\frac{100}{21}=\frac{2400}{525}=\frac{32}{7}$$

So $\lambda'=\frac{32}{7}\lambda$.

The **5→ 2** jump has a smaller energy gap than **5→ 1**, so its wavelength is the longer one, consistent with a multiplying factor greater than 1.

Hence the wavelength of the **5→ 2** photon is $\frac{32}{7}\lambda$**, which is option (D).**

CONCEPT

Longest jump down (**51**) has the biggest energy gap and the shortest wavelength; the smaller-gap **52** jump must give a longer wavelength.

Final answer. The wavelength of the 5 to 2 photon is (32/7) lambda, so the answer is option **(D).**

← Back to Q39

Q40

Specify the transition of electron in the wavelength of the line in the Bohr model of hydrogen atom which gives rise to the spectral line of highest wavelength.

(A) $n=3$ to $n=1$

(B) $n=3$ to $n=2$

(C) $n=4$ to $n=1$

(D) $n=4$ to $n=2$

[2023 • Set 55-2-2]

SOLUTION

The wavelength of an emitted line is inversely related to the energy gap of the transition: $\lambda = \frac{hc}{\Delta E}$. So the highest wavelength corresponds to the smallest energy gap. Compute the gap for each option using $\Delta E = 13.6\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)\text{eV}$.

For **3→ 1**:

$$\Delta E = 13.6\left(1 - \frac{1}{9}\right) = 12.09 \text{ eV}$$

For **3→ 2**:

$$\Delta E = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = 1.89 \text{ eV}$$

For **4→ 1**:

$$\Delta E = 13.6\left(1 - \frac{1}{16}\right) = 12.75 \text{ eV}$$

For **4→ 2**:

$$\Delta E = 13.6\left(\frac{1}{4} - \frac{1}{16}\right) = 2.55 \text{ eV}$$

The smallest energy gap is for the **3→ 2** transition (1.89 eV), which therefore gives the longest wavelength. This is the H_α line of the Balmer series (**$\lambda \approx$ 656 nm**, red).

Hence the transition giving the highest wavelength is ***n*=3** to ***n*=2**, which is **option (B)**.

Final answer. The smallest energy gap, ***n*=3→ *n*=2**, gives the highest wavelength, so the answer is option **(B)**.

← Back to Q40

Q41

The ratio of maximum frequency and minimum frequency of light emitted in Balmer

series of hydrogen spectrum, in Bohr's model, is:

(A) $\frac{9}{5}$

(B) $\frac{5}{9}$

(C) $\frac{16}{7}$

(D) $\frac{7}{16}$

[2023 • Set 55-3-1]

SOLUTION

In the Balmer series of hydrogen, all transitions end on the level $n_f=2$, while the electron jumps in from $n_i=3,4,5$, The frequency of the emitted light is proportional to the energy gap, $\nu \propto \left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$.

The minimum frequency comes from the smallest energy gap, which is the closest jump $n_i=3 \to n_f=2$.

$$\nu_{min} \propto \frac{1}{2^2}-\frac{1}{3^2}=\frac{1}{4}-\frac{1}{9}=\frac{9-4}{36}=\frac{5}{36}$$

The maximum frequency comes from the largest energy gap, which is the series limit $n_i \to \infty \to n_f=2$.

$$\nu_{max} \propto \frac{1}{2^2}-\frac{1}{\infty}=\frac{1}{4}-0=\frac{1}{4}=\frac{9}{36}$$

Taking the ratio, the common factor $\frac{1}{36}$ cancels.

$$\frac{\nu_{max}}{\nu_{min}}=\frac{9/36}{5/36}=\frac{9}{5}$$

So the **ratio of maximum to minimum frequency is 9:5**, which is option (A). The other options either invert this ratio (B) or belong to a different pair of levels (C, D) and so do not match.

Final answer. The ratio is $\frac{9}{5}$, so the answer is option **(A)**.

← *Back to Q41*

Q42

The potential energy of an electron in the second excited state in hydrogen atom is:

(A) −3.4 eV

(B) **−3.02 eV**

(C) −1.51 eV

(D) −6.8 eV

[2023 • Set 55-4-1]

SOLUTION

The ground state is $n=1$, the first excited state is $n=2$, so the **second excited state is $n=3$**.

The total energy of the electron in the nth Bohr orbit of hydrogen is $\mathbf{E_n = -\frac{13.6}{n^2}}$ **eV**.

$$E_3 = -\frac{13.6}{3^2} = -\frac{13.6}{9} = -1.51 \text{ eV}$$

In Bohr's model the potential energy is twice the total energy, that is $\mathbf{U_n = 2E_n}$ (this follows because the kinetic energy equals $-E_n$, and $\mathbf{U = E_n - KE = 2E_n}$).

$$U_3 = 2 \times (-1.51) = -3.02 \text{ eV}$$

So the **potential energy in the second excited state is −3.02 eV**, which is option (B). Option (A) is the total energy of $n=2$, option (C) is the total energy of $n=3$ itself, and option (D) does not correspond to any of these.

CONCEPT

Remember the energy split: $\mathbf{U_n = 2E_n}$ and $\mathbf{KE_n = -E_n}$, so U is always negative and double the total energy.

Final answer. The potential energy is **−3.02 eV**, so the answer is option **(B)**.

← *Back to Q42*

Q43

The ratio of the magnitude of magnetic moment to that of angular momentum, of an electron revolving around the nucleus in a hydrogen atom is:

(A) Zero

(B) 9.27×10^{-24} C/kg

(C) $\mathbf{8.8 \times 10^{10}}$ **C/kg**

(D) 6.6×10^{-12} C/kg

[2023 • Set 55-4-2]

SOLUTION

An electron of charge e moving in an orbit acts like a tiny current loop, so it has a magnetic moment $\boldsymbol{\mu} = \dfrac{e}{2m_e}\mathbf{L}$, where L is its orbital angular momentum and m_e is the electron mass.

Therefore the required ratio is independent of which orbit the electron is in.

$$\frac{\mu}{L} = \frac{e}{2m_e}$$

Substituting $e = 1.6 \times 10^{-19}\ C$ and $m_e = 9.1 \times 10^{-31}$ kg:

$$\frac{\mu}{L} = \frac{1.6 \times 10^{-19}}{2 \times 9.1 \times 10^{-31}}$$

$$\frac{\mu}{L} = \frac{1.6 \times 10^{-19}}{1.82 \times 10^{-30}}$$

$$\frac{\mu}{L} = 8.8 \times 10^{10}\ C/\text{kg}$$

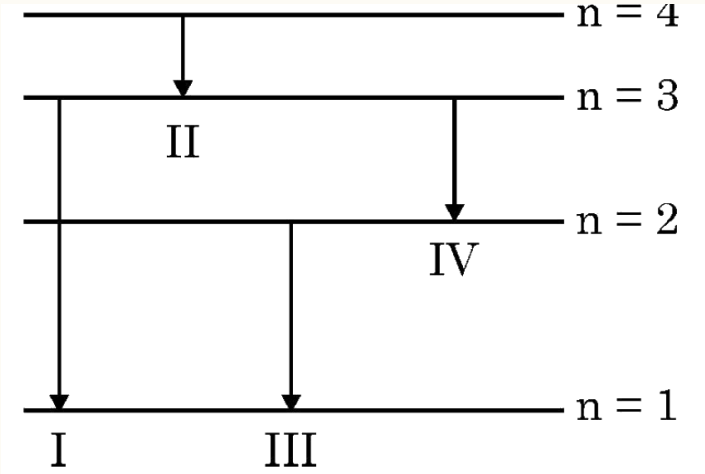
So the **ratio is 8.8×10^{10} C/kg**, which is option (C). This constant is the orbital gyromagnetic ratio of the electron. Option (A) is wrong because the ratio is clearly non-zero, and option (B) is the value of the Bohr magneton μ_B (a magnetic moment in J/T, not this ratio), while option (D) has the wrong order of magnitude.

Final answer. The ratio is $\mathbf{8.8 \times 10^{10}}$ **C/kg**, so the answer is option **(C)**.

← Back to Q43

Q44

The diagram shows four energy level of an electron in Bohr model of hydrogen atom. Identify the transition in which the emitted photon will have the highest energy.



(A) I

(B) II

(C) III

(D) IV

[2023 • Set 55-5-1]

SOLUTION

When an electron jumps from a higher level to a lower level, the energy of the emitted photon equals the gap between the two levels, $\mathbf{E_{photon} = E_i - E_f}$.

So the photon with the highest energy comes from the transition with the **largest gap between its two energy levels**.

In the given diagram the four levels are drawn with the lowest (most negative) energy at the bottom and the higher levels above it. Transition III is the one that starts at the highest shown level and ends at the lowest level, so it covers the biggest vertical drop on the diagram.

Since $\boldsymbol{E \propto \Delta E}$, the largest drop gives the largest E_{photon}.

Therefore **transition III emits the highest-energy photon**, which is option (C). Transitions I, II and IV each span a smaller energy gap on the diagram, so the photons they emit carry less energy.

Final answer. Transition III has the largest energy drop, so the answer is option **(C)**.

← *Back to Q44*

Q45

In Geiger-Marsden scattering experiment, thin gold foil is used to scatter alpha particles because alpha particles will
(A) not suffer more than one scattering and gold nucleus is 50 times heavier than alpha particle.
(B) not suffer more than one scattering and gold nucleus is lighter than alpha particle.
(C) not suffer more than few scatterings and gold nucleus is 25 times heavier than alpha particle.
(D) suffer more than one scattering and gold nucleus is 25 times heavier than alpha particle.

[2021]

SOLUTION

A very thin gold foil is chosen in the Geiger-Marsden experiment for two clear reasons.

First, because the foil is extremely thin, each α-particle passes through only a few atomic layers. This makes sure an α-particle does not suffer more than one scattering, so the deflection it shows is due to a single nucleus. This keeps the analysis simple, with no need to correct for repeated collisions.

Second, gold has a heavy nucleus (Z**=79**, mass number A**=197**). It is about 50 times heavier than the α-particle (A**=4**). Because the gold nucleus is so much heavier, it stays almost fixed during the collision, and the α-particle simply scatters off the strong Coulomb field of this nearly stationary nucleus.

These two points together match option (A): the α-particles **will not suffer more than one scattering, and the gold nucleus is about 50 times heavier**

than the α-particle.

Options (B), (C) and (D) are wrong because the gold nucleus is far heavier (not lighter) than the α-particle, the mass ratio is about 50 (not 25), and a thin foil avoids (rather than causes) multiple scattering.

Final answer. Correct statement is option **(A)**: no more than one scattering and the gold nucleus is about 50 times heavier than the **α**-particle.

← *Back to Q45*

Q46

According to Bohr's atomic model, the circumference of the electron orbit is always an _____ multiple of de Broglie wavelength.

[2020 • Set 55-1-1]

SOLUTION

According to de Broglie, the revolving electron behaves like a wave, and a stable orbit can exist only if this matter wave forms a standing wave around the orbit. For that, a whole number of wavelengths must fit exactly into the circumference.

$$2\pi r = n\lambda,\ n = 1, 2, 3,$$

Here $2\pi r$ is the circumference of the orbit and λ is the de Broglie wavelength of the electron. So the circumference is always an **integral (whole-number) multiple** of the de Broglie wavelength.

If n were not a whole number, the wave would overlap out of step with itself and interfere destructively, so no stable orbit could form. This is exactly the condition that gives Bohr's quantisation of angular momentum, $\mathbf{mvr} = \dfrac{\mathbf{nh}}{\mathbf{2\pi}}$.

Final answer. The blank is filled by integral (whole-number / integer).

← *Back to Q46*

Q47

Which of the following statements is not correct according to Rutherford model?

(A) Most of the space inside an atom is empty.

(B) The electrons revolve around the nucleus under the influence of coulomb force acting on them.

(C) Most part of the mass of the atom and its positive charge are concentrated at its centre.

(D) The stability of atom was established by the model.

[2020 • Set 55-1-1]

SOLUTION

Let us check each statement against Rutherford's nuclear model of the atom.

(A) Most of the space inside an atom is empty. This is correct, since the tiny nucleus holds nearly all the mass while electrons occupy a huge outer volume.

(B) Electrons revolve around the nucleus under the Coulomb force of attraction. This is correct and is exactly what Rutherford proposed.

(C) Most of the mass and all the positive charge are concentrated at the centre (the nucleus). This is correct, and was the main conclusion of the scattering experiment.

(D) The stability of the atom was established by the model. This is **not correct**. A revolving electron is an accelerating charge, and by classical electromagnetism it must continuously radiate energy. Losing energy, it would spiral into the nucleus in about 10^{-8} s, so the atom could not be stable.

Hence the statement that is **not correct is option (D)**. This failure was later removed by Bohr's quantum model.

Final answer. The incorrect statement is option **(D)**: Rutherford's model could not explain atomic stability.

← Back to Q47

Q48

In Bohr's model of hydrogen atom, the total energy of the electron in n^{th} *discrete orbit is proportional to*

(A) n

(B) $\frac{1}{n}$

(C) n^2

(D) $\mathbf{\frac{1}{n^2}}$

[2020 • Set 55-3-1]

SOLUTION

The total energy of the electron in the nth Bohr orbit of hydrogen is given by the standard result of Bohr's model.

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

The numerator 13.6 eV is a fixed constant, so the only variable part is $\frac{1}{n^2}$.

$$E_n \propto \frac{1}{n^2}$$

So the **total energy is proportional to** $\mathbf{\frac{1}{n^2}}$, which is option (D). The energy values $\mathbf{E_1 = -13.6}$, $\mathbf{E_2 = -3.4}$, $\mathbf{E_3 = -1.51}$, $\mathbf{E_4 = -0.85\ eV}$ confirm that the levels get closer together as n grows, exactly as a $1/n^2$ dependence predicts. Options (A), (B) and (C) do not give this rapidly narrowing spacing.

Final answer. The total energy varies as $\mathbf{\frac{1}{n^2}}$, so the answer is option **(D)**.

← *Back to Q48*

Q49

In the α-particle scattering experiment, the shape of the trajectory of the scattered α-particles depend upon:

(A) only on impact parameter.

(B) only on the source of α-particles.

(C) both impact parameter and source of α-particles.

(D) impact parameter and the screen material of the detector.

[2020 • Set 55-4-1]

SOLUTION

In Rutherford's experiment, each α-particle moves in the Coulomb field of a target nucleus and follows a hyperbolic path. For a beam of α-particles with a fixed energy hitting a given target, the shape of this trajectory is decided by the impact parameter b, which is the perpendicular distance between the initial line of approach and the centre of the nucleus.

A large impact parameter means the α-particle passes far from the nucleus, feels a weaker force and is deflected only slightly. A very small impact parameter means a close approach, a stronger repulsion and a large deflection. A head-on approach with $\boldsymbol{b=0}$ gives the maximum, 180° back-scattering.

The 'source' of the α-particles (radium, polonium, etc.) does not itself decide the path; only the particle's own charge, mass and kinetic energy matter, and these are fixed here. The detector screen also plays no part in shaping the trajectory.

Hence, for a given energy, the trajectory shape **depends only on the impact parameter**, which is option (A). Options (B), (C) and (D) wrongly bring in the source or the detector material.

Final answer. The trajectory shape depends only on the impact parameter, so the answer is option **(A)**.

← *Back to Q49*

Q50

Paschen series of atomic spectrum of hydrogen gas lies in:

(A) Infrared region

(B) Ultraviolet region

(C) Visible region

(D) Partly in ultraviolet and partly in visible region

[2020 • Set 55-4-1]

SOLUTION

The hydrogen line series are named by the lower level n_f on which the electron lands. The Paschen series consists of all transitions $\mathbf{n_i = 4, 5, 6, \rightarrow n_f = 3}$.

Because these jumps end on the relatively high level $\mathbf{n_f=3}$, the energy gaps are small, so the emitted photons have low energy and long wavelength.

$$\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n_i^2}\right)$$

These wavelengths come out around 0.82 μm to 1.87 μm, which is well beyond the red end of visible light. Such wavelengths lie in the **infrared region**.

So the Paschen series lies in the infrared region, which is option (A). For reference, the Lyman series ($\mathbf{n_f=1}$) lies in the ultraviolet and the Balmer series ($\mathbf{n_f=2}$) in the visible, so options (B), (C) and (D) do not apply to Paschen.

TIP

Lyman ends on $n=1$ (UV), Balmer on $n=2$ (visible), Paschen on $n=3$ (IR). The higher the landing level, the longer the wavelength.

Final answer. The Paschen series lies in the infrared region, so the answer is option **(A)**.

← *Back to Q50*

Q51

Define ionisation energy. What is its value for a hydrogen atom?

[2012]

SOLUTION

Ionisation energy is the minimum energy required to completely remove the most loosely bound electron from an isolated atom (initially in its ground state) and take it to infinity, where it has zero kinetic energy.

In other words, it is the energy needed to ionise the atom, raising the electron from the lowest energy level $n=1$ to the level $n=\infty$.

$$E_{ion} = E_{\infty} - E_1$$

$$E_{ion} = 0 - E_1 = |E_1|$$

For a hydrogen atom, the ground-state energy in Bohr's model is $\mathbf{E_1 = -13.6\ eV}$.

$$E_{ion} = |-13.6\ eV| = 13.6\ eV$$

Hence the **ionisation energy of hydrogen is 13.6 eV**, which matches the experimentally measured value very closely.

Final answer. The ionisation energy of a hydrogen atom is 13.6 eV.

← *Back to Q51*

Q52

What is the ratio of radii of the orbits corresponding to first excited state and ground state in a hydrogen atom?

[2010]

SOLUTION

In Bohr's model of the hydrogen atom, the radius of the nth allowed orbit is given by $\mathbf{r_n = n^2\, a_0}$, where $\mathbf{a_0 = 0.53\ Å}$ is the Bohr radius.

So the orbit radius is directly proportional to n^2.

The ground state corresponds to $\mathbf{n_1 = 1}$ and the first excited state corresponds to $\mathbf{n_2 = 2}$.

$$\frac{r_2}{r_1} = \frac{(2)^2\, a_0}{(1)^2\, a_0} = \frac{4\, a_0}{a_0}$$

$$\frac{r_2}{r_1} = 4$$

Hence the **ratio of the radii is 4:1**.

Final answer. The ratio of the radii (first excited state to ground state) is 4:1.

← *Back to Q52*

Q53

Prove that, in Bohr model of hydrogen atom, the time period of revolution of an electron in n^{th} *orbit is proportional to* n^3.

[2025 • Set 55-1-1]

SOLUTION

The electron in the nth Bohr orbit moves in a circle of radius r_n with speed v_n, so the time period of one revolution is the circumference divided by the speed.

$$T_n = \frac{2\pi\, r_n}{v_n}$$

From Bohr's theory, the radius of the nth orbit is proportional to n^2.

$$r_n = n^2\, a_0$$

and the speed of the electron is inversely proportional to n.

$$v_n = \frac{v_1}{n}$$

Substituting these into the expression for T_n:

$$T_n = \frac{2\pi\,(n^2\,a_0)}{\left(\frac{v_1}{n}\right)}$$

$$T_n = \frac{2\pi\,a_0\,n^3}{v_1}$$

Since a_0 and v_1 are constants, this gives

$$T_n \propto n^3$$

Hence the **time period of revolution of the electron is proportional to n^3**, which is what was to be proved.

Final answer. Proved: $\mathbf{T_n = \frac{2\pi\,a_0\,n^3}{v_1} \propto n^3}$.

← *Back to Q53*

Q54

The energy of an electron in an orbit of Bohr hydrogen atom is −3.4 eV. *Find its angular momentum.*

[2025 • Set 55-1-2]

SOLUTION

The energy of the electron in the nth Bohr orbit of hydrogen is given by $\mathbf{E_n = -\frac{13.6}{n^2}\ eV}$.

First find the orbit number n for the given energy.

$$-3.4 = -\frac{13.6}{n^2}$$

$$n^2 = \frac{13.6}{3.4} = 4$$

$$n = 2$$

The angular momentum of the electron in the nth orbit is quantised according to Bohr's postulate.

$$L = \frac{nh}{2\pi}$$

$$L = \frac{2 \times 6.63 \times 10^{-34}}{2\pi}$$

$$L = \frac{6.63 \times 10^{-34}}{\pi}$$

$$L = 2.11 \times 10^{-34}\ J\ s$$

Hence the **angular momentum of the electron is $2.11 \times 10^{-34}\ J\ s$**.

Final answer. The angular momentum of the electron is about **2.11×10^{-34}** J s (for $n = 2$).

← *Back to Q54*

Q55

An electron in Bohr model of hydrogen atom makes a transition from energy level −1.51 eV *to* −3.40 eV. *Calculate the change in the radius of its orbit. The radius of orbit of electron in its ground state is* 0.53 .

[2025 • Set 55-1-3]

SOLUTION

The energy of the electron in the nth orbit of hydrogen is $\mathbf{E_n = -\frac{13.6}{n^2}\ eV}$. First identify the two orbit numbers from the given energies.

For the level −3.40 eV:

$$n_1^2 = \frac{13.6}{3.40} = 4 \Rightarrow n_1 = 2$$

For the level −1.51 eV:

$$n_2^2 = \frac{13.6}{1.51} = 9 \Rightarrow n_2 = 3$$

So the transition is from $n = 3$ to $n = 2$.

The radius of the nth orbit is $r_n = n^2 a_0$ with $a_0 = 0.53$ Å.

$$r_3 = (3)^2 \times 0.53 = 9 \times 0.53 = 4.77 \text{ Å}$$

$$r_2 = (2)^2 \times 0.53 = 4 \times 0.53 = 2.12 \text{ Å}$$

The change in radius is

$$\Delta r = r_2 - r_3 = 2.12 - 4.77 = -2.65 \text{ Å}$$

The negative sign shows the radius decreases. Hence the **radius of the orbit decreases by $2.65 \text{ Å} = 2.65 \times 10^{-10}$ *m*.**

Final answer. The orbit radius decreases by **$2.65 \text{ Å} = 2.65 \times 10^{-10}$** m.

← Back to Q55

Q56

Find the ratio of minimum to maximum wavelength of radiations emitted when electron jumps from higher energy state into ground state of hydrogen atom.

[2025 • Set 55-4-2]

SOLUTION

When the electron jumps from a higher energy state into the ground state ($\mathbf{n_f = 1}$), the emitted radiations belong to the **Lyman series**. The wavelength is given by the Rydberg formula.

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n_i^2}\right) = R\left(1 - \frac{1}{n_i^2}\right)$$

The maximum wavelength corresponds to the smallest energy gap, i.e. the transition $\mathbf{n_i = 2 \to 1}$.

$$\frac{1}{\lambda_{max}} = R\left(1 - \frac{1}{4}\right) = \frac{3R}{4}$$

The minimum wavelength corresponds to the largest energy gap, i.e. the transition $\mathbf{n_i = \infty \to 1}$.

$$\frac{1}{\lambda_{min}} = R(1 - 0) = R$$

Taking the ratio:

$$\frac{\lambda_{min}}{\lambda_{max}} = \frac{1/R}{4/(3R)} = \frac{3R}{4R} = \frac{3}{4}$$

Hence the **ratio of minimum to maximum wavelength is 3:4**.

Final answer. The ratio of minimum to maximum wavelength is **3:4**.

← *Back to Q56*

Q57

In Bohr's model of hydrogen atom, find the percentage change in the radius of its orbit when an electron makes a transition from $n = 3$ *state to* $n = 2$ *state.*

[2025 • Set 55-4-3]

SOLUTION

In Bohr's model, the radius of the nth orbit is $r_n = n^2 a_0$, where a_0 is the Bohr radius.

For the two states involved:

$$r_3 = (3)^2 a_0 = 9\, a_0$$

$$r_2 = (2)^2 a_0 = 4\, a_0$$

As the electron makes a transition from $n = 3$ to $n = 2$, the radius changes from $9a_0$ to $4a_0$ (a decrease). The percentage change is calculated with respect to the initial radius r_3.

$$\text{Percentage change} = \frac{r_2 - r_3}{r_3} \times 100$$

$$= \frac{4a_0 - 9a_0}{9a_0} \times 100$$

$$= -\frac{5a_0}{9a_0} \times 100$$

$$= -\frac{5}{9} \times 100 = -55.6\%$$

The negative sign indicates a decrease. Hence the **radius of the orbit decreases by about 55.6%**.

Final answer. The orbit radius decreases by about **55.6%**.

← *Back to Q57*

Q58

Define the term, 'distance of closest approach'. A proton of 3.95 MeV *energy approaches a target nucleus* $Z = 79$ *in head-on position. Calculate its distance of closest approach.*

[2025 • Set 55-5-3]

SOLUTION

The **distance of closest approach** is the smallest distance to which a charged projectile (such as a proton or an α-particle) can approach the target nucleus in a head-on collision. At this point the projectile is momentarily brought to rest by the Coulomb repulsion, and its entire kinetic energy has been converted into electrostatic potential energy.

At the closest approach, by conservation of energy, the kinetic energy K equals the Coulomb potential energy between the proton (charge e) and the nucleus (charge Ze).

$$K = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(e)(\mathrm{Ze})}{r_0}$$

Solving for r_0:

$$r_0 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\mathrm{Ze}^2}{K} = \frac{\mathrm{kZe}^2}{K}$$

Convert the kinetic energy into joules.

$$K = 3.95\ \mathrm{MeV} = 3.95 \times 10^6 \times 1.6 \times 10^{-19} = 6.32 \times 10^{-13}\ J$$

Substitute the values with $k = 9 \times 10^9\ N\,m^2C^{-2}$, $Z = 79$, $e = 1.6 \times 10^{-19}\ C$.

$$r_0 = \frac{(9 \times 10^9)(79)(1.6 \times 10^{-19})^2}{6.32 \times 10^{-13}}$$

$$r_0 = \frac{(9 \times 10^9)(79)(2.56 \times 10^{-38})}{6.32 \times 10^{-13}}$$

$$r_0 = \frac{1.82 \times 10^{-26}}{6.32 \times 10^{-13}}$$

$$r_0 = 2.88 \times 10^{-14}\ m$$

Hence the **distance of closest approach is $2.88 \times 10^{-14}\ m$, i.e. about 28.8 fm**.

Final answer. The distance of closest approach is $\mathbf{2.88 \times 10^{-14}}$ m (≈ **28.8** fm).

← *Back to Q58*

Q59

An alpha particle and a deuterium ion are accelerated through the same potential difference. These are then directed towards a target nucleus to make head-on collision. It is observed that their distance of closest approach is the same. Justify it theoretically.

[2025 • Set 55-6-3]

SOLUTION

At the distance of closest approach, the entire kinetic energy of the projectile is converted into Coulomb potential energy between it and the target nucleus. By conservation of energy:

$$K = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Z_{tar}\, e \cdot q}{r_0}$$

where q is the charge of the projectile and $Z_{tar}\, e$ is the charge of the target nucleus.

A projectile of charge $\boldsymbol{q = ze}$ accelerated through a potential difference V gains kinetic energy

$$K = qV = zeV$$

Substituting this K into the energy-balance equation:

$$zeV = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Z_{tar}\, e \cdot (ze)}{r_0}$$

$$r_0 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Z_{tar}\, e \cdot ze}{zeV}$$

The projectile charge factor ze cancels out:

$$r_0 = \frac{Z_{tar}\, e}{4\pi\varepsilon_0\, V}$$

Thus r_0 depends only on the target's atomic number Z_{tar} and the accelerating potential V, and is independent of the projectile's charge and mass.

For the α-particle ($z = 2$) and the deuterium ion ($z = 1$), both are accelerated through the same V and strike the same target nucleus. Therefore the **distance of closest approach is the same for both**, exactly as observed.

CONCEPT

The accelerating energy $K = qV$ carries the projectile charge, and the Coulomb energy also carries it, so the charge always cancels in r_0.

Final answer. Since $\mathbf{r_0 = \frac{Z_{tar}\, e}{4\pi\varepsilon_0\, V}}$ is independent of projectile charge, $\mathbf{r_0}$ is the same for the α-particle and the deuteron.

← *Back to Q59*

Q60

Define the 'distance of closest approach'. An α-particle of kinetic energy K is bombarded on a thin gold foil. Derive an expression for the 'distance of closest approach'.

[2025 • Set 55-7-3]

SOLUTION

The **distance of closest approach** is the minimum distance to which an α-particle approaches the nucleus in a head-on collision before being momentarily brought to rest and turned back by the strong Coulomb repulsion. At this point all of its kinetic energy has been converted into electrostatic potential energy.

Let the α-particle have charge 2e (two protons) and kinetic energy K far away from the gold nucleus of charge Ze, where the potential energy is taken as zero.

As it approaches, the repulsive Coulomb force decelerates it. At the closest approach, a distance d from the nucleus, the particle is instantaneously at rest, so its kinetic energy is fully stored as Coulomb potential energy.

By conservation of energy:

$$K = U(d) = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(2e)(Ze)}{d}$$

$$K = \frac{2Ze^2}{4\pi\varepsilon_0 d}$$

Rearranging for d:

$$d = \frac{2Ze^2}{4\pi\varepsilon_0 K}$$

or equivalently, writing $\boldsymbol{k = \frac{1}{4\pi\varepsilon_0}}$:

$$d = \frac{2kZe^2}{K}$$

where Z is the atomic number of the target (gold, $\boldsymbol{Z = 79}$), e is the elementary charge and K is the initial kinetic energy of the α-particle. This is the required **expression for the distance of closest approach**.

Final answer. The distance of closest approach is $\boldsymbol{d = \frac{2Ze^2}{4\pi\varepsilon_0 K} = \frac{2kZe^2}{K}}$.

← *Back to Q60*

Q61

In Bohr model of hydrogen atom, find the ratio of period of revolution of the electron in the orbit $\boldsymbol{n=2}$ *to that in the orbit* $\boldsymbol{n=1}$.

[2024 • Set 55-2-3]

SOLUTION

In Bohr's model, the time period of revolution of the electron in the nth orbit is the circumference of the orbit divided by the orbital speed.

$$T_n = \frac{2\pi r_n}{v_n}$$

The radius of the nth orbit and the orbital speed depend on n as follows.

$$r_n \propto n^2 \text{ and } v_n \propto \frac{1}{n}$$

Substituting these dependences into the expression for T_n, we get the variation of the period with n.

$$T_n \propto \frac{r_n}{v_n} \propto \frac{n^2}{1/n} = n^3$$

Therefore the required ratio of the period in the n=2 orbit to that in the n=1 orbit is found by taking the cube of the orbit numbers.

$$\frac{T_2}{T_1} = \frac{(2)^3}{(1)^3} = \frac{8}{1}$$

Hence the **ratio of the periods is** $T_2 : T_1 = 8 : 1$, meaning the electron in the second orbit takes eight times as long to complete one revolution as it does in the first orbit.

CONCEPT

Remember the three Bohr scalings: $r_n \propto n^2$, $v_n \propto 1/n$, and so $T_n \propto n^3$. These let you answer ratio questions in one line without computing absolute values.

Final answer. The ratio of the periods is $T_2 : T_1 = 8 : 1$.

← *Back to Q61*

Q62

(a) Two energy levels of an electron in hydrogen atom are separated by 2.55 eV. *Find the wavelength of radiation emitted when the electron makes transition from the*

higher energy level to the lower energy level.
(b) In which series of hydrogen spectrum this line shall fall?

[2024 • Set 55-5-1]

SOLUTION

(a) When an electron jumps from a higher energy level to a lower energy level, the energy difference is carried away as a photon. The wavelength of this emitted radiation is given by the relation between photon energy and wavelength.

$$\Delta E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{\Delta E}$$

Here the energy gap is $\Delta E = 2.55$ **eV**, and it is convenient to use the constant $hc = 1240$ **eV nm**.

$$\lambda = \frac{1240 \text{ eV nm}}{2.55 \text{ eV}}$$

$$\lambda = 486.3 \text{ nm} \approx 486 \text{ nm}$$

So the **wavelength of the emitted radiation is about** 486 nm, which lies in the blue-green part of the visible spectrum.

(b) To identify the series, we find the levels giving $\Delta E = 2.55$ **eV** using $E_n = -\frac{13.6}{n^2}$ **eV**.

$$E_4 - E_2 = -0.85 - (-3.4) = 2.55 \text{ eV}$$

The transition is therefore $n = 4 \to n = 2$. Since it ends at $n = 2$, this line lies in the **Balmer series**, which falls in the visible region.

Final answer. The wavelength is $\lambda \approx 486$ **nm**, and the line lies in the Balmer series (visible region).

← *Back to Q62*

Q63

The earth revolves around the sun in an orbit of radius 1.5×10^{11} *m with orbital speed* 30 km/*s. Find the quantum number that characterises its revolution using Bohr's model in this case (mass of earth* $= 6.0 \times 10^{24}$ kg*).*

[2024 • Set 55-5-1]

SOLUTION

Bohr's quantisation condition states that the angular momentum of a body in orbit is an integral multiple of $\frac{h}{2\pi}$. Applying this to the Earth, treated as a particle revolving around the Sun, gives the quantum number n.

$$L = M\,v\,r = \frac{n\,h}{2\pi}$$

Rearranging to solve for the quantum number n.

$$n = \frac{2\pi\,M\,v\,r}{h}$$

The given data, converted to SI units, are mass $\boldsymbol{M = 6.0 \times 10^{24}\ kg}$, orbital speed $\boldsymbol{v = 30\ km/s = 3.0 \times 10^{4}\ m/s}$, radius $\boldsymbol{r = 1.5 \times 10^{11}\ m}$, and $\boldsymbol{h = 6.626 \times 10^{-34}\ J\,s}$.

First evaluate the product $M\,v\,r$, which is the angular momentum of the Earth.

$$M\,v\,r = (6.0 \times 10^{24})(3.0 \times 10^{4})(1.5 \times 10^{11}) = 2.7 \times 10^{40}\ J\,s$$

Now substitute into the expression for n.

$$n = \frac{2\pi\,(2.7 \times 10^{40})}{6.626 \times 10^{-34}}$$

$$n = \frac{1.696 \times 10^{41}}{6.626 \times 10^{-34}}$$

$$n \approx 2.56 \times 10^{74}$$

Thus the **quantum number characterising the Earth's revolution is** $\boldsymbol{n \approx 2.6 \times 10^{74}}$. Such an enormous value confirms the correspondence principle: for very large n the orbits are so closely spaced that the motion appears continuous and behaves exactly like classical motion.

Final answer. The quantum number is $n \approx 2.6 \times 10^{74}$.

← Back to Q63

Q64

A proton of energy 1.6 MeV *approaches a gold nucleus* **(Z = 79)**. *Find the distance of its closest approach.*

[2024 • Set 55-5-2]

SOLUTION

At the distance of closest approach, the proton momentarily comes to rest because all of its kinetic energy has been converted into the electrostatic potential energy of the proton and the gold nucleus. Equating the initial kinetic energy to the potential energy at that distance gives the closest approach d.

$$K = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(Ze)(e)}{d} = \frac{k\,Z\,e^2}{d}$$

Rearranging to make d the subject.

$$d = \frac{k\,Z\,e^2}{K}$$

The data are $k = 9 \times 10^9\ N\,m^2C^{-2}$, atomic number $Z = 79$, $e = 1.6 \times 10^{-19}\ C$, and kinetic energy $K = 1.6$ MeV. Converting the energy to joules.

$$K = 1.6\ \text{MeV} = 1.6 \times 10^6 \times 1.6 \times 10^{-19}\ J = 2.56 \times 10^{-13}\ J$$

Substituting all values into the expression for d.

$$d = \frac{(9 \times 10^9)(79)(1.6 \times 10^{-19})^2}{2.56 \times 10^{-13}}$$

$$d = \frac{(9 \times 10^9)(79)(2.56 \times 10^{-38})}{2.56 \times 10^{-13}}$$

$$d = \frac{1.82 \times 10^{-26}}{2.56 \times 10^{-13}}$$

$$d \approx 7.1 \times 10^{-14}\ m$$

Therefore the **distance of closest approach is about** $7.1 \times 10^{-14}\ m$. (Note: the proton carries a single charge, so $\mathbf{Z_{proton} = 1}$ and the factor of 2 used for an alpha particle does not appear here.)

COMMON MISTAKE

The standard $\boldsymbol{d = 2kZe^2/K}$ *formula has a factor of 2 because an alpha particle carries charge* 2e. *A proton carries only e, so use* $\boldsymbol{d = kZe^2/K}$ *here, with no factor of 2.*

Final answer. The distance of closest approach is $\boldsymbol{d \approx 7.1 \times 10^{-14}\ m}$.

← *Back to Q64*

Q65

The electron in hydrogen atom is revolving with the speed of $2.2 \times 10^6\ m/s$ *in an orbit of radius* 0.53 . *Calculate the initial frequency of light emitted by the electron using classical physics.*

[2024 • Set 55-5-3]

SOLUTION

According to classical electromagnetism, an electron moving in a circular orbit is an accelerating charge and so it radiates electromagnetic waves at the frequency of its revolution. This frequency equals the reciprocal of the time period of one revolution.

$$\nu = \frac{1}{T} = \frac{v}{2\pi r}$$

The given quantities are the orbital speed $\boldsymbol{v = 2.2 \times 10^6\ m/s}$ and the orbital radius $\boldsymbol{r = 0.53}$ **angstrom** $\boldsymbol{= 0.53 \times 10^{-10}\ m}$.

Substituting these values into the expression for the frequency.

$$\nu = \frac{2.2 \times 10^6}{2\pi\,(0.53 \times 10^{-10})}$$

$$\nu = \frac{2.2 \times 10^6}{3.33 \times 10^{-10}}$$

$$\nu \approx 6.6 \times 10^{15}\ \text{Hz}$$

Hence the **classically predicted frequency of emitted light is about** 6.6×10^{15} Hz, which lies in the ultraviolet region. Importantly, classical theory says the electron would keep radiating, lose energy, and spiral into the nucleus, whereas Bohr's model resolves this by postulating that an electron in a stationary orbit does not radiate at all.

Final answer. The classical frequency of emitted light is $\boldsymbol{\nu \approx 6.6 \times 10^{15}\ \text{Hz}}$.

← Back to Q65

Q66

The ground state energy of hydrogen atom is −13.6 eV. What is the potential energy and kinetic energy of an electron in the third excited state?

[2023 • Set 55-1-1]

SOLUTION

The third excited state corresponds to $\boldsymbol{n = 4}$, because the ground state is the first state ($\boldsymbol{n = 1}$), the first excited state is $\boldsymbol{n = 2}$, the second excited state is $\boldsymbol{n = 3}$, and the third excited state is $\boldsymbol{n = 4}$.

The total energy of the electron in the nth orbit of hydrogen is given by the following relation.

$$E_n = \frac{-13.6}{n^2}\ \text{eV}$$

For $\boldsymbol{n = 4}$, the total energy is obtained by substituting the value of n.

$$E_4 = \frac{-13.6}{(4)^2} = \frac{-13.6}{16} = -0.85\ \text{eV}$$

In any hydrogen orbit the potential energy is twice the total energy, and the kinetic energy is the negative of the total energy.

$$U_n = 2\,E_n \text{ and } K_n = -E_n$$

The potential energy in the third excited state is therefore as follows.

$$U_4 = 2 \times (-0.85) = -1.70 \text{ eV}$$

The kinetic energy in the third excited state is as follows.

$$K_4 = -(-0.85) = +0.85 \text{ eV}$$

Hence in the third excited state the **potential energy is** −1.70 eV and the **kinetic energy is** +0.85 eV.

COMMON MISTAKE

The third excited state is $n = 4$*, not* $n = 3$*. Count the ground state as* $n = 1$*, then each excited state adds one to n.*

Final answer. In the third excited state, $U = -1.70$ **eV** and $K = +0.85$ **eV**.

← Back to Q66

Q67

Calculate the wavelength of the second line of Lyman series in a spectrum of hydrogen atom. (Take Rydberg constant, $R = 1.1 \times 10^7\ m^{-1}$*.)*

[2023 • Set 55-1-2]

SOLUTION

The Lyman series consists of all transitions ending at the ground level, so $n_f = 1$. The first line is the transition $n_i = 2 \to 1$, and therefore the second line of the Lyman series is the transition $n_i = 3 \to 1$.

The wavenumber for a hydrogen transition is given by Rydberg's formula.

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

Substituting $n_f = 1$ and $n_i = 3$.

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = R\left(1 - \frac{1}{9}\right) = R \cdot \frac{8}{9}$$

Putting in the value $R = 1.1 \times 10^7\ m^{-1}$.

$$\frac{1}{\lambda} = (1.1 \times 10^7) \times \frac{8}{9} = 9.78 \times 10^6 \text{ m}^{-1}$$

Taking the reciprocal to find the wavelength.

$$\lambda = \frac{1}{9.78 \times 10^6} \approx 1.023 \times 10^{-7}\, m$$

Therefore the **wavelength of the second line of the Lyman series is $\lambda \approx 1.023 \times 10^{-7}\, m = 102.3$ nm**, which lies in the ultraviolet region of the spectrum.

TIP

For the Lyman series fix $\mathbf{n_f = 1}$. The first line uses $\mathbf{n_i = 2}$, the second line uses $\mathbf{n_i = 3}$, the third line uses $\mathbf{n_i = 4}$, and so on.

Final answer. The wavelength of the second Lyman line is $\boldsymbol{\lambda \approx 1.023 \times 10^{-7}\, m = 102.3}$ **nm.**

← *Back to Q67*

Q68

The ground state energy of hydrogen atom is −13.6 *eV in Bohr model. An electron makes a transition from an energy level of* −1.51 *eV to that of* −3.4 *eV. Calculate the wavelength of the spectral line emitted.*

[2023 • Set 55-1-3]

SOLUTION

When the electron drops from a higher level to a lower level, a photon is emitted whose energy equals the difference between the two levels. The wavelength of the emitted spectral line is related to this energy difference.

$$\lambda = \frac{hc}{\Delta E}$$

The energy released in the transition is the difference between the initial and final energy levels.

$$\Delta E = E_i - E_f = (-1.51) - (-3.4) = 1.89 \text{ eV}$$

Using the convenient constant **hc = 1240 eV nm**, the wavelength is found by substitution.

$$\lambda = \frac{1240 \text{ eV nm}}{1.89 \text{ eV}}$$

$$\lambda \approx 656 \text{ nm}$$

Therefore the **wavelength of the spectral line emitted is about** 656 nm, which is red light. The level −1.51 eV corresponds to $n = 3$ and the level −3.4 eV corresponds to $n = 2$, so this is the $n = 3 \to n = 2$ transition, the H_α line of the Balmer series.

Final answer. The wavelength of the emitted spectral line is $\lambda \approx 656$ nm (red, Balmer H_α).

← *Back to Q68*

Q69

What is meant by ionisation energy? Write its value for hydrogen atom.

[2023 • Set 55-2-1]

SOLUTION

Ionisation energy is the minimum energy required to remove the most loosely bound electron completely from a free atom in its ground state, taking it from the atom to rest at infinity, so that the neutral atom becomes a positive ion and a free electron.

Since the electron must be raised from the ground level E_1 to the level at infinity where the energy is zero, the ionisation energy equals the magnitude of the ground-state energy.

$$E_{ion} = E_\infty - E_1 = 0 - E_1 = |E_1|$$

For the hydrogen atom in the Bohr model, the ground-state energy is the following value.

$$E_1 = -13.6 \text{ eV}$$

Therefore the ionisation energy of hydrogen is the magnitude of this energy.

$$E_{ion} = |-13.6\ eV| = 13.6\ eV$$

Hence the **ionisation energy of the hydrogen atom is** 13.6 eV, which agrees with the experimentally measured value.

Final answer. Ionisation energy is the minimum energy to free the most loosely bound electron; for hydrogen it is **13.6 eV**.

← *Back to Q69*

Q70

Draw the graph showing the variation of the number (N) of scattered alpha particles with scattering angle (θ) in Geiger-Marsden experiment. Infer two conclusions from the graph.

[2022 • Set 55-1-1]

SOLUTION

Graph: On the horizontal axis plot the scattering angle θ (ranging from 0° to 180°) and on the vertical axis plot the number N of scattered alpha particles detected per unit area of the screen. The curve starts very high near small angles, then falls extremely steeply as θ increases, and flattens to a very small, nearly zero value as θ approaches 180°. The curve never touches zero; a tiny but non-zero number of particles is still detected even at **θ = 180°** (complete back-scattering). The shape obeys the relation predicted by Rutherford.

$$N \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$

First conclusion: Most of the alpha particles are scattered through very small angles, showing that **most of the space inside an atom is empty**, so the particles pass through almost undeflected.

Second conclusion: A very small fraction of alpha particles are scattered through large angles, some even bouncing back near 180°. This is possible only

if the entire positive charge and almost the whole mass of the atom are concentrated in a very small, dense central region, which is the **nucleus**.

Final answer. The N-versus-θ curve falls steeply with angle; it shows the atom is mostly empty space with a tiny dense positively charged nucleus at the centre.

← *Back to Q70*

Q71

(i) Define the terms: 'impact parameter' and 'distance of closest approach' for an α*-particle in Geiger-Marsden scattering experiment.*
(ii) What will be the value of the impact parameter for scattering angle (I) $\theta = 0°$ *and (II)* $\theta = 180°$*?*

[2022 • Set 55-2-1]

SOLUTION

In the Geiger-Marsden experiment, two geometric quantities describe how an α-particle approaches the nucleus.

(i) Impact parameter (b) is the perpendicular distance between the initial velocity vector of the α-particle (its line of approach when far from the nucleus) and a parallel line drawn through the centre of the nucleus.

Distance of closest approach (r_0) is the smallest distance to which a head-on α-particle (one fired directly at the nucleus, $b = 0$) can come before it momentarily stops and turns back. At this point its entire kinetic energy has converted into electrostatic potential energy, so

$$\frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0}$$

(ii) The impact parameter is related to the scattering angle θ by Rutherford's relation $b \propto \cot\left(\frac{\theta}{2}\right)$.

(I) For $\theta = 0°$: $\cot(0°2) = \cot 0° \to \infty$, so $b \to \infty$. The particle passes very far from the nucleus and is **not deflected**.

(II) For $\theta = 180°$: $\cot(180°2) = \cot 90° = 0$, so $b = 0$. This is a **head-on collision** and the particle is scattered straight back.

COMMON MISTAKE

Use cot(theta/2), not cot(theta). At theta = 180 degrees the half-angle is 90 degrees, whose cotangent is zero, giving a head-on hit.

Final answer. For $\theta = 0°$, $b \to \infty$; for $\theta = 180°$, $b = 0$.

← Back to Q71

Q72

Name the spectral series for a hydrogen atom which lies in the visible region. Find the ratio of the maximum to the minimum wavelengths of this series.

[2022 • Set 55-4-1]

SOLUTION

The spectral series of hydrogen that lies in the **visible region** is the **Balmer series**, in which the electron makes transitions to the final level $n_f = 2$.

The wavelengths of this series are given by the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n_i^2}\right)$$

The maximum wavelength corresponds to the smallest energy jump, i.e. the transition $n_i = 3 \to n_f = 2$:

$$\frac{1}{\lambda_{max}} = R\left(\frac{1}{4} - \frac{1}{9}\right) = R \cdot \frac{9-4}{36} = \frac{5R}{36}$$

The minimum wavelength corresponds to the series limit, i.e. $n_i \to \infty \to n_f = 2$:

$$\frac{1}{\lambda_{min}} = R\left(\frac{1}{4} - 0\right) = \frac{R}{4} = \frac{9R}{36}$$

Taking the ratio,

$$\frac{\lambda_{max}}{\lambda_{min}} = \frac{1/\lambda_{min}}{1/\lambda_{max}} = \frac{9R/36}{5R/36} = \frac{9}{5}$$

Hence $\lambda_{max} : \lambda_{min} = 9 : 5$ for the Balmer series.

Final answer. The Balmer series lies in the visible region, with $\lambda_{max} : \lambda_{min} = 9 : 5$.

← *Back to Q72*

Q73

What results do you expect if α-particle scattering experiment is repeated using a thin sheet of hydrogen in place of a gold foil? Explain. (Hydrogen is a solid at temperature below 14 K)

[2022 • Set 55-5-1]

SOLUTION

In Rutherford's original experiment, gold has a large atomic number ($Z = 79$), so its nucleus carries a large positive charge and is far heavier than an α-particle. This produces a strong Coulomb repulsion, deflecting some α-particles through large angles, with a few even bouncing straight back ($\theta \approx 180°$). This large-angle and back-scattering was the key evidence for a small, massive, positively charged nucleus.

If gold is replaced by a thin sheet of solid hydrogen ($Z = 1$), two changes occur.

First, the Coulomb force between the α-particle and the hydrogen nucleus is much weaker, because the nuclear charge is only 1e instead of 79e. So the deflections produced are very small.

Second, and more important, the hydrogen nucleus (a proton, mass $\approx 1\ u$) is far **lighter** than the α-particle (mass $\approx 4\ u$). By the laws of mechanics, a heavy moving body cannot be bounced backward by a much lighter stationary one. The light proton simply gets pushed forward and the α-particle continues almost undeviated.

Therefore, with a hydrogen target, **almost no large-angle scattering and no back-scattering would be observed**. The α-particles would pass through with only tiny deflections, and the experiment would fail to reveal a concentrated heavy nucleus.

Final answer. Almost all α-particles pass straight through with tiny deflections; no large-angle or back-scattering occurs.

← *Back to Q73*

Q74

Using Bohr's atomic model, derive the expression for the radius of nth *orbit of the revolving electron in a hydrogen atom.*

[2020 • Set 55-1-1]

SOLUTION

By Bohr's first postulate, the electron revolves in certain stable circular orbits without radiating energy; these are the **stationary orbits**. The electrostatic (Coulomb) attraction between the nucleus (charge $+e$) and the electron (charge $-e$) supplies the centripetal force needed for this circular motion:

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r^2} \quad ...(1)$$

By Bohr's second (quantisation) postulate, the angular momentum of the electron is an integral multiple of $\frac{h}{2\pi}$:

$$mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi\, mr} \quad ...(2)$$

Substituting v from (2) into (1):

$$\frac{m}{r}\left(\frac{nh}{2\pi\, mr}\right)^2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r^2}$$

$$\frac{n^2 h^2}{4\pi^2 m\, r^3} = \frac{e^2}{4\pi\varepsilon_0\, r^2}$$

Solving for r (writing it as r_n for the nth orbit):

$$r_n = \frac{n^2 h^2 \varepsilon_0}{\pi\, m\, e^2}$$

Thus $\mathbf{r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2} = n^2 a_0}$, where $\mathbf{a_0 = \frac{h^2 \varepsilon_0}{\pi m e^2} = 0.529\ \text{Å}}$ is the **Bohr radius**.
Since $\mathbf{r_n \propto n^2}$, the orbit radii increase rapidly and are not equally spaced.

Final answer. The radius of the $\mathbf{n^t h}$ orbit is $\mathbf{r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2} = n^2 a_0}$, so $\mathbf{r_n \propto n^2}$.

← *Back to Q74*

Q75

Using Bohr's atomic model, derive the expression for the velocity of electron revolving in the nth *orbit of hydrogen atom.*

[2020 • Set 55-1-3]

SOLUTION

The Coulomb attraction between the nucleus and the orbiting electron provides the centripetal force for circular motion. Writing $k = \frac{1}{4\pi\varepsilon_0}$:

$$\frac{mv_n^2}{r_n} = \frac{ke^2}{r_n^2} \Rightarrow v_n^2 = \frac{ke^2}{m r_n} \quad ...(1)$$

By Bohr's quantisation of angular momentum,

$$m v_n r_n = \frac{nh}{2\pi} \Rightarrow r_n = \frac{nh}{2\pi m v_n} \quad ...(2)$$

Substituting r_n from (2) into (1):

$$v_n^2 = \frac{ke^2}{m} \cdot \frac{2\pi m v_n}{nh} = \frac{2\pi k e^2 v_n}{nh}$$

Dividing both sides by v_n:

$$v_n = \frac{2\pi k e^2}{nh}$$

Putting back $k = \frac{1}{4\pi\varepsilon_0}$:

$$v_n = \frac{2\pi e^2}{nh} \cdot \frac{1}{4\pi\varepsilon_0} = \frac{e^2}{2\varepsilon_0 n h}$$

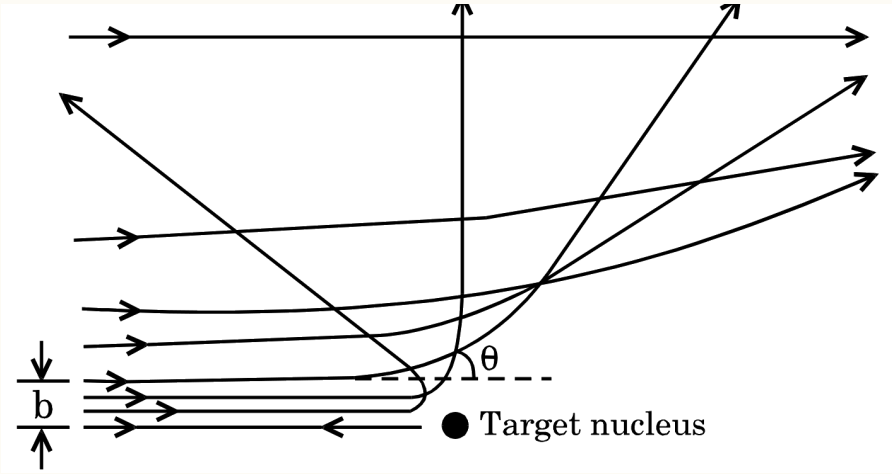
Hence $v_n = \frac{e^2}{2\varepsilon_0 n h}$, showing that $v_n \propto \frac{1}{n}$. The electron moves slower in higher orbits; for the ground state of hydrogen, $v_1 \approx 2.18 \times 10^6\ m\ s^{-1}$.

Final answer. The speed in the nth orbit is $v_n = \frac{e^2}{2\varepsilon_0 n h}$, so $v_n \propto \frac{1}{n}$.

← *Back to Q75*

Q76

In Geiger-Marsden scattering experiment, the trajectory of α-particles in Coulomb's field of a heavy nucleus is shown in the figure.



(a) What do 'b' and 'θ' represent in the figure?
(b) What will be the value of 'b' for (i) $\theta = 0°$*, and (ii)* $\theta = 180°$*?*

[2020 • Set 55-2-1]

SOLUTION

(a) In the figure, the two symbols describe the geometry of the α-particle's path past the nucleus.

b is the **impact parameter**: the perpendicular distance from the nucleus to the initial line of motion of the α-particle (i.e. how far off-centre the particle is aimed as it comes in from far away).

θ is the **scattering angle**: the angle between the final direction of the deflected α-particle and its original direction of incidence.

(b) Rutherford's scattering relation links the two: $b \propto \cot\left(\frac{\theta}{2}\right)$.

(i) For $\theta = 0°$: $\cot(0°2) = \cot 0° \to \infty$, so $b \to \infty$. The α-particle travels far from the nucleus and goes through essentially undeflected.

(ii) For $\theta = 180°$: $\cot(180°2) = \cot 90° = 0$, so $b = 0$. This is a **head-on collision**, and the particle retraces its path straight back.

Hence $b(\theta = 0°) \to \infty$ **and** $b(\theta = 180°) = 0$.

Final answer. b is the impact parameter and θ the scattering angle; $b \to \infty$ at $\theta = 0°$ and $b = 0$ at $\theta = 180°$.

← *Back to Q76*

Q77

A hydrogen atom is in its third excited state.
(a) How many spectral lines can be emitted by it before coming to the ground state? Show these transitions in the energy level diagram.
(b) In which of the above transitions will the spectral line of shortest wavelength be emitted?

[2020 • Set 55-3-1]

SOLUTION

The third excited state of hydrogen corresponds to $n = 4$ (the ground state is $n = 1$, so the first, second and third excited states are $n = 2, 3, 4$).

(a) The maximum number of spectral lines emitted as the atom de-excites from level n to the ground state is

$$N = \frac{n(n-1)}{2}$$

$$N = \frac{4(4-1)}{2} = \frac{4 \times 3}{2} = 6$$

So **6 spectral lines** are emitted, arising from the transitions **43**, **42**, **41**, **32**, **31** and **21**.

Energy level diagram: draw four horizontal lines, the lowest labelled ***n*=1** (ground state, −13.6 eV) at the bottom, then ***n*=2** (−3.4 eV), ***n*=3** (−1.51 eV) and ***n*=4** (−0.85 eV) higher up, with the gaps decreasing towards the top. Draw downward arrows for each of the six transitions: three arrows from ***n*=4** (to ***n*=3**, 2 and 1), two from ***n*=3** (to ***n*=2** and 1) and one from ***n*=2** (to ***n*=1**).

(b) The shortest wavelength goes with the largest energy difference, which is the transition between the most widely separated levels, **4 → 1**:

$$\Delta E = E_4 - E_1 = (-0.85) - (-13.6) = 12.75 \text{ eV}$$

$$\lambda_{min} = \frac{hc}{\Delta E} = \frac{1240 \text{ eV nm}}{12.75 \text{ eV}} \approx 97.3 \text{ nm}$$

Hence the **shortest-wavelength line is emitted in the 4 → 1 transition** (Lyman series), of about 97 nm.

COMMON MISTAKE

Third excited state means n = 4, not n = 3. Count the ground state as n = 1, so excited states start at n = 2.

Final answer. Six spectral lines are emitted; the shortest wavelength (**≈ 97 nm**) comes from the **4 → 1** transition.

← Back to Q77

Q78

Use Bohr's model of hydrogen atom to obtain the relationship between the angular momentum and the magnetic moment of the revolving electron.

[2020 • Set 55-5-1]

SOLUTION

An electron of charge e moving with speed v in a circular orbit of radius r completes one revolution in period $T = \frac{2\pi r}{v}$. This is equivalent to a tiny current loop carrying current

$$I = \frac{e}{T} = \frac{e\,v}{2\pi\,r}$$

The magnetic moment of a current loop is $\mu = I\,A$, where $A = \pi\,r^2$ is the area enclosed:

$$\mu = \frac{e\,v}{2\pi\,r} \cdot \pi\,r^2 = \frac{e\,v\,r}{2}$$

The orbital angular momentum of the electron is

$$L = m\,v\,r$$

Taking the ratio of magnetic moment to angular momentum:

$$\frac{\mu}{L} = \frac{e\,v\,r\,/\,2}{m\,v\,r} = \frac{e}{2m}$$

Hence $\mu = \frac{e}{2m}\,L$. The ratio $\frac{e}{2m}$ is called the **gyromagnetic ratio** and is a constant for the electron.

Using Bohr's quantisation $L = \frac{nh}{2\pi}$, the magnetic moment is quantised as $\mu_n = \frac{neh}{4\pi\,m} = n\mu_B$, where $\mu_B = \frac{eh}{4\pi\,m} = 9.27\times 10^{-24}\,A\,m^2$ is the Bohr magneton.

Final answer. The magnetic moment and angular momentum are related by $\mu = \frac{e}{2m}\,L$.

← *Back to Q78*

Q79

Write shortcomings of Rutherford atomic model. Explain how these were overcome by the postulates of Bohr's atomic model.

[2020 • Set 55-5-3]

SOLUTION

Rutherford's nuclear model successfully explained α-scattering but had two serious shortcomings.

(1) Stability of the atom: A revolving electron is continuously accelerated (centripetal acceleration). By classical electromagnetic theory an accelerating charge radiates energy continuously. The electron would therefore lose energy, spiral inward, and fall into the nucleus in about 10^{-8} *s*. This predicts an **unstable atom**, contradicting the observed stability of matter.

(2) Nature of the spectrum: As the spiralling electron lost energy continuously, it would emit radiation of continuously changing frequency, giving a **continuous spectrum**. But atoms are observed to emit sharp **line (discrete) spectra** characteristic of each element.

Bohr removed these defects with his postulates.

His first postulate (stationary orbits) states that the electron revolves only in certain allowed orbits in which it does **not** radiate energy. This makes the atom stable and removes the stability problem.

His second postulate quantises the angular momentum, $L = \mathbf{mvr} = \dfrac{\mathbf{nh}}{2\pi}$, which permits only discrete orbits with discrete energies $\mathbf{E}_n = -\dfrac{13.6}{\mathbf{n}^2}$ **eV**.

His third postulate (frequency condition) states that radiation is emitted only when the electron jumps from a higher orbit to a lower one, with $\mathbf{h\nu} = \mathbf{E}_i - \mathbf{E}_f$. Because the energies are discrete, only definite frequencies are emitted, giving the observed **line spectrum**.

Final answer. Rutherford's model failed on atomic stability and predicted a continuous spectrum; Bohr's stationary-orbit, quantisation and frequency postulates fixed both.

← *Back to Q79*

Q80

State Bohr's quantization condition of angular momentum. Calculate the shortest wavelength of the Brackett series and state to which part of the electromagnetic spectrum does it belong.

[2019 • Set 55-1-1]

SOLUTION

Bohr's quantisation condition states that an electron can revolve only in those orbits in which its orbital angular momentum is an integral multiple of $\frac{h}{2\pi}$:

$$L = m_e v r_n = \frac{nh}{2\pi},\ n = 1, 2, 3, \ldots$$

For the Brackett series the electron falls to the level $\mathbf{n_f = 4}$, with $\mathbf{n_i = 5, 6, 7, \ldots}$ The wavelengths follow

$$\frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{n_i^2}\right)$$

The shortest wavelength corresponds to the series limit, $\mathbf{n_i \to \infty}$:

$$\frac{1}{\lambda_{min}} = R\left(\frac{1}{16} - 0\right) = \frac{R}{16}$$

$$\lambda_{min} = \frac{16}{R} = \frac{16}{1.097 \times 10^7\ m^{-1}}$$

$$\lambda_{min} = 1.46 \times 10^{-6}\ m = 1460\ nm \approx 1.46\ \mu m$$

This wavelength lies in the **infrared region** of the electromagnetic spectrum.

Final answer. The shortest Brackett wavelength is $\boldsymbol{\lambda_{min} = \frac{16}{R} \approx 1.46\ \mu m}$, in the infrared region.

← Back to Q80

Q81

Calculate the orbital period of the electron in the first excited state of hydrogen atom.

[2019 • Set 55-1-1]

SOLUTION

The first excited state of hydrogen corresponds to $\boldsymbol{n = 2}$.

The period of revolution of the electron in the nth Bohr orbit is the time taken to complete one circular path:

$$T_n = \frac{2\pi r_n}{v_n}$$

In Bohr's model the orbit radius and speed vary with n as $\mathbf{r_n = n^2 a_0}$ and $\mathbf{v_n = \frac{v_1}{n}}$, where $\mathbf{a_0 = 0.53\times 10^{-10}\ m}$ and $\mathbf{v_1 = 2.18\times 10^{6}\ m\ s^{-1}}$.

Substituting these:

$$T_n = \frac{2\pi (n^2 a_0)}{v_1/n} = \frac{2\pi a_0 n^3}{v_1}$$

So the period grows as $\mathbf{T_n \propto n^3}$. For the ground state ($\boldsymbol{n = 1}$):

$$T_1 = \frac{2\pi a_0}{v_1} = \frac{2\pi (0.53\times 10^{-10})}{2.18\times 10^{6}}$$

$$T_1 \approx 1.52\times 10^{-16}\ s$$

For the first excited state ($\boldsymbol{n = 2}$):

$$T_2 = 2^3 T_1 = 8\times 1.52\times 10^{-16}\ s$$

$$T_2 \approx 1.22\times 10^{-15}\ s$$

Hence the electron in the first excited state completes one revolution in about $\mathbf{1.22\times 10^{-15}\ s}$.

TIP

Remember the scalings $\mathbf{r_n \propto n^2}$, $\mathbf{v_n \propto 1/n}$, so $\mathbf{T_n \propto n^3}$; once you know T_1 you can read off any orbit instantly.

Final answer. The orbital period in the first excited state is $\mathbf{T_2 \approx 1.22\times 10^{-15}\ s}$.

← *Back to Q81*

Q82

Obtain the expression for the ratio of the de-Broglie wavelengths associated with the electron orbiting in the second and third excited states of hydrogen atom.

[2019 • Set 55-1-3]

SOLUTION

The second excited state corresponds to $n = 3$ and the third excited state to $n = 4$ (ground state is $n=1$, first excited $n=2$).

According to de Broglie's hypothesis, a stable Bohr orbit contains a whole number of electron wavelengths around its circumference:

$$2\pi\, r_n = n\lambda_n$$

Therefore the de Broglie wavelength in the nth orbit is:

$$\lambda_n = \frac{2\pi\, r_n}{n}$$

The Bohr radius varies as $r_n = n^2\, a_0$, so substituting:

$$\lambda_n = \frac{2\pi\,(n^2\, a_0)}{n} = 2\pi\, a_0\, n$$

This shows the de Broglie wavelength is directly proportional to n:

$$\lambda_n \propto n$$

Taking the ratio for the second ($n=3$) and third ($n=4$) excited states:

$$\frac{\lambda_3}{\lambda_4} = \frac{3}{4}$$

Hence the required ratio is $\lambda_3 : \lambda_4 = 3 : 4$.

Final answer. The ratio of de Broglie wavelengths is $\lambda_3 : \lambda_4 = 3 : 4$.

← *Back to Q82*

Q83

A photon emitted during the de-excitation of electron from a state n to the first excited state in a hydrogen atom, irradiates a metallic cathode of work function 2 eV*, in a photo cell, with a stopping potential of* 0.55 *V. Obtain the value of the quantum number of the state n.*

OR

A hydrogen atom in the ground state is excited by an electron beam of 12.5 eV *energy. Find out the maximum number of lines emitted by the atom from its excited state.*

[2019 • Set 55-2-1]

SOLUTION

The energy of the incident photon is shared between the work function of the metal and the maximum kinetic energy of the ejected photoelectron. Using Einstein's photoelectric equation with stopping potential V_0:

$$E_\gamma = \phi + eV_0$$

$$E_\gamma = 2 + (1)(0.55) = 2.55 \text{ eV}$$

This photon is emitted when the electron de-excites from level n to the first excited state, i.e. $\boldsymbol{n \to 2}$. The transition energy for hydrogen is:

$$\Delta E = 13.6\left(\frac{1}{2^2} - \frac{1}{n^2}\right) \text{ eV} = 2.55 \text{ eV}$$

$$\frac{1}{4} - \frac{1}{n^2} = \frac{2.55}{13.6} = 0.1875 = \frac{3}{16}$$

$$\frac{1}{n^2} = \frac{1}{4} - \frac{3}{16} = \frac{4-3}{16} = \frac{1}{16}$$

$$n^2 = 16 \Rightarrow n = 4$$

Hence the upper state is $\boldsymbol{n = 4}$.

OR

A ground-state hydrogen atom is hit by an electron beam of energy 12.5 eV. The atom can be excited only up to a level whose excitation energy does not exceed the beam energy:

$$\Delta E = 13.6\left(1 - \frac{1}{n^2}\right) \leq 12.5 \text{ eV}$$

For $\boldsymbol{n = 3}$: $\boldsymbol{\Delta E = 13.6\left(1 - \frac{1}{9}\right) = 12.09 \text{ eV}}$ (allowed).

For $\boldsymbol{n = 4}$: $\boldsymbol{\Delta E = 13.6\left(1 - \frac{1}{16}\right) = 12.75 \text{ eV}}$ (exceeds 12.5 eV, not reachable).

So the highest state reached is $n = 3$. The maximum number of spectral lines emitted as the atom de-excites is:

$$N = \frac{n(n-1)}{2} = \frac{3 \times 2}{2} = 3$$

These are the transitions **3→ 2, 3→ 1, 2→ 1**, giving **3 spectral lines**.

Final answer. The quantum number of the upper state is $n = 4$; in the OR part the maximum number of lines emitted is **3**.

← *Back to Q83*

Q84

Obtain Bohr's quantisation condition for angular momentum of electron orbiting in n^{th} *orbit in hydrogen atom on the basis of the wave picture of an electron using de Broglie hypothesis.*

[2019 • Set 55-3-1]

SOLUTION

According to de Broglie, a moving electron of mass m and speed v behaves as a matter wave of wavelength:

$$\lambda = \frac{h}{mv}$$

For the electron to occupy a stable, non-radiating orbit, the matter wave must close on itself after one revolution and form a standing wave. This requires the circumference of the orbit to contain a whole number of wavelengths:

$$2\pi r = n\lambda,\ n = 1, 2, 3, \ldots$$

Substituting $\lambda = \frac{h}{mv}$:

$$2\pi r = \frac{n h}{mv}$$

Rearranging to isolate the angular momentum mvr:

$$mvr = \frac{nh}{2\pi}$$

This is exactly **Bohr's quantisation condition for angular momentum**: the angular momentum of the orbiting electron is an integral multiple of $\frac{h}{2\pi}$.

Diagram (describe): draw a circular orbit with the nucleus at the centre. Along the circle draw a smooth wavy line that completes an exact whole number of crests and troughs and joins back onto itself with no break, representing $2\pi r = n\lambda$. A second smaller sketch can show a wave that does not close (a mismatch at the join), labelled "destructive interference, not allowed." If the number of wavelengths is not an integer the wave interferes destructively with itself on successive revolutions and dies out, so only the discrete values $n = 1, 2, 3, \ldots$ give stable stationary orbits.

Final answer. The standing-wave condition $2\pi r = n\lambda$ gives Bohr's quantisation $\mathbf{mvr = \frac{nh}{2\pi}}$.

← *Back to Q84*

Q85

An electron in a hydrogen atom in the second excited state jumps to the first and ground state of the atom. Find the ratio of the wavelengths emitted during this process.

[2019 • Set 55-3-2]

SOLUTION

The second excited state of hydrogen corresponds to $n = 3$. From this state the electron makes two jumps: to the first excited state ($n = 2$) and to the ground state ($n = 1$).

The wavelength of each emitted line is given by the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For the transition $3 \rightarrow 2$ (to the first excited state):

$$\frac{1}{\lambda_{32}} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$$

For the transition $3 \to 1$ (to the ground state):

$$\frac{1}{\lambda_{31}} = R\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = R\left(1 - \frac{1}{9}\right) = \frac{8R}{9} = \frac{32R}{36}$$

Taking the ratio of the two wavelengths:

$$\frac{\lambda_{32}}{\lambda_{31}} = \frac{1/\lambda_{31}}{1/\lambda_{32}} = \frac{32R/36}{5R/36} = \frac{32}{5}$$

Hence the ratio of the wavelengths emitted is $\boldsymbol{\lambda_{32} : \lambda_{31} = 32 : 5}$.

COMMON MISTAKE

When forming a wavelength ratio, invert the $\frac{1}{\lambda}$ values carefully; the larger $\frac{1}{\lambda}$ (bigger energy gap) gives the shorter wavelength.

Final answer. The ratio of the emitted wavelengths is $\boldsymbol{\lambda_{3\to 2} : \lambda_{3\to 1} = 32 : 5}$.

← *Back to Q85*

Q86

Define the distance of closest approach. An α-particle of kinetic energy 'K' is bombarded on a thin gold foil. The distance of the closest approach is 'r'. What will be the distance of closest approach for an α-particle of double the kinetic energy?

OR

Write two important limitations of Rutherford nuclear model of the atom.

[2017]

SOLUTION

Distance of closest approach: it is the minimum distance to which an α-particle (or any positively charged projectile) moving directly towards a nucleus can approach it before its velocity becomes zero, after which it is pushed back by the Coulomb repulsion. At this point the entire kinetic energy of the α-particle has been converted into electrostatic potential energy.

At the closest approach, by conservation of energy, kinetic energy equals the Coulomb potential energy between the α-particle (charge 2e) and the gold nucleus (charge Ze):

$$K = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(2e)(Ze)}{d} = \frac{2kZe^2}{d}$$

Solving for the distance of closest approach:

$$d = \frac{2kZe^2}{K} \Rightarrow d \propto \frac{1}{K}$$

For the first α-particle, $\boldsymbol{d = r}$ at kinetic energy K. When the kinetic energy is doubled, $\boldsymbol{K' = 2K}$:

$$\frac{d'}{r} = \frac{K}{K'} = \frac{K}{2K} = \frac{1}{2}$$

$$d' = \frac{r}{2}$$

Hence on doubling the kinetic energy the new distance of closest approach is $\boldsymbol{\frac{r}{2}}$.

OR

Two important limitations of Rutherford's nuclear model:

(i) Instability of the atom: according to classical electromagnetic theory an electron revolving around the nucleus is accelerating and must continuously radiate energy. It would therefore spiral inward and fall into the nucleus within about 10^{-8} s. This contradicts the observed stability of atoms.

(ii) Failure to explain line spectra: as the electron spirals in, its frequency of revolution would change continuously, so it would emit a continuous spectrum. But atoms actually emit sharp, discrete line spectra, which Rutherford's model cannot explain.

Final answer. The new distance of closest approach is $\boldsymbol{r/2}$; (OR) the model fails to explain atomic stability and the observed discrete line spectra.

← *Back to Q86*

Q87

Find out the wavelength of the electron orbiting in the ground state of hydrogen atom.

[2017]

SOLUTION

For the ground state of hydrogen the principal quantum number is $n = 1$, and the orbit radius is the Bohr radius $r_1 = a_0 = 0.53$.

According to de Broglie's hypothesis, a stable Bohr orbit must contain a whole number of electron wavelengths:

$$2\pi r_1 = n\lambda$$

For the ground state $n = 1$, so:

$$\lambda = 2\pi r_1$$

Substituting $r_1 = 0.53$:

$$\lambda = 2\pi \times 0.53 = 2 \times 3.14 \times 0.53$$

$$\lambda \approx 3.33 = 0.333 \text{ nm}$$

As a cross-check, using $\lambda = \dfrac{h}{m_e v_1}$ with $v_1 = 2.18 \times 10^6\ m\ s^{-1}$:

$$\lambda = \frac{6.63 \times 10^{-34}}{(9.1 \times 10^{-31})(2.18 \times 10^6)} \approx 3.34 \times 10^{-10}\ m$$

which agrees. Hence the de Broglie wavelength of the electron in the ground state is $\lambda \approx 3.33$ **(0.333 nm)**.

Final answer. The de Broglie wavelength in the ground state is $\lambda \approx 3.33 = 0.333$ **nm**.

← *Back to Q87*

Q88

A 12.5 eV *electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the wavelengths and the corresponding series of the lines emitted.*

[2017]

SOLUTION

A 12.5 eV electron beam can transfer up to 12.5 eV to a ground-state hydrogen atom by collision. The atom can reach the highest level n for which the excitation energy from $n=1$ does not exceed the beam energy.

The energy of the nth level is $\mathbf{E_n = -\frac{13.6}{n^2}\ eV}$, with $\mathbf{E_1 = -13.6\ eV}$. The condition is:

$$E_n - E_1 \le 12.5\ \text{eV} \Rightarrow |E_n| \ge 13.6 - 12.5 = 1.1\ \text{eV}$$

$$\frac{13.6}{n^2} \ge 1.1 \Rightarrow n^2 \le \frac{13.6}{1.1} = 12.36 \Rightarrow n \le 3.5$$

So the highest state reached is $n = 3$. On de-excitation the possible transitions are **3→ 1**, **3→ 2** and **2→ 1**, giving three spectral lines. Using $\lambda(\text{nm}) = \frac{1240}{\Delta E(\text{eV})}$ with $E_1 = -13.6$, $E_2 = -3.40$, $E_3 = -1.51$ eV:

For **3→ 1**: $\Delta E = -1.51 - (-13.6) = 12.09$ eV

$\lambda = \frac{1240}{12.09} \approx 102.5$ nm (Lyman series, ultraviolet)

For **2→ 1**: $\Delta E = -3.40 - (-13.6) = 10.20$ eV

$\lambda = \frac{1240}{10.20} \approx 121.5$ nm (Lyman series, ultraviolet)

For **3→ 2**: $\Delta E = -1.51 - (-3.40) = 1.89$ eV

$\lambda = \frac{1240}{1.89} \approx 656$ nm (Balmer series, visible/red)

Hence three lines are emitted: **102.5 nm and 121.5 nm in the Lyman series (UV), and 656 nm in the Balmer series (visible)**.

Final answer. Three lines are emitted: **102.5 nm** and **121.5 nm** (Lyman, UV) and **656 nm** (Balmer, visible).

← *Back to Q88*

Q89

The short wavelength limit for the Lyman series of the hydrogen spectrum is 913.4 . *Calculate the short wavelength limit for Balmer series of the hydrogen spectrum.*

[2017]

SOLUTION

The short-wavelength (series) limit of a spectral series corresponds to the electron jumping from $\mathbf{n_i \to \infty}$ down to the lowest level of that series, which gives the smallest wavelength (largest energy) in the series.

The wavelengths are given by the Rydberg formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For the Lyman series $\mathbf{n_f = 1}$, and at its short-wavelength limit $\mathbf{n_i \to \infty}$:

$$\frac{1}{\lambda_L} = R\left(\frac{1}{1^2} - 0\right) = R$$

$$\therefore R = \frac{1}{\lambda_L} = \frac{1}{913.4}$$

For the Balmer series $\mathbf{n_f = 2}$, and at its short-wavelength limit $\mathbf{n_i \to \infty}$:

$$\frac{1}{\lambda_B} = R\left(\frac{1}{2^2} - 0\right) = \frac{R}{4}$$

$$\lambda_B = \frac{4}{R} = 4\,\lambda_L = 4\times 913.4$$

$$\lambda_B = 3653.6 = 365.36 \text{ nm}$$

Hence the short-wavelength limit of the Balmer series is $\mathbf{\lambda_B \approx 3653.6 = 365.36}$ **nm** (ultraviolet).

CONCEPT

At the series limit $n_i \to \infty$, so $\frac{1}{n_i^2} \to 0$; the Balmer limit is simply four times the Lyman limit because n_f^2 goes from 1 to 4.

Final answer. The short-wavelength limit of the Balmer series is $\lambda_B \approx 3653.6 = 365.36$ **nm**.

← *Back to Q89*

Q90

The ground state energy of hydrogen atom is −13.6 eV. If an electron makes a transition from an energy level −1.51 eV to −3.4 eV, calculate the wavelength of the spectral line emitted and name the series of hydrogen spectrum to which it belongs.

[2017]

SOLUTION

The two given energy levels are identified first. Using $E_n = -\frac{13.6}{n^2}$ **eV**: $E = -1.51$ **eV** corresponds to $n = 3$, and $E = -3.4$ **eV** corresponds to $n = 2$. So the electron jumps from $n = 3$ to $n = 2$.

The energy of the emitted photon equals the difference of the two levels:

$$\Delta E = E_3 - E_2 = (-1.51) - (-3.4) = 1.89 \text{ eV}$$

The wavelength of the emitted spectral line is:

$$\lambda = \frac{hc}{\Delta E}$$

Using the convenient form $hc = 1240$ **eV nm**:

$$\lambda = \frac{1240}{1.89} \text{ nm}$$

$$\lambda \approx 656 \text{ nm}$$

Since the transition ends at $n_f = 2$, the line belongs to the **Balmer series** of the hydrogen spectrum (it is the H_α line, in the red region of visible light).

Hence the wavelength of the emitted line is **$\lambda \approx 656$ nm, belonging to the Balmer series**.

Final answer. The emitted line has wavelength **$\lambda \approx 656$ nm** and belongs to the Balmer series.

← *Back to Q90*

Q91

Define ionization energy.
How would the ionization energy change when electron in hydrogen atom is replaced by a particle of mass 200 *times that of the electron but having the same charge?*
OR
Calculate the shortest wavelength of the spectral lines emitted in Balmer series.
[Given Rydberg constant, $R = 10^7\ m^{-1}$*]*

[2016]

SOLUTION

Ionization energy is the minimum energy required to remove the electron completely from the atom, i.e. to take it from its ground state (n=1) to infinity (n=∞, where the total energy is zero).

For a hydrogen atom in the ground state the energy is $E_1 = -13.6$ eV, so the ionization energy is $E_{ion} = 13.6$ eV.

The energy of a Bohr level depends on the mass m of the orbiting particle as $E_n = -\dfrac{m e^4}{8\varepsilon_0^2 n^2 h^2}$, so $E_n \propto m$ when the charge is unchanged.

Replacing the electron by a particle of mass 200 m_e with the same charge multiplies every energy level (and hence the ionization energy) by 200.

$E_{ion}' = 200 \times 13.6$ eV $= 2720$ eV $= 2.72$ keV.

So the ionization energy becomes **2.72 keV**, that is 200 times larger.

OR

The Balmer series corresponds to transitions ending at $n_f=2$. The shortest wavelength is the series limit, emitted when the electron jumps from $n_i=\infty$ to $n_f=2$.

$$\frac{1}{\lambda}=R\left(\frac{1}{2^2}-\frac{1}{\infty^2}\right)=\frac{R}{4}.$$

$$\frac{1}{\lambda}=\frac{10^7}{4}=2.5\times 10^6\ \text{m}^{-1}.$$

$$\lambda=\frac{4}{R}=\frac{4}{10^7}=4\times 10^{-7}\ m=400\ \text{nm}.$$

So the shortest Balmer wavelength is **400 nm**.

Final answer. Ionization energy becomes 2.72 keV; OR shortest Balmer wavelength = 400 nm.

← *Back to Q91*

Q92

Show mathematically how Bohr's postulate of quantization of orbital angular momentum in hydrogen atom is explained by de-Broglie's hypothesis.

[2016]

SOLUTION

According to de Broglie's hypothesis, a particle of mass m moving with speed v has an associated matter wave of wavelength $\lambda=\frac{h}{mv}$.

For the electron's orbit to be stable, its matter wave must close on itself after one revolution, forming a standing wave. This requires the circumference of the orbit to contain a whole number of wavelengths.

$$2\pi\, r_n=n\lambda,\ n=1,2,3,$$

Substituting $\lambda=\frac{h}{mv}$ gives $2\pi\, r_n=\frac{nh}{mv}$.

Rearranging, $mvr_n=\frac{nh}{2\pi}$.

This is exactly **Bohr's postulate of quantization of orbital angular momentum**, $L = mvr_n = \frac{nh}{2\pi}$.

If $2\pi r$ were not an integer multiple of λ, the wave would not join smoothly and would destructively interfere with itself, so no stable orbit could exist. Hence de Broglie's standing-wave picture explains why only the discrete orbits with $L = \frac{nh}{2\pi}$ are allowed.

Final answer. The standing-wave condition $2\pi r = n\lambda$ leads directly to $mvr = \frac{nh}{2\pi}$, which is Bohr's quantization rule.

← *Back to Q92*

Q93

A hydrogen atom initially in its ground state absorbs a photon and is in the excited state with energy 12.5 *eV. Calculate the longest wavelength of the radiation emitted and identify the series to which it belongs. [Take Rydberg constant* $R = 1.1 \times 10^7\ m^{-1}$*]*

OR

Use Bohr model of hydrogen atom to calculate the speed of the electron in the first excited state.

[2016]

SOLUTION

The energy of the nth level of hydrogen is $E_n = -\frac{13.6}{n^2}$ eV.

The atom starts in the ground state, $E_1 = -13.6$ eV. The excited level reached must satisfy $E_n = E_1 + \Delta E$, so the absorbed energy must exactly match a level difference.

Energy needed to reach $n = 3$ is $E_3 - E_1 = -1.51 - (-13.6) = 12.09$ eV, and to reach $n = 4$ is $E_4 - E_1 = -0.85 - (-13.6) = 12.75$ eV.

A supply of 12.5 eV is more than enough to reach $n = 3$ but less than the 12.75 eV needed for $n = 4$, so the highest level the atom can be excited to is $n = 3$.

From $n=3$ the longest wavelength corresponds to the smallest energy gap, which is the transition $n=3 \to n=2$ (the H_α line of the **Balmer series**).

$$\frac{1}{\lambda}=R\left(\frac{1}{2^2}-\frac{1}{3^2}\right)=R\left(\frac{1}{4}-\frac{1}{9}\right)=\frac{5R}{36}.$$

$$\frac{1}{\lambda}=\frac{5\times 1.1\times 10^7}{36}=1.53\times 10^6\ \text{m}^{-1}.$$

$$\lambda=\frac{36}{5\times 1.1\times 10^7}=6.55\times 10^{-7}\ m=655\ \text{nm}.$$

So the longest emitted wavelength is **655 nm**, belonging to the Balmer series.

OR

The speed of the electron in the nth Bohr orbit is $v_n=\frac{e^2}{2\varepsilon_0 nh}=\frac{c}{137\,n}$.

The first excited state is $n=2$.

$$v_2=\frac{c}{137\times 2}=\frac{3\times 10^8}{274}.$$

$$v_2=1.09\times 10^6\ m/s.$$

So the electron's speed in the first excited state is $1.09\times 10^6\ m/s$.

> **TIP**
>
> The longest wavelength always comes from the smallest energy gap, which is the transition to the next-lower level.

Final answer. The 12.5 eV input excites the atom only up to n=3 (n=4 needs 12.75 eV), so the longest wavelength is the n=3 to n=2 line at 655 nm (Balmer series); OR speed in the first excited state = 1.09 x 10^6 m/s.

← *Back to Q93*

Q94

When is H_α line in the emission spectrum of hydrogen atom obtained? Calculate the frequency of the photon emitted during this transition.

[2016]

SOLUTION

The H_α line is obtained when the electron in a hydrogen atom jumps from $n=3$ to $n=2$. It is the first (lowest-energy, longest-wavelength) line of the **Balmer series** and lies in the red part of the visible spectrum.

Using the Rydberg formula with $n_f=2$ and $n_i=3$:

$$\frac{1}{\lambda}=R\left(\frac{1}{2^2}-\frac{1}{3^2}\right)=R\left(\frac{1}{4}-\frac{1}{9}\right)=\frac{5R}{36}.$$

The frequency is $\nu=\frac{c}{\lambda}=cR\left(\frac{1}{4}-\frac{1}{9}\right)=\frac{5cR}{36}$.

$$\nu=\frac{5\times(3\times 10^8)\times(1.1\times 10^7)}{36}.$$

$$\nu=\frac{1.65\times 10^{16}}{36}=4.58\times 10^{14}\ \text{Hz}.$$

So the frequency of the photon emitted in the H_α transition is 4.58×10^{14} Hz.

Final answer. The H_α line comes from the $n=3\to n=2$ transition, with frequency 4.58×10^{14} Hz.

← *Back to Q94*

Q95

Calculate the wavelength of radiation emitted when electron in a hydrogen atom jumps from $n=\infty$ to $n=1$.

[2016]

SOLUTION

The transition $n=\infty\to n=1$ is the series limit of the **Lyman series**. It corresponds to the electron falling from rest at infinity into the ground state, releasing an energy equal to the ground-state binding energy, 13.6 eV.

Using the Rydberg formula with $n_f=1$ and $n_i=\infty$:

$$\frac{1}{\lambda}=R\left(\frac{1}{1^2}-\frac{1}{\infty^2}\right)=R.$$

$$\frac{1}{\lambda}=1.097\times 10^7\ \text{m}^{-1}.$$

$$\lambda=\frac{1}{R}=\frac{1}{1.097\times 10^{7}}.$$

$$\lambda=9.11\times 10^{-8}\ m=91.1\ \text{nm}=911\ .$$

So the emitted radiation has wavelength **91.1 nm** (the Lyman series limit, in the far ultraviolet region).

Final answer. Wavelength of radiation = 9.11×10^{-8} m, i.e. about 91 nm (Lyman series limit).

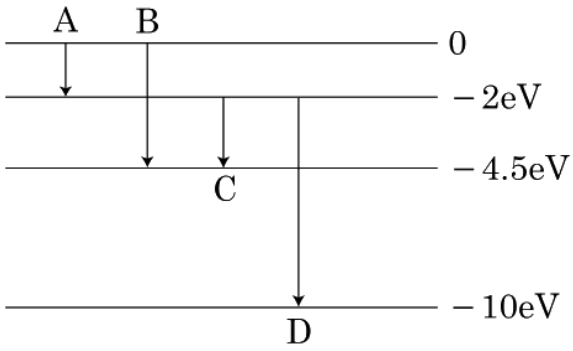
← *Back to Q95*

Q96

In the ground state of hydrogen atom, its Bohr radius is given as 5.3×10^{-11} *m. The atom is excited such that the radius becomes* 21.2×10^{-11} *m. Find (i) the value of the principal quantum number and (ii) the total energy of the atom in this excited state.*

OR

The energy levels of a hypothetical atom are shown below. Which of the shown transitions will result in the emission of photon of wavelength 275 nm*?*



[2016]

SOLUTION

(i) The radius of the nth Bohr orbit is $\mathbf{r_n=n^2\,a_0}$, where a_0 is the Bohr radius (ground-state radius).

$$n^2=\frac{r_n}{a_0}=\frac{21.2\times 10^{-11}}{5.3\times 10^{-11}}=4.$$

$n=\sqrt{4}=2$.

So the principal quantum number is $n=2$.

(ii) The total energy of the nth level is $E_n=-\frac{13.6}{n^2}$ eV.

$E_2=-\frac{13.6}{2^2}=-\frac{13.6}{4}$.

$E_2=-3.4$ eV.

So the total energy of the atom in this excited state is **−3.4 eV**.

OR

The energy of a 275 nm photon is $E_\gamma=\frac{hc}{\lambda}=\frac{1240\text{ eV nm}}{\lambda\text{ (in nm)}}$.

$E_\gamma=\frac{1240}{275}=4.51$ eV.

Among the given energy levels, the only pair whose difference equals about 4.5 eV is the one separating the 0 eV level and the −4.5 eV level.

So the transition that emits the 275 nm photon is the one whose arrow goes from the **0 eV state down to the −4.5 eV state**.

Final answer. $n=2$ and $E_2=-3.4$ eV; OR the 275 nm photon comes from the transition with energy gap ≈ **4.5** eV (the **0** eV → **−4.5** eV transition).

← *Back to Q96*

Q97

How does one explain, using de Broglie hypothesis, Bohr's second postulate of quantization of orbital angular momentum?

[2015]

SOLUTION

According to de Broglie's hypothesis, every moving particle has an associated matter wave of wavelength $\lambda=\frac{h}{mv}$, where m is the mass and v the speed of the particle.

For an electron revolving in a circular orbit to be stable, its matter wave must form a standing wave around the orbit. This is possible only if the circumference of the orbit equals a whole number of wavelengths.

$$2\pi r = n\lambda,\ n = 1, 2, 3,$$

Putting $\lambda = \frac{h}{mv}$:

$2\pi r = \frac{nh}{mv}$.

Rearranging gives $mvr = \frac{nh}{2\pi}$.

This is **Bohr's second postulate**, that the orbital angular momentum is an integer multiple of $\frac{h}{2\pi}$.

If the circumference were not an integer multiple of λ, the wave would interfere destructively with itself on successive revolutions and the orbit would be unstable. Hence only the discrete orbits satisfying $2\pi r = n\lambda$ are allowed, which is the physical basis of Bohr's quantization of angular momentum.

Final answer. The standing-wave condition $2\pi r = n\lambda$ gives $mvr = \frac{nh}{2\pi}$, which is Bohr's second postulate.

← *Back to Q97*

Q98

Show that the radius of the orbit in hydrogen atom varies as n^2, where n is the principal quantum number of the atom.

[2015]

SOLUTION

By Bohr's first postulate, the electrostatic (Coulomb) attraction between the nucleus and the electron supplies the centripetal force for circular motion.

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} = \frac{mv^2}{r} \quad \ldots(1)$$

By Bohr's second postulate, the angular momentum is quantized.

$$mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi\, m\, r} \quad ...(2)$$

Substituting (2) into (1):

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} = \frac{m}{r}\left(\frac{nh}{2\pi\, m\, r}\right)^2 = \frac{n^2h^2}{4\pi^2\, m\, r^3}.$$

Cancelling and solving for r:

$$\mathbf{r_n} = \frac{n^2\, h^2\, \varepsilon_0}{\pi\, m\, e^2}.$$

All the quantities h, ε_0, m, e are constants, so $\mathbf{r_n \propto n^2}$.

Hence the radius of the orbit in the hydrogen atom **varies as $\mathbf{n^2}$**. The constant factor $\mathbf{a_0 = \frac{h^2\varepsilon_0}{\pi\, m\, e^2} = 0.529}$ is the Bohr radius, so $\mathbf{r_n = n^2\, a_0}$ and successive orbits get farther apart as n increases.

Final answer. Solving the force-balance and quantization equations gives $\mathbf{r_n = \frac{n^2\, h^2\, \varepsilon_0}{\pi\, m\, e^2} = n^2\, a_0}$, so $\mathbf{r_n \propto n^2}$.

← *Back to Q98*

Q99

Using Rutherford model of the atom, derive the expression for the total energy of the electron in hydrogen atom. What is the significance of total negative energy possessed by the electron ?

OR

Using Bohr's postulates of the atomic model, derive the expression for radius of nth *electron orbit. Hence obtain the expression for Bohr's radius.*

[2014]

SOLUTION

In the Rutherford model the electron moves in a circular orbit, with the Coulomb attraction of the nucleus providing the centripetal force.

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}=\frac{m_e v^2}{r} \Rightarrow m_e v^2=\frac{e^2}{4\pi\varepsilon_0 r}.$$

The kinetic energy of the electron is

$$K=\frac{1}{2}m_e v^2=\frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}.$$

The potential energy of the electron in the field of the nucleus is

$$U=-\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}.$$

The total energy is the sum.

$$E=K+U=\frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}-\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}.$$

$$E=-\frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}=-\frac{e^2}{8\pi\varepsilon_0 r}.$$

So the total energy is $E=-\frac{e^2}{8\pi\varepsilon_0 r}$, which is negative.

The negative total energy shows that the electron is **bound** to the nucleus; an amount of energy $|E|$ must be supplied to remove it (ionize the atom). A positive total energy would mean a free, unbound electron.

OR

Using the same force balance, $\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}=\frac{m_e v^2}{r}$ ...(1).

By Bohr's quantization condition, $m_e v r=\frac{nh}{2\pi} \Rightarrow v=\frac{nh}{2\pi m_e r}$ ...(2).

Substituting (2) into (1) and solving for r:

$$r_n=\frac{\varepsilon_0 n^2 h^2}{\pi m_e e^2}.$$

Putting $n=1$ gives the Bohr radius.

$$a_0=\frac{\varepsilon_0 h^2}{\pi m_e e^2}=0.529\ .$$

So $r_n=n^2 a_0$, with $a_0\approx 0.529$.

Final answer. Total energy $E=-\frac{e^2}{8\pi\varepsilon_0 r}$ (negative, so the electron is bound); OR $r_n=\frac{\varepsilon_0 n^2 h^2}{\pi m_e e^2}$ with Bohr radius $a_0=0.529$ \AA.

← Back to Q99

Q100

(a) State Bohr's second postulate and mention its significance.
(b) Prove that, in Bohr model of hydrogen atom as principal quantum number n becomes large, the energy levels get closer and closer.

[2026 • Set 55-2-1]

SOLUTION

(a) **Bohr's second postulate** states that an electron revolves only in those orbits for which its orbital angular momentum is an integral multiple of $\frac{h}{2\pi}$.

$$L=m_e v r_n=\frac{nh}{2\pi},\ n=1,2,3,$$

Significance: this quantization picks out only certain discrete (stationary) orbits out of all possible ones. It explains the stability of the atom and the appearance of sharp, discrete line spectra. According to de Broglie, it is equivalent to fitting a whole number of electron matter-wavelengths around the orbit, $2\pi r_n=n\lambda$.

(b) The energy of the nth level of hydrogen is

$$E_n=-\frac{13.6}{n^2}\ \text{eV}.$$

The spacing between two consecutive levels is

$$\Delta E=E_{n+1}-E_n=13.6\left(\frac{1}{n^2}-\frac{1}{(n+1)^2}\right)\ \text{eV}.$$

Combining the fractions:

$$\Delta E=13.6\left(\frac{(n+1)^2-n^2}{n^2(n+1)^2}\right)=13.6\left(\frac{2n+1}{n^2(n+1)^2}\right)\ \text{eV}.$$

For large n, $2n+1\approx 2n$ and $(n+1)^2\approx n^2$, so

$$\Delta E\approx 13.6\times\frac{2n}{n^4}=\frac{2\times 13.6}{n^3}\ \text{eV}.$$

Thus $\Delta E\propto\frac{1}{n^3}$, which tends to zero as $n\to\infty$.

Hence, as n becomes large the **energy levels get closer and closer** together and crowd towards the ionization limit at 0 eV.

Final answer. Bohr's second postulate gives $L = \frac{nh}{2\pi}$; and since $\Delta E \approx \frac{2 \times 13.6}{n^3}$ eV $\propto 1/n^3 \to 0$, the levels crowd together for large n.

← *Back to Q100*

Q101

Calculate the de Broglie wavelength associated with an electron revolving in the second excited state of hydrogen atom. The ground state energy of the hydrogen atom is −13·6 eV. *Take* $m_e = 9 \times 10^{-31}$ kg, $h = 6{\cdot}6 \times 10^{-34}$ J s.

[2026 • Set 55-2-2]

SOLUTION

The second excited state corresponds to $n = 3$ (ground state is n=1, first excited is n=2).

The energy of the electron in the nth state of hydrogen is $E_n = \frac{E_1}{n^2}$, where $E_1 = -13.6$ eV.

$$E_3 = \frac{-13.6}{3^2} = \frac{-13.6}{9} = -1.51 \text{ eV}$$

The kinetic energy equals the magnitude of the total energy: $K = |E_3| = 1.51$ eV.

Converting to joules: $K = 1.51 \times 1.6 \times 10^{-19} = 2.42 \times 10^{-19}$ J.

The de Broglie wavelength is related to kinetic energy through the momentum $p = m_e v = \sqrt{2 m_e K}$.

$$\lambda = \frac{h}{\sqrt{2 m_e K}}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times (9 \times 10^{-31}) \times (2.42 \times 10^{-19})}}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{4.356 \times 10^{-49}}}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{6.6 \times 10^{-25}}$$

$$\lambda \approx 1.0 \times 10^{-9}\ m = 1.0\ \text{nm}$$

As a check, the circumference of the n=3 orbit equals exactly 3 de Broglie wavelengths, consistent with the standing wave picture of Bohr's orbits. So the **de Broglie wavelength is about 1.0 nm**.

Final answer. The de Broglie wavelength of the electron in the second excited state is about $\mathbf{1.0 \times 10^{-9}\ m}$ (1.0 nm).

← *Back to Q101*

Q102

A deuterium nucleus and an alpha particle approach a target nucleus separately in head-on position. Find the ratio of their distance of closest approach to the nucleus, when both have:
(a) the same velocity, and
(b) the same momentum.

[2026 • Set 55-3-3]

SOLUTION

At the distance of closest approach d, the projectile momentarily stops, so its entire initial kinetic energy is converted into electrostatic potential energy with the target nucleus of charge Ze.

$$K = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(Ze)(qe)}{d}$$

where q is the charge number of the projectile. Solving for d:

$$d = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Ze \cdot qe}{K}$$

So $\boldsymbol{d \propto \frac{q}{K}}$ for a fixed target. For a deuterium nucleus, $\mathbf{q_d = 1}$; for an alpha particle, $\mathbf{q_\alpha = 2}$. Their masses are $\mathbf{m_d = 2u}$ and $\mathbf{m_\alpha = 4u}$.

(a) Same velocity v: the kinetic energy is $K = \frac{1}{2}mv^2$, so $K \propto m$.

$$\frac{d_d}{d_\alpha} = \frac{q_d}{q_\alpha} \cdot \frac{K_\alpha}{K_d} = \frac{q_d}{q_\alpha} \cdot \frac{m_\alpha}{m_d}$$

$$\frac{d_d}{d_\alpha} = \frac{1}{2} \times \frac{4}{2} = \frac{1}{2} \times 2 = 1$$

So for the same velocity, **the two distances of closest approach are equal**, $d_d : d_\alpha = 1:1$.

(b) Same momentum p: the kinetic energy is $K = \frac{p^2}{2m}$, so $K \propto \frac{1}{m}$.

$$\frac{d_d}{d_\alpha} = \frac{q_d}{q_\alpha} \cdot \frac{K_\alpha}{K_d} = \frac{q_d}{q_\alpha} \cdot \frac{m_d}{m_\alpha}$$

$$\frac{d_d}{d_\alpha} = \frac{1}{2} \times \frac{2}{4} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

So for the same momentum, $d_d : d_\alpha = 1:4$.

COMMON MISTAKE

Always decide whether $K \propto m$ (same v) or $K \propto 1/m$ (same p) before substituting; mixing these up is the commonest error here.

Final answer. For equal velocity $d_d : d_\alpha = 1:1$; for equal momentum $d_d : d_\alpha = 1:4$.

← Back to Q102

Q103

In Bohr model of hydrogen atom, calculate:
(a) the radius of the innermost electron orbit, and
(b) the speed of electron in this orbit.

[2026 • Set 55-4-2]

SOLUTION

In Bohr's model the Coulomb attraction between the proton and electron supplies the centripetal force, and the angular momentum is quantised as **mvr**

$= \dfrac{nh}{2\pi}$. The innermost orbit is the ground state, $n = 1$.

(a) The radius of the nth orbit is

$$r_n = \frac{\varepsilon_0 n^2 h^2}{\pi m_e e^2}$$

For $n = 1$ this is the Bohr radius:

$$r_1 = \frac{\varepsilon_0 h^2}{\pi m_e e^2}$$

$$r_1 = \frac{(8.85 \times 10^{-12})(6.63 \times 10^{-34})^2}{\pi (9.1 \times 10^{-31})(1.6 \times 10^{-19})^2}$$

$$r_1 \approx 5.3 \times 10^{-11}\ m = 0.53$$

(b) The speed of the electron in the nth orbit follows from $mvr = \dfrac{nh}{2\pi}$, giving

$$v_n = \frac{e^2}{2\varepsilon_0 n h}$$

For $n = 1$:

$$v_1 = \frac{e^2}{2\varepsilon_0 h} = \frac{(1.6 \times 10^{-19})^2}{2(8.85 \times 10^{-12})(6.63 \times 10^{-34})}$$

$$v_1 \approx 2.18 \times 10^6\ m/s$$

So the **radius of the innermost orbit is 0.53** and the **speed of the electron in it is about 2.18×10^6 m/s**, which is about 1/137 of the speed of light.

Final answer. The innermost orbit radius is **0.53** and the electron speed in it is about **2.18×10^6 m/s**.

← *Back to Q103*

Q104

(b) (i) Draw the number of scattered particles versus the scattering angle graph for scattering of alpha particles by a thin foil. Write two important conclusions that can

be drawn from this plot.

(ii) If Bohr's quantization postulate (angular momentum = $\frac{nh}{2\pi}$) is a basic law of nature, it should be equally valid for the case of planetary motion also. Why, then, do we never speak of quantization of orbits of planets around the Sun? Explain.

[2026 • Set 55-5-1]

SOLUTION

(b)(i) Graph of number of scattered particles N versus scattering angle θ: Plot N on the vertical axis and θ on the horizontal axis. The curve starts very high near $\theta = 0°$ and falls extremely steeply as θ increases, becoming very small for large angles and only a tiny non-zero value near $\theta = 180°$ (back-scattering). The curve follows $N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$, so it is a sharply decreasing curve.

Two important conclusions from this plot:

First, most alpha particles suffer very small deflection, which shows that the atom is mostly empty space.

Second, a very small number of alpha particles are scattered through large angles (some even bounce back), which shows that the entire positive charge and almost all the mass of the atom are concentrated in a very small, dense region called the **nucleus**.

(b)(ii) Bohr's quantisation condition gives angular momentum $L = \frac{nh}{2\pi}$, so the quantum number is $n = \frac{2\pi L}{h}$.

For a planet such as the Earth, the orbital angular momentum is enormous, of the order $L \sim 2.7 \times 10^{40}\ J\ s$, while $\frac{h}{2\pi} \sim 10^{-34}\ J\ s$.

$$n = \frac{L}{h/2\pi} \sim \frac{2.7 \times 10^{40}}{10^{-34}} \approx 10^{74}$$

The quantum number is so huge ($n \sim 10^{74}$) that the spacing between two successive allowed orbits is utterly negligible compared with the orbit size. The orbits therefore appear continuous and the quantisation is completely unobservable. This is the **correspondence principle**: quantum results merge into classical behaviour for very large quantum numbers, which is why we never speak of quantisation of planetary orbits.

Final answer. Most alphas pass straight (atom is mostly empty) and a few back-scatter (dense nucleus); planetary orbits have $n \sim 10^{74}$, so quantisation is unobservable.

← Back to Q104

Q105

How is the necessary force provided to an electron to keep it moving in a circular orbit according to Bohr model of hydrogen atom? Derive an expression for the total energy of an electron moving in an orbit of radius r in hydrogen atom. Give the significance of negative sign in this expression.

[2025 • Set 55-6-1]

SOLUTION

The force needed to keep the electron in its circular orbit is the centripetal force, and in the Bohr model this is supplied entirely by the electrostatic Coulomb attraction between the negatively charged electron and the positively charged proton in the nucleus.

Equating the Coulomb force to the centripetal force, with $k = \dfrac{1}{4\pi\varepsilon_0}$:

$$\frac{k\,e^2}{r^2} = \frac{m\,v^2}{r}$$

This gives $m\,v^2 = \dfrac{k\,e^2}{r}$, so the kinetic energy is

$$K = \frac{1}{2}\,m\,v^2 = \frac{k\,e^2}{2r}$$

Taking the potential energy as zero at infinite separation, the electrostatic potential energy of the electron-proton system is

$$U = -\frac{k\,e^2}{r}$$

The total energy is the sum of kinetic and potential energy:

$$E = K + U = \frac{k\,e^2}{2r} - \frac{k\,e^2}{r}$$

$$E = -\frac{k\,e^2}{2r}$$

or in full, $E = -\dfrac{1}{4\pi\varepsilon_0} \cdot \dfrac{e^2}{2r}$

Significance of the negative sign: the total energy is negative, which means the electron is in a **bound state**. It cannot leave the atom unless an external energy equal to $|E|$ is supplied to take it to $E = 0$ (ionisation). A positive total energy would mean the electron is free and unbound.

Final answer. The Coulomb attraction provides the centripetal force, and the total energy is $E = -\dfrac{k\,e^2}{2r}$; the negative sign shows the electron is bound to the nucleus.

← *Back to Q105*

Q106

Describe briefly Geiger-Marsden scattering experiment. Depict the graph showing the variation of number of scattered particles detected with the scattering angle. How did this graph lead to the discovery of the nucleus?

[2025 • Set 55-6-2]

SOLUTION

Geiger-Marsden (Rutherford) scattering experiment: A narrow, collimated beam of fast alpha particles, obtained from a radioactive source such as radium and passed through a lead slit, is directed at a very thin gold foil of thickness about 10^{-7} m. A movable detector consisting of a zinc-sulphide (ZnS) fluorescent screen with a microscope is placed so that it can be set at different scattering angles θ around the foil. Each alpha particle striking the screen produces a tiny flash (scintillation), and the number of flashes per unit time is counted for each angle.

Graph of N versus θ: Plot the number of scattered particles N (vertical axis) against the scattering angle θ (horizontal axis). The curve is very high at small θ and falls off extremely rapidly as θ increases, becoming very small at large angles and almost flat near $\theta = 180°$. It obeys

$$N(\theta) \propto \frac{1}{\sin^4(\frac{\theta}{2})}$$

How the graph led to the discovery of the nucleus: The graph shows that the overwhelming majority of alpha particles pass through almost undeflected (small θ), proving that the atom is mostly empty space. However, a small but definite number are scattered through very large angles, and about 1 in 8000 is turned back through more than 90°. Such large deflections need a very strong repulsive force, which is possible only if all the positive charge and nearly all the mass of the atom are packed into an extremely small, dense central region. This region is the **nucleus**. A diffuse positive charge, as in Thomson's model, could never produce such large-angle scattering. Hence the experiment established the nuclear model of the atom.

Final answer. Alpha particles fired at thin gold foil scatter as $N \propto 1/\sin^4(\theta/2)$; the rare large-angle deflections proved the atom has a tiny, dense, positively charged nucleus.

← *Back to Q106*

Q107

In Bohr model of hydrogen atom, an electron is revolving in second orbit. Find the value of:
(i) angular momentum of electron,
(ii) radius of the orbit, and
(iii) kinetic energy of electron.
Take radius of first orbit of hydrogen atom as 0·5 Å.

[2025 • Set 55-6-3]

SOLUTION

Here the electron is in the second orbit, so $n = 2$, and the first-orbit radius is given as $a_0 = 0.5$.

(i) Angular momentum: by Bohr's quantisation postulate,

$$L_n = \frac{n\,h}{2\pi}$$

$$L_2 = \frac{2 \times 6.626 \times 10^{-34}}{2\pi}$$

$$L_2 \approx 2.11 \times 10^{-34}\ J\,s$$

(ii) Radius of the orbit: the radius scales as $\mathbf{r_n = n^2\,a_0}$.

$$r_2 = 2^2 \times 0.5 = 4 \times 0.5$$

$$r_2 = 2.0 = 2.0 \times 10^{-10}\ m$$

(iii) Kinetic energy: for hydrogen the kinetic energy equals the magnitude of the total energy,

$$K_n = \frac{13.6}{n^2}\ \text{eV}$$

$$K_2 = \frac{13.6}{2^2} = \frac{13.6}{4} = 3.4\ \text{eV}$$

In joules: $\mathbf{K_2 = 3.4 \times 1.6 \times 10^{-19} = 5.44 \times 10^{-19}\ J}$.

So in the second orbit, the **angular momentum is $2.11 \times 10^{-34}\ J\,s$, the radius is 2.0 , and the kinetic energy is 3.4 eV**.

Final answer. For n=2: $L = 2.11 \times 10^{-34}\ J\,s$, $r = 2.0$, and $K = 3.4$ eV.

← *Back to Q107*

Q108

Write the mathematical forms of three postulates of Bohr's theory of the hydrogen atom. Using them prove that, for an electron revolving in the nth *orbit,*
(a) *the radius of the orbit is proportional to* n^2*, and*
(b) *the total energy of the atom is proportional to* $(-\frac{1}{n^2})$.

[2025 • Set 55-7-1]

SOLUTION

Bohr's three postulates in mathematical form:

The electron revolves only in certain stable orbits in which the Coulomb attraction provides the centripetal force, $\frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r^2}=\frac{m\,v^2}{r}$.

The angular momentum in these orbits is quantised, $L = m\,v\,r = \frac{n\,h}{2\pi}$.

When the electron jumps from a higher to a lower level, a photon is emitted with $h\nu = E_i - E_f$.

(a) To show $r \propto n^2$: from the force equation,

$$v^2 = \frac{e^2}{4\pi\varepsilon_0\, m\, r}$$

From the quantisation condition,

$$v = \frac{n\,h}{2\pi\, m\, r} \Rightarrow v^2 = \frac{n^2\,h^2}{4\pi^2\, m^2\, r^2}$$

Equating the two expressions for v^2:

$$\frac{n^2\,h^2}{4\pi^2\, m^2\, r^2} = \frac{e^2}{4\pi\varepsilon_0\, m\, r}$$

$$r_n = \frac{\varepsilon_0\, n^2\, h^2}{\pi\, m\, e^2}$$

Since all other quantities are constants, $r_n \propto n^2$.

(b) To show $E \propto -\frac{1}{n^2}$: the kinetic energy is

$$K = \frac{1}{2}\, m\, v^2 = \frac{1}{8\pi\varepsilon_0}\cdot\frac{e^2}{r}$$

The potential energy is

$$U = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r}$$

So the total energy is

$$E = K + U = -\frac{1}{8\pi\varepsilon_0}\cdot\frac{e^2}{r}$$

Substituting $r = r_n = \frac{\varepsilon_0\, n^2\, h^2}{\pi\, m\, e^2}$:

$$E_n = -\frac{m\,e^4}{8\varepsilon_0{}^2\,h^2\,n^2} = -\frac{13.6}{n^2}\ \text{eV}$$

Hence $E_n \propto -\frac{1}{n^2}$.

Final answer. From the three postulates, $r_n = \frac{\varepsilon_0\,n^2\,h^2}{\pi\,m\,e^2} \propto n^2$ and $E_n = -\frac{m\,e^4}{8\varepsilon_0{}^2\,h^2\,n^2} \propto -\frac{1}{n^2}$.

← *Back to Q108*

Q109

Write the drawbacks of Rutherford's atomic model. How did Bohr remove them? Show that different orbits in Bohr's atom are not equally spaced.

[2024 • Set 55-1-1]

SOLUTION

Drawbacks of Rutherford's atomic model:

First, according to classical electromagnetic theory, an electron revolving in a circular orbit is accelerating and must continuously radiate energy. It would therefore lose energy, spiral inward, and fall into the nucleus in about 10^{-8} *s*. This means the atom should be unstable, which contradicts the observed stability of atoms.

Second, as the electron spirals in, its orbital frequency would change continuously, so it should emit radiation of continuously varying frequency, giving a continuous spectrum. But atoms are actually observed to emit a discrete line spectrum.

How Bohr removed them: Bohr postulated that electrons revolve only in certain special stationary orbits in which they do not radiate energy, so the atom is stable. He also postulated that emission occurs only when an electron jumps from a higher orbit to a lower one, with $h\nu = E_i - E_f$, which gives sharp, discrete spectral lines instead of a continuous spectrum.

Orbits are not equally spaced: The radius of the *n*th Bohr orbit is

$$r_n = n^2 r_1$$

where $\mathbf{r_1 = 0.53}$ is the first-orbit radius. So $\mathbf{r_1 = r_1}$, $\mathbf{r_2 = 4\,r_1}$, $\mathbf{r_3 = 9\,r_1}$, $\mathbf{r_4 = 16\,r_1}$, and so on.

The gap between successive orbits is

$$r_{n+1} - r_n = [(n+1)^2 - n^2]\, r_1 = (2n + 1)\, r_1$$

Since this spacing $(2n+1)r_1$ increases with n, **the orbits get farther apart as n increases and are therefore not equally spaced**.

Final answer. Rutherford's model failed on atomic stability and discrete spectra; Bohr fixed both with non-radiating stationary orbits, and since $\mathbf{r_n = n^2\, r_1}$ the spacing $\mathbf{(2n+1)r_1}$ grows, so orbits are not equally spaced.

← *Back to Q109*

Q110

State the three postulates of Bohr's theory of hydrogen atom.
A hydrogen atom de-excites from level n to level $(n-1)$. Show that, according to Bohr's theory, the frequency of radiation emitted $\mathbf{\nu = \frac{C}{n^3}}$ *for large values of n, where C is a constant. This result exactly agrees with that obtained from classical physics - one of the successes of Bohr's theory.*

[2024 • Set 55-1-3]

SOLUTION

Bohr's three postulates:

The electron revolves only in certain stationary orbits without radiating energy, in which the Coulomb force provides the centripetal force.

The angular momentum of the electron in these orbits is an integral multiple of $\frac{h}{2\pi}$, i.e. $\boldsymbol{m\,v\,r = \frac{n\,h}{2\pi}}$.

When an electron jumps from a higher energy level E_i to a lower one E_f, a photon is emitted with frequency given by $\mathbf{h\nu = E_i - E_f}$.

The energy of the nth level of hydrogen is $\mathbf{E_n = -\frac{13.6}{n^2}\ eV}$. For a transition from level n to level $(n-1)$, the energy emitted is

$$\Delta E = E_n - E_{n-1} = 13.6\left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right] \text{ eV}$$

Taking the common denominator:

$$\Delta E = 13.6 \cdot \frac{n^2 - (n-1)^2}{n^2\,(n-1)^2} = 13.6 \cdot \frac{2n-1}{n^2\,(n-1)^2}$$

For large n, we approximate $\mathbf{2n-1 \approx 2n}$ and $\mathbf{(n-1)^2 \approx n^2}$, so

$$\Delta E \approx 13.6 \cdot \frac{2n}{n^2 \cdot n^2} = \frac{27.2}{n^3} \text{ eV}$$

The frequency of emitted radiation is

$$\nu = \frac{\Delta E}{h} \approx \frac{27.2}{h\,n^3}$$

$$\nu = \frac{C}{n^3}$$

where $\mathbf{C = \frac{27.2 \times 1.6 \times 10^{-19}}{h}}$ is a constant. Thus for large n the frequency varies as $\frac{1}{n^3}$, which **exactly matches the classical orbital frequency**, illustrating Bohr's correspondence principle.

TIP

When asked to reach a $1/n^3$ form, apply the large-n approximations $\mathbf{2n-1 \approx 2n}$ and $\mathbf{(n-1)^2 \approx n^2}$ only after combining the fractions, not before.

Final answer. For a level $\boldsymbol{n \to (n-1)}$ transition at large $\boldsymbol{n}$, $\mathbf{\nu = \frac{C}{n^3}}$ with $\mathbf{C = \frac{27.2\ eV}{h}}$, agreeing with the classical result.

← *Back to Q110*

Q111

Using Bohr's postulates, derive the expression for the radius of the nth orbit of an electron in a hydrogen atom. Also find the numerical value of Bohr's radius a_0.

[2024 • Set 55-2-1]

SOLUTION

Bohr based his model on two key postulates. The electron revolves in certain stable circular orbits without radiating energy, and the angular momentum in such an orbit is quantised as $\mathbf{mvr}=\frac{\mathbf{nh}}{\mathbf{2\pi}}$, where $\boldsymbol{n}=1,2,3,...$

For the electron of mass m and charge $-e$ moving in a circle of radius r around the proton of charge $+e$, the electrostatic Coulomb attraction supplies the necessary centripetal force.

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}=\frac{mv^2}{r} \quad ...(1)$$

From this, $\mathbf{mv^2}=\frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_0}}\frac{\mathbf{e^2}}{\boldsymbol{r}}$

The quantisation of angular momentum gives

$$mvr=\frac{nh}{2\pi} \quad ...(2)$$

From equation (2), $\boldsymbol{v}=\frac{\mathbf{nh}}{\mathbf{2\pi}\,\boldsymbol{m}\,\boldsymbol{r}}$

Substituting this value of v into equation (1),

$$m\left(\frac{nh}{2\pi\, m\, r}\right)^2=\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

$$\frac{n^2h^2}{4\pi^2\, m\, r^2}=\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

Solving for r, the **radius of the *n*th orbit** is

$$r_n=\frac{n^2h^2\varepsilon_0}{\pi\, m\, e^2} \quad ...(3)$$

This shows $\mathbf{r_n}\propto\mathbf{n^2}$, so the orbits get more widely spaced as n increases.

For the first orbit, $\boldsymbol{n}=\mathbf{1}$, the radius is called the **Bohr radius** a_0.

$$a_0=\frac{h^2\varepsilon_0}{\pi\, m\, e^2}$$

$$a_0=\frac{(6.626\,10^{-34})^2(8.854\,10^{-12})}{\pi(9.11\,10^{-31})(1.6\,10^{-19})^2}$$

$$a_0\approx 5.291\,0^{-11}\ m=0.529\ \text{Å}$$

Hence the radius of the nth orbit is $\mathbf{r_n=\frac{n^2\,h^2\varepsilon_0}{\pi\, m\, e^2}}$ and the Bohr radius is $\mathbf{a_0\approx 0.529}$ **Å**.

Final answer. $\mathbf{r_n=\frac{n^2\,h^2\varepsilon_0}{\pi\, m\, e^2}}$ and the Bohr radius $\mathbf{a_0\approx 5.291\,0^{-11}\ m=0.529}$ **Å**.

← *Back to Q111*

Q112

Draw a schematic arrangement of Geiger-Marsden experiment. Briefly explain the conclusions drawn from the observations about the structure of an atom. Define 'distance of closest approach' in this case.

[2024 • Set 55-2-3]

SOLUTION

The schematic arrangement of the Geiger-Marsden (Rutherford) scattering experiment is as follows. A radioactive source of α-particles (such as $^{214}_{83}$Bi, polonium or radium) is enclosed in a lead block with a small opening, so that a narrow beam emerges. This beam passes through a lead collimator that produces a fine, parallel pencil of α-particles. The collimated beam strikes a very thin gold foil (about 100 nm thick). Surrounding the foil is a movable detector consisting of a zinc sulphide (ZnS) fluorescent screen backed by a microscope, which can be rotated to any scattering angle θ. Each α-particle hitting the ZnS screen produces a tiny flash (scintillation) that is counted through the microscope.

Labelled flow of the diagram: α-source (in lead box) → lead collimator (gives narrow beam) → thin gold foil (target) → rotatable ZnS screen + microscope detector measuring count N at angle θ.

The observations led to the following conclusions about the structure of the atom.

Most of the α-particles passed straight through with little or no deflection. This means **most of the atom is empty space**.

A small fraction were deflected through large angles, and about 1 in 8000 was scattered backwards (through more than 90°). Such large deflections are possible only if the α-particle meets a heavy, highly concentrated charge. Hence almost the **entire mass and the whole positive charge of the atom are concentrated in a tiny central region called the nucleus** (size ~ 10^{-15} m, while the atom is ~ 10^{-10} m).

The electrons revolve around this nucleus in the surrounding empty space.

Distance of closest approach: For a head-on collision, the α-particle moves directly towards the nucleus, slows down, momentarily stops, and is then turned back. The minimum distance to which the α-particle approaches the centre of the nucleus before reversing its direction is called the distance of closest approach r_0. At this point its entire kinetic energy has been converted into electrostatic potential energy.

$$\frac{1}{2} m_\alpha v^2 = \frac{1}{4\pi\varepsilon_0} \frac{(2e)(Ze)}{r_0}$$

$$r_0 = \frac{1}{4\pi\varepsilon_0} \frac{2Ze^2}{K_\alpha} \quad K_\alpha = \frac{1}{2} m_\alpha v^2$$

Final answer. The experiment establishes the nuclear model, and the distance of closest approach is $\mathbf{r_0 = \frac{1}{4\pi\varepsilon_0} \frac{2Ze^2}{K_\alpha}}$.

← *Back to Q112*

Q113

State Bohr's first and second postulates. Use them to derive an expression for the radius of the nth orbit in a hydrogen atom.

[2024 • Set 55-3-1]

SOLUTION

Bohr's first postulate: An electron in an atom can revolve only in certain stable circular orbits around the nucleus without radiating energy, even though it is accelerating. These special orbits are called stationary orbits (or stationary states).

Bohr's second postulate: The electron revolves only in those orbits for which its angular momentum is an integral multiple of $\frac{h}{2\pi}$, that is, the angular momentum is quantised.

$$L=\text{mvr}=\frac{\text{nh}}{2\pi},\ n=1,2,3,...$$

Derivation of the radius of the nth orbit:

The Coulomb force of attraction between the nucleus (charge $+e$) and the electron (charge $-e$) provides the centripetal force needed for the circular motion.

$$\frac{\text{mv}^2}{r}=\frac{1}{4\pi\varepsilon_0}\frac{\text{e}^2}{\text{r}^2}\ ...(1)$$

From the second postulate,

$$v=\frac{\text{nh}}{2\pi\, m\, r}\ ...(2)$$

Substituting equation (2) into equation (1),

$$m\left(\frac{\text{nh}}{2\pi\, m\, r}\right)^2\frac{1}{r}=\frac{1}{4\pi\varepsilon_0}\frac{\text{e}^2}{\text{r}^2}$$

$$\frac{\text{n}^2\,\text{h}^2}{4\pi^2\, m\,\text{r}^3}=\frac{\text{e}^2}{4\pi\varepsilon_0\,\text{r}^2}$$

Solving for r,

$$\text{r}_n=\frac{\text{n}^2\,\text{h}^2\,\varepsilon_0}{\pi\, m\,\text{e}^2}=\text{n}^2\,\text{a}_0$$

where $\mathbf{a_0=\frac{h^2\varepsilon_0}{\pi\, m\, e^2}=0.529\ \text{Å}}$ is the **Bohr radius**, the radius of the first orbit.

Since $\mathbf{r_n \propto n^2}$, the successive orbits are not equally spaced; the spacing between them grows as n increases.

TIP

Always quote both postulates word-for-word before deriving, as the marking scheme awards separate marks for stating the postulates and for the derivation.

Final answer. The radius of the nth orbit is $\mathbf{r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2} = n^2 a_0}$, with $\mathbf{a_0 = 0.529\ \text{Å}}$.

← *Back to Q113*

Q114

Explain the origin of the spectral lines of hydrogen atom using Bohr's theory. Draw the energy level diagram of hydrogen atom showing its various spectral series.

[2024 • Set 55-3-2]

SOLUTION

Origin of the spectral lines: According to Bohr's third postulate, an electron in a higher stationary orbit (quantum number n_i) can jump to a lower orbit (quantum number n_f). The atom then loses energy, which is emitted as a single photon of frequency ν.

$$h\nu = E_{n_i} - E_{n_f}$$

The energy of an electron in the nth orbit of hydrogen is

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

Therefore the energy of the emitted photon is

$$h\nu = E_{n_i} - E_{n_f} = 13.6\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)\ \text{eV}$$

The corresponding wavelength is given by

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

where $R = 1.09710^7\ \text{m}^{-1}$ is the Rydberg constant. Since n_i and n_f take only discrete integer values, only certain definite frequencies (wavelengths) are

emitted. This is why the hydrogen spectrum consists of **discrete, sharp spectral lines** rather than a continuous band.

When the lines are grouped by the final orbit n_f, they form distinct spectral series.

Lyman series: transitions to $\mathbf{n_f=1}$, lying in the ultraviolet region.

Balmer series: transitions to $\mathbf{n_f=2}$, lying in the visible region.

Paschen series: transitions to $\mathbf{n_f=3}$, lying in the infrared region.

Brackett series: transitions to $\mathbf{n_f=4}$, lying in the infrared region.

Pfund series: transitions to $\mathbf{n_f=5}$, lying in the far-infrared region.

Energy level diagram (described): Draw a set of horizontal lines, one for each value of n, with energy plotted vertically (most negative at the bottom). The lowest line is the ground state $\boldsymbol{n=1}$ at $\mathbf{E_1=-13.6\ eV}$; above it are $\boldsymbol{n=2}$ at −3.4 eV, $\boldsymbol{n=3}$ at −1.51 eV, $\boldsymbol{n=4}$ at −0.85 eV, $\boldsymbol{n=5}$ at −0.54 eV, and the line $\boldsymbol{n=\infty}$ at 0 eV at the top (lines crowd together near the top). Draw downward arrows from the upper levels to $\boldsymbol{n=1}$ and label them Lyman (UV); arrows ending on $\boldsymbol{n=2}$ labelled Balmer (visible); arrows ending on $\boldsymbol{n=3}$ labelled Paschen (IR); arrows to $\boldsymbol{n=4}$ Brackett; arrows to $\boldsymbol{n=5}$ Pfund.

Final answer. Spectral lines arise from electron transitions $\mathbf{h\nu=13.6\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)\ eV}$, grouped into the Lyman, Balmer, Paschen, Brackett and Pfund series.

← *Back to Q114*

Q115

State Bohr's second postulate for his theory of hydrogen atom. Prove that the radius of Bohr's orbit of a hydrogen atom is directly proportional to n^2*, where n is the principal quantum number.*

[2024 • Set 55-3-3]

SOLUTION

Bohr's second postulate: The electron revolves around the nucleus only in those orbits for which its angular momentum is an integral multiple of $\frac{h}{2\pi}$ (where h is Planck's constant). In other words, the angular momentum is quantised.

$$L=\text{mvr}=\frac{\text{nh}}{2\pi},\ n=1,2,3,...$$

Proof that $\mathbf{r_n \propto n^2}$:

The electrostatic Coulomb force between the nucleus ($+e$) and the electron ($-e$) provides the centripetal force for the circular orbit.

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}=\frac{mv^2}{r} \quad ...(1)$$

From equation (1),

$$v^2=\frac{e^2}{4\pi\varepsilon_0 m r} \quad ...(1a)$$

From the second postulate,

$$v=\frac{nh}{2\pi m r} \quad ...(2)$$

Squaring equation (2),

$$v^2=\frac{n^2 h^2}{4\pi^2 m^2 r^2} \quad ...(2a)$$

Equating (1a) and (2a),

$$\frac{n^2 h^2}{4\pi^2 m^2 r^2}=\frac{e^2}{4\pi\varepsilon_0 m r}$$

Solving for r,

$$r=\frac{n^2 h^2 \varepsilon_0}{\pi m e^2}$$

In this expression h, ε_0, m, e and π are all constants, so

$$r_n=n^2 a_0 \Rightarrow \mathbf{r_n \propto n^2}$$

where $a_0=\frac{h^2\varepsilon_0}{\pi m e^2}=0.53$ Å is the Bohr radius. Hence the radius of a Bohr orbit is directly proportional to the square of the principal quantum number.

Final answer. The orbit radius is $r_n=\frac{n^2 h^2\varepsilon_0}{\pi m e^2}=n^2 a_0$, proving $r_n \propto n^2$.

← *Back to Q115*

Q116

Draw the graph showing variation of scattered particles detected (N) with the scattering angle (θ) in Geiger-Marsden experiment. Write two conclusions that you can draw from this graph. Obtain the expression for the distance of closest approach in this experiment.

[2024 • Set 55-4-1]

SOLUTION

Graph of N versus θ (described): Plot the number of scattered α-particles N (vertical axis) against the scattering angle θ (horizontal axis, from 0° to 180°). The curve is extremely high (sharp peak) for small angles near $\theta \to 0$, falls very steeply as θ increases, and becomes very small and almost flat beyond 90°, with only a tiny number of particles reaching 180°. The count follows the relation $N \propto \frac{1}{\sin^4(\theta/2)}$.

Two conclusions drawn from this graph:

A very large number of particles are scattered through small angles (the peak near $\theta \to 0$). This shows that **most of the atom is empty space**, through which the α-particles pass almost undeflected.

Only a very small number (about 1 in 8000) are scattered through large angles ($\theta > 90°$). This shows that **the entire positive charge and almost all the mass of the atom are concentrated in a very small, dense central nucleus**.

Distance of closest approach: Consider a head-on collision in which the α-particle (charge 2e) travels straight towards the nucleus (charge Ze). As it approaches, the repulsive Coulomb force decelerates it. At the closest point r_0

the particle momentarily comes to rest, so its entire kinetic energy has been converted into electrostatic potential energy.

$$\frac{1}{2}m_\alpha v^2=\frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0}$$

Writing the kinetic energy as $K_\alpha=\frac{1}{2}m_\alpha v^2$ and solving for r_0,

$$r_0=\frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K_\alpha}$$

This is the required **expression for the distance of closest approach**.

Final answer. The distance of closest approach is $\mathbf{r_0=\frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K_\alpha}}$.

← *Back to Q116*

Q117

Draw the energy level diagram for hydrogen atom. Mark the transitions corresponding to the series lying in the ultraviolet region, visible region and infrared region.

[2023 • Set 55-1-1]

SOLUTION

Energy level diagram of hydrogen (described): Draw a set of horizontal lines representing the allowed energy levels, with energy increasing upwards (more negative at the bottom). Using $\mathbf{E_n=-\frac{13.6}{n^2}\ eV}$, the levels are placed as follows.

$E_1=-13.6$ eV ($n=1$, ground state)

$E_2=-3.40$ eV ($n=2$, first excited state)

$E_3=-1.51$ eV ($n=3$, second excited state)

$E_4=-0.85$ eV ($n=4$)

$E_5=-0.54$ eV ($n=5$)

$E_\infty=0$ eV ($n=\infty$, ionisation level, lines crowd together here)

The lowest line ($n=1$) is widely separated from the others; as n increases the lines become more closely spaced and converge towards $E=0$ at the top. Now mark the transitions with downward arrows.

Ultraviolet region (Lyman series): Draw arrows from $n=2,3,4,\ldots$ down to $n=1$. These transitions release the most energy and lie in the ultraviolet region.

Visible region (Balmer series): Draw arrows from $n=3,4,5,\ldots$ down to $n=2$. These lines (such as H_α, H_β) lie in the visible region.

Infrared region (Paschen series): Draw arrows from $n=4,5,6,\ldots$ down to $n=3$. These transitions lie in the infrared region. (The Brackett series, ending on $n=4$, and the Pfund series, ending on $n=5$, also lie in the infrared region.)

Final answer. The Lyman series (to $n=1$) is ultraviolet, the Balmer series (to $n=2$) is visible, and the Paschen series (to $n=3$) is infrared.

← Back to Q117

Q118

Briefly explain Geiger-Marsden experiment. Show the variation of the number of particles scattered (N) with scattering angle (θ) in this experiment. What is the main conclusion that can be inferred from this plot?

[2023 • Set 55-4-1]

SOLUTION

Geiger-Marsden experiment: A narrow, collimated beam of fast α-particles (obtained from a radioactive source such as $^{214}_{83}Bi$) is directed at a very thin gold foil, about 100 nm thick. The α-particles that pass through, or are scattered by, the foil strike a zinc sulphide (ZnS) fluorescent screen, producing tiny flashes called scintillations. A microscope attached to the screen can be rotated to count the number of particles N scattered at any chosen angle θ.

Observations:

Most of the α-particles passed through the foil almost undeflected.

A small fraction were deflected through appreciable angles.

Very few (about 1 in 8000) were deflected by more than 90°, some bouncing almost straight back.

Variation of N with θ (described): Plot N (number of scattered particles) on the vertical axis against the scattering angle θ on the horizontal axis. The curve shows a very sharp, high peak near $\theta \to 0$, then falls off steeply as θ increases, becoming extremely small for large angles, with only a tiny tail beyond 90°. The variation follows $N \propto \frac{1}{\sin^4(\theta/2)}$.

Main conclusion: Since most particles pass through nearly undeflected, most of the atom is empty space; and since a tiny fraction are scattered through very large angles, **the entire positive charge and almost the whole mass of the atom must be concentrated in an extremely small, dense central core called the nucleus**. This led Rutherford to propose the nuclear model of the atom.

Final answer. The plot shows nearly all the atom's mass and positive charge are concentrated in a tiny dense nucleus (Rutherford's nuclear model).

← *Back to Q118*

Q119

(a) Differentiate between 'impact parameter' and 'distance of closest approach' in alpha particle scattering experiment. Identify, situations when impact parameter is (i) minimum, and (ii) very large.
(b) An electron cannot revolve around the nucleus in a hydrogen atom at an arbitrary distance. Explain.

[2023 • Set 55-4-2]

SOLUTION

(a) Difference between impact parameter and distance of closest approach:

The impact parameter b is the perpendicular distance of the initial velocity vector of the α-particle from the centre of the nucleus, measured when the particle is still far away (before scattering begins). It decides how large the scattering angle will be.

The distance of closest approach d is the actual minimum distance to which the α-particle approaches the centre of the nucleus during its trajectory, where it is momentarily closest before moving away again.

Relation linking the impact parameter to the scattering angle θ:

$$b=\frac{1}{4\pi\varepsilon_0}\frac{Ze^2\cot(\theta/2)}{K}$$

(i) Minimum impact parameter: When $b=0$, the α-particle is aimed directly at the centre of the nucleus. This is a **head-on collision**, and the particle is scattered straight back through $\theta=180°$. In this case the distance of closest approach is smallest.

(ii) Very large impact parameter: When b is very large, the α-particle passes far from the nucleus, feels almost no Coulomb force, and is **scattered through a very small angle** ($\theta\to 0$), that is, it goes almost undeflected.

(b) Why the electron cannot orbit at an arbitrary distance: According to Bohr's second postulate, the angular momentum of the electron is quantised.

$$mvr=\frac{nh}{2\pi},\ n=1,2,3,...$$

Only orbits that satisfy this condition are allowed (stationary) orbits. Combining this with the Coulomb-centripetal balance gives the permitted radii

$$r_n=\frac{n^2\,h^2\,\varepsilon_0}{\pi\,m\,e^2}=n^2\,a_0$$

Since n can take only the discrete integer values 1,2,3,..., the radius can take only certain **discrete (quantised) values**. An arbitrary radius would require a non-integer n, which is not allowed. Hence the electron cannot revolve at an arbitrary distance from the nucleus.

Final answer. Impact parameter is the perpendicular offset of the incoming path; closest approach is the actual minimum gap. Only quantised radii $\mathbf{r_n=n^2\,a_0}$ are permitted, so an arbitrary orbit is forbidden.

← *Back to Q119*

Q120

(a) (i) Differentiate between 'distance of closest approach' and 'impact parameter'.
(ii) Determine the distance of closest approach when an alpha particle of kinetic energy 3.95 MeV approaches a nucleus of $Z = 79$*, stops and reverses its directions.*
OR
(b) (i) State three postulates of Bohr's theory of hydrogen atom.
(ii) Find the angular momentum of an electron revolving in the second orbit in Bohr's hydrogen atom.

[2023 • Set 55-5-1]

SOLUTION

(a)(i) Distance of closest approach versus impact parameter:

The distance of closest approach r_0 is the minimum distance from the centre of the target nucleus that the α-particle reaches in a head-on $(b=0)$ collision, where it momentarily stops before reversing. It is given by $\mathbf{r_0=\frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}}$ and is the same for all head-on shots of the same energy.

The impact parameter b is the perpendicular distance between the initial line of motion of the α-particle and a parallel line passing through the centre of the nucleus. It varies from particle to particle and controls the scattering angle θ through $\boldsymbol{b=\frac{1}{4\pi\varepsilon_0}\frac{Ze^2\cot(\theta/2)}{K}}$.

(a)(ii) Numerical: Here the α-particle stops and reverses, so this is a head-on collision and the distance of closest approach formula applies.

Given: $K=3.95$ **MeV**, $Z=79$, charge of α-particle $=2e$.

Converting the kinetic energy to joules,

$$K=3.95 10^6 1.610^{-19}=6.3210^{-13}\,J$$

$$r_0=\frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}=\frac{(910^9)(2)(79)(1.610^{-19})^2}{6.3210^{-13}}$$

$$r_0=\frac{3.6410^{-26}}{6.3210^{-13}}$$

$$r_0\approx 5.7610^{-14}\,m=57.6\text{ fm}$$

OR (b)(i) Three postulates of Bohr's theory:

First postulate: An electron revolves only in certain stable circular orbits (stationary orbits) without radiating energy; in these orbits the Coulomb attraction provides the centripetal force.

Second postulate: The angular momentum of the electron in a stationary orbit is quantised, $\mathbf{mvr=\frac{nh}{2\pi}}$, with $\boldsymbol{n}$**=1,2,3,...**

Third postulate: When an electron jumps from a higher energy orbit (E_i) to a lower one (E_f), a photon is emitted (or absorbed for the reverse jump) with $\mathbf{h\nu=E_i-E_f}$.

OR (b)(ii) Angular momentum in the second orbit: Using $L=\frac{nh}{2\pi}$ with $n=2$,

$$L_2=\frac{2h}{2\pi}=\frac{h}{\pi}$$

$$L_2=\frac{6.62610^{-34}}{\pi}\approx 2.1110^{-34}\ J\ s$$

COMMON MISTAKE

When a projectile stops and reverses, it is by definition a head-on collision, so use the distance-of-closest-approach formula, not the impact-parameter formula.

Final answer. (a)(ii) $\mathbf{r_0\approx 5.7610^{-14}\ m=57.6\ fm}$; OR (b)(ii) $\mathbf{L_2=\frac{h}{\pi}\approx 2.1110^{-34}\ J\ s}$.

← Back to Q120

Q121

(a) (i) Write the limitations of Rutherford's model of atom.
(ii) The wavelength of the second line of the Balmer series in the hydrogen spectrum is 4861 x212B. *Calculate the wavelength of the first line of the same series.*
OR
(b) (i) Increase in the intensity of the radiation causing photo-electric emission from a surface, does not affect the maximum K.E. of the photo electrons. Explain.
(ii) The photon emitted during the de-excitation from the first excited level to the

ground state of hydrogen atom is used to irradiate a photo cathode in which stopping potential is 5 V. Calculate the work function of the cathode used.

[2023 • Set 55-5-2]

SOLUTION

This question has two alternatives; both are solved below in full.

(a)(i) Limitations of Rutherford's nuclear model:

According to classical electromagnetic theory, an electron revolving around the nucleus is accelerating, so it must continuously radiate electromagnetic energy. Losing energy, the electron would spiral inward and fall into the nucleus, making the atom unstable. But atoms are stable, which Rutherford's model cannot explain.

As the electron spirals in, its frequency of revolution changes continuously, so it would emit a continuous spectrum. In reality atoms emit a sharp line spectrum of fixed wavelengths, which the model again cannot explain.

(a)(ii) For the Balmer series the electron falls to $\mathbf{n_f = 2}$. The wavenumber is $\frac{\mathbf{1}}{\lambda} = R(\frac{\mathbf{1}}{\mathbf{2^2}} - \frac{\mathbf{1}}{\mathbf{n_i^2}})$.

Second line of Balmer means $\mathbf{n_i = 4}$, so:

$$\frac{1}{\lambda_2} = R(\frac{1}{4} - \frac{1}{16}) = \frac{3R}{16}$$

First line of Balmer means $\mathbf{n_i = 3}$, so:

$$\frac{1}{\lambda_1} = R(\frac{1}{4} - \frac{1}{9}) = \frac{5R}{36}$$

Dividing the two to cancel R:

$$\frac{\lambda_1}{\lambda_2} = \frac{3R/16}{5R/36} = \frac{3}{16} \times \frac{36}{5} = \frac{108}{80} = \frac{27}{20}$$

$$\lambda_1 = \frac{27}{20} \times 4861 = 6562.35 \text{ Å}$$

So the first Balmer line has wavelength about **6563 Å** (the red H_α line).

OR

(b)(i) In the photoelectric effect, each photon carries energy $E = h\nu$ which depends only on the frequency, not the intensity. Increasing the intensity increases the number of photons striking the surface per second, so more electrons are emitted (larger current). But the energy given to each electron stays the same, since one photon ejects one electron. Hence the maximum kinetic energy, $K_{max} = h\nu - \phi_0$, is unchanged by intensity and depends only on the frequency of the incident light.

(b)(ii) The photon comes from de-excitation $n = 2 \rightarrow n = 1$ in hydrogen.

$$E = E_2 - E_1 = -3.4 - (-13.6) = 10.2 \text{ eV}$$

By Einstein's photoelectric equation, $E = \phi_0 + eV_0$, where $V_0 = 5\ V$ is the stopping potential.

$$\phi_0 = E - eV_0 = 10.2 - 5 = 5.2 \text{ eV}$$

So the work function of the cathode is 5.2 eV.

TIP

For the wavelength ratio, divide one wavenumber equation by the other so the unknown constant R cancels; you never need R's numerical value.

Final answer. (a) First Balmer line wavelength = 6563 Å; OR (b) work function = 5.2 eV.

← *Back to Q121*

Q122

Define the term — Distance of closest approach. How will it be affected, for an α-particle, if kinetic energy of the particle is doubled?

[2022 • Set 55-1-1]

SOLUTION

Distance of closest approach: In an alpha-scattering experiment, the distance of closest approach is the minimum distance from the centre of the target nucleus

to which an α-particle, fired directly head-on, can come before the repulsive Coulomb force stops it momentarily and pushes it back.

At this point the α-particle has come to rest, so its entire initial kinetic energy has been converted into electrostatic potential energy of the α-particle, nucleus system.

Writing kinetic energy = Coulomb potential energy at separation r_0:

$$K = \frac{1}{4\pi\varepsilon_0}\,\frac{(2e)(Ze)}{r_0}$$

Solving for the distance of closest approach:

$$r_0 = \frac{1}{4\pi\varepsilon_0}\,\frac{2Ze^2}{K}$$

Here Z is the atomic number of the target nucleus and K is the kinetic energy of the α-particle. From this relation:

$$r_0 \propto \frac{1}{K}$$

Effect of doubling the kinetic energy: If $\boldsymbol{K \to 2K}$, then

$$r_0' = \frac{1}{4\pi\varepsilon_0}\,\frac{2Ze^2}{2K} = \frac{r_0}{2}$$

So the distance of closest approach is **halved**. A faster (more energetic) α-particle can penetrate closer to the nucleus before being turned back.

Final answer. r0 = 2kZe^2/K, so r0 is inversely proportional to KE; doubling the KE halves the distance of closest approach.

← *Back to Q122*

Q123

(a) Sketch the energy level diagram for hydrogen atom.
(b) Find the ratio of the longest and the shortest wavelength in Lyman series in hydrogen atom.

[2022 • Set 55-1-2]

SOLUTION

(a) Energy level diagram for the hydrogen atom:

Draw a vertical energy axis (energy in eV increasing upward, all values negative). Mark horizontal lines for the allowed energies $\mathbf{E_n = -\frac{13.6}{n^2}}$ **eV**:

$\mathbf{E_1 = -13.6}$ **eV** (lowest line, the ground state, n**=1**),

$\mathbf{E_2 = -3.40}$ **eV** (n**=2**),

$\mathbf{E_3 = -1.51}$ **eV** (n**=3**),

$\mathbf{E_4 = -0.85}$ **eV** (n**=4**), and so on, getting closer together as n rises, up to $\mathbf{E_\infty = 0}$ **eV** (the topmost line, the ionisation limit).

Show downward arrows between levels to denote emission. Arrows ending on n**=1** form the Lyman series (ultraviolet); arrows ending on n**=2** form the Balmer series (visible); arrows ending on n**=3** form the Paschen series (infrared).

(b) Lyman series has the electron falling to $\mathbf{n_f = 1}$:

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n_i^2}\right)$$

The longest wavelength corresponds to the smallest energy jump, i.e. $\mathbf{n_i = 2}$:

$$\frac{1}{\lambda_{max}} = R\left(1 - \frac{1}{4}\right) = \frac{3R}{4}$$

The shortest wavelength is the series limit, $\mathbf{n_i \to \infty}$:

$$\frac{1}{\lambda_{min}} = R(1 - 0) = R$$

Taking the ratio:

$$\frac{\lambda_{max}}{\lambda_{min}} = \frac{R}{3R/4} = \frac{4}{3}$$

So the ratio of the longest to the shortest wavelength in the Lyman series is 4 : 3.

Final answer. Longest : shortest wavelength in the Lyman series = 4 : 3.

← *Back to Q123*

Q124

The energy of a hydrogen atom in the first excited state is −3.4 eV. Find:
(a) the radius of this orbit. (Take Bohr radius **= 0.53** *\AA)*
(b) the angular momentum of the electron in the orbit.
(c) the kinetic and potential energy of the electron in the orbit.

[2022 • Set 55-2-1]

SOLUTION

The first excited state of hydrogen is $n = 2$, and the given energy $E_2 = -3.4$ **eV** confirms this.

(a) Radius of the orbit. The Bohr radius scales as $r_n = n^2 a_0$, where $a_0 = 0.53$ **Å**:

$$r_2 = (2)^2 \times 0.53 = 4 \times 0.53 = 2.12 \text{ Å}$$

$$r_2 = 2.12 \times 10^{-10}\ m$$

(b) Angular momentum. By Bohr's quantisation condition, $L_n = \frac{nh}{2\pi}$, with $h = 6.626 \times 10^{-34}\ J\ s$:

$$L_2 = \frac{2 \times 6.626 \times 10^{-34}}{2 \times 3.14}$$

$$L_2 = \frac{1.325 \times 10^{-33}}{6.28} = 2.11 \times 10^{-34}\ J\ s$$

(c) Kinetic and potential energy. For the hydrogen atom the total energy relates to the kinetic energy K and potential energy U by $K = -E_n$ and $U = 2E_n$ (with $E_n = K + U$).

$$K_2 = -E_2 = -(-3.4) = +3.4 \text{ eV}$$

$$U_2 = 2E_2 = 2 \times (-3.4) = -6.8 \text{ eV}$$

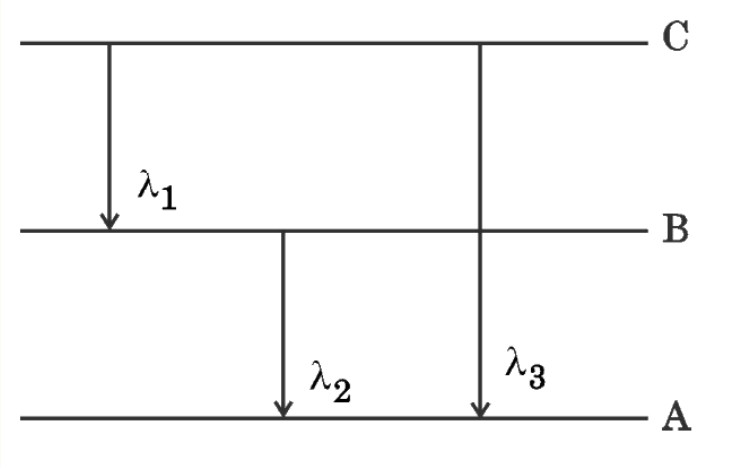
So in the first excited state: radius **2.12 Å**, angular momentum $2.11 \times 10^{-34}\ J\ s$, kinetic energy **+3.4 eV** and potential energy **−6.8 eV**.

Final answer. r2 = 2.12 Å, L2 = 2.11 x 10^-34 J s, K = +3.4 eV, U = −6.8 eV.

← Back to Q124

Q125

(i) State Bohr's quantization condition for defining stationary orbits.
(ii) Use the energy level diagram shown below to obtain the relation between three wavelengths λ_1, λ_2 and λ_3 emitted due to the transition of electron from the energy states C and B.



[2022 • Set 55-2-2]

SOLUTION

(i) Bohr's quantisation condition for stationary orbits: An electron can revolve only in those orbits in which its orbital angular momentum is an integral multiple of $\frac{h}{2\pi}$, where h is Planck's constant.

$$L = m_e v r_n = \frac{nh}{2\pi}, n = 1, 2, 3,$$

These are the allowed stationary (non-radiating) orbits.

(ii) From the energy level diagram, the three levels are A (lowest), B (middle) and C (highest). The transitions are: $C \rightarrow B$ giving wavelength λ_1, $B \rightarrow A$ giving wavelength λ_2, and $C \rightarrow A$ giving wavelength λ_3.

The photon energy in each transition equals the energy gap:

$$E_C - E_B = \frac{hc}{\lambda_1}$$

$$E_B - E_A = \frac{hc}{\lambda_2}$$

$$E_C - E_A = \frac{hc}{\lambda_3}$$

Since the energy gap from C to A is the sum of the gaps $C \to B$ and $B \to A$:

$$E_C - E_A = (E_C - E_B) + (E_B - E_A)$$

$$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

Cancelling hc throughout:

$$\frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$

This is the required relation between the three wavelengths.

Final answer. $1/\lambda 3 = 1/\lambda 1 + 1/\lambda 2$, where $\lambda 3$ is the direct C→A transition.

← *Back to Q125*

Q126

An electron is in the third excited state in a hydrogen atom. It undergoes transitions to the lower energy states.
(a) What is the maximum number of spectral lines that can be emitted?
(b) Calculate the minimum wavelength of the spectral lines emitted.

[2022 • Set 55-2-3]

SOLUTION

The third excited state corresponds to $n = 4$ (ground n=1, first excited n=2, second excited n=3, third excited n=4).

(a) Maximum number of spectral lines. The number of distinct lines emitted when an electron de-excites from level n to all lower levels is:

$$N = \frac{n(n-1)}{2}$$

$$N = \frac{4 \times 3}{2} = 6$$

So a maximum of **6** spectral lines can be emitted.

(b) Minimum wavelength. The minimum wavelength corresponds to the maximum energy difference, which is the transition from $n = 4$ to $n = 1$:

$$\Delta E = 13.6\left(\frac{1}{1^2} - \frac{1}{4^2}\right) \text{ eV}$$

$$\Delta E = 13.6\left(1 - \frac{1}{16}\right) = 13.6 \times \frac{15}{16} = 12.75 \text{ eV}$$

Using $\lambda = \frac{hc}{\Delta E}$ with $hc = 1240$ eV nm:

$$\lambda_{min} = \frac{1240}{12.75} = 97.25 \text{ nm}$$

So the minimum wavelength emitted is about **97.3 nm**, which lies in the Lyman (ultraviolet) series.

CONCEPT

Minimum wavelength always pairs with the maximum energy gap, so pick the jump from the starting level straight down to n = 1.

Final answer. Maximum 6 spectral lines; minimum wavelength ≈ 97.3 nm.

← *Back to Q126*

Q127

State Bohr's postulate to explain stable orbits in a hydrogen atom. Prove that the speed with which the electron revolves in n^{th} *orbit is proportional to* $(1/n)$.

[2022 • Set 55-3-1]

SOLUTION

Bohr's postulate for stable orbits: An electron in an atom can revolve in certain stable orbits without emitting radiant energy, contrary to the predictions of

classical electromagnetic theory. These are the allowed stationary orbits, and in them the electron's orbital angular momentum is quantised as $m\,v\,r = \dfrac{nh}{2\pi}$.

Proof that $v_n \propto 1/n$:

The electrostatic Coulomb attraction between the nucleus (charge $+e$) and the electron (charge $-e$) provides the centripetal force for circular motion:

$$\frac{1}{4\pi\varepsilon_0}\,\frac{e^2}{r_n^{\,2}} = \frac{m\,v_n^{\,2}}{r_n}$$

Writing $k = \dfrac{1}{4\pi\varepsilon_0}$ and simplifying:

$$v_n^{\,2} = \frac{k\,e^2}{m\,r_n} \quad \ldots(1)$$

From Bohr's quantisation condition, $m\,v_n\,r_n = \dfrac{nh}{2\pi}$, so:

$$r_n = \frac{nh}{2\pi\,m\,v_n} \quad \ldots(2)$$

Substituting (2) into (1):

$$v_n^{\,2} = \frac{k\,e^2}{m}\cdot\frac{2\pi\,m\,v_n}{nh} = \frac{2\pi\,k\,e^2\,v_n}{nh}$$

Dividing both sides by v_n:

$$v_n = \frac{2\pi\,k\,e^2}{nh} = \frac{e^2}{2\varepsilon_0\,n\,h}$$

Since e, ε_0 and h are all constants:

$$v_n \propto \frac{1}{n}$$

Hence the speed of the electron in the nth orbit is inversely proportional to n, which is the required result.

Final answer. vn = e^2/(2ε0 n h), hence vn ∝ 1/n is proved.

← *Back to Q127*

Q128

A narrow beam of protons, each having 4.1 MeV *energy is approaching a sheet of lead* **(Z = 82)**. *Calculate:*

(i) the speed of a proton in the beam, and

(ii) the distance of its closest approach.

[2022 • Set 55-3-1]

SOLUTION

(i) Speed of a proton. First convert the kinetic energy to joules:

$$K = 4.1 \text{ MeV} = 4.1 \times 10^6 \times 1.6 \times 10^{-19} = 6.56 \times 10^{-13}\, J$$

Using $K = \frac{1}{2} m_p v^2$, so $v = \sqrt{\frac{2K}{m_p}}$, with $m_p = 1.67 \times 10^{-27}$ kg:

$$v = \sqrt{\frac{2 \times 6.56 \times 10^{-13}}{1.67 \times 10^{-27}}}$$

$$v = \sqrt{7.86 \times 10^{14}}$$

$$v = 2.80 \times 10^7\, m\, s^{-1}$$

(ii) Distance of closest approach. At the closest point all the kinetic energy becomes Coulomb potential energy. For a proton (charge +*e*) and lead nucleus (charge +Ze):

$$K = \frac{1}{4\pi\varepsilon_0} \frac{Z e^2}{r_0} \Rightarrow r_0 = \frac{k Z e^2}{K}$$

Substituting $k = 9 \times 10^9$, $Z = 82$, $e = 1.6 \times 10^{-19}\, C$:

$$r_0 = \frac{(9 \times 10^9)(82)(1.6 \times 10^{-19})^2}{6.56 \times 10^{-13}}$$

$$r_0 = \frac{(9 \times 10^9)(82)(2.56 \times 10^{-38})}{6.56 \times 10^{-13}}$$

$$r_0 = \frac{1.889 \times 10^{-26}}{6.56 \times 10^{-13}}$$

$$r_0 = 2.88 \times 10^{-14}\, m$$

So the proton's speed is $\mathbf{2.80 \times 10^{7}\ m\ s^{-1}}$ and its distance of closest approach is $\mathbf{2.88 \times 10^{-14}\ m}$ (about 28.8 fm).

Final answer. v ≈ 2.80 x 10^7 m/s; distance of closest approach ≈ 2.88 x 10^-14 m (≈ 28.8 fm).

← *Back to Q128*

Q129

(a) The energy of hydrogen atom in an orbit is −1.51 eV. *What are kinetic and potential energies of the electron in this orbit?*
(b) The electron in a hydrogen atom is typically found at a distance of about 5.3×10^{-11} *m from the nucleus which has a diameter of about* 1.0×10^{-15} *m. Assuming the hydrogen atom to be a sphere of radius* 5.3×10^{-11} *m, what fraction of its volume is occupied by the nucleus?*

[2022 • Set 55-3-3]

SOLUTION

(a) For a hydrogen atom the total energy E relates to kinetic energy K and potential energy U by $\boldsymbol{K = -E}$ and $\boldsymbol{U = 2E}$.

Given $\boldsymbol{E = -1.51\ eV}$:

$$K = -E = -(-1.51) = +1.51\ \text{eV}$$

$$U = 2E = 2 \times (-1.51) = -3.02\ \text{eV}$$

So the kinetic energy is **+1.51 eV** and the potential energy is **−3.02 eV**.

(b) Fraction of atomic volume occupied by the nucleus. Both nucleus and atom are taken as spheres, so volume $\propto \mathbf{r^3}$ and the fraction is $\left(\frac{r_{nucleus}}{r_{atom}}\right)^3$.

Nuclear radius = $\frac{\mathbf{1.0 \times 10^{-15}}}{\mathbf{2}} = \mathbf{0.5 \times 10^{-15}\ m}$, atomic radius = $\mathbf{5.3 \times 10^{-11}\ m}$.

$$\frac{r_{nucleus}}{r_{atom}} = \frac{0.5 \times 10^{-15}}{5.3 \times 10^{-11}} = 9.43 \times 10^{-6}$$

$$\frac{V_{nucleus}}{V_{atom}} = (9.43 \times 10^{-6})^3$$

$$\frac{V_{nucleus}}{V_{atom}} = 8.4 \times 10^{-16}$$

So the nucleus occupies only about $\mathbf{8.4 \times 10^{-16}}$ of the atom's volume, showing that the atom is almost entirely empty space.

COMMON MISTAKE

Use the nuclear radius (half the given diameter), not the diameter, when cubing the ratio of sizes.

Final answer. K = +1.51 eV, U = −3.02 eV; nucleus occupies ≈ 8.4 x 10^-16 of the atomic volume.

← Back to Q129

Q130

(a) In Geiger-Marsden experiment, calculate the distance of closest approach for an alpha particle with energy 2.56×10^{-12} *J. Consider that the particle approaches the gold nucleus (***Z = 79***) in head-on position.*
(b) If the above experiment is repeated with a proton of the same energy, then what will be the value of the distance of closest approach?

[2022 • Set 55-4-1]

SOLUTION

(a) Distance of closest approach for the α-particle. At the closest point the kinetic energy is fully converted into Coulomb potential energy. For an α-particle (charge 2e) and gold nucleus (charge Ze):

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0} \Rightarrow r_0 = \frac{2\,k\,Z\,e^2}{K}$$

Substituting $k = 9 \times 10^9$, $Z = 79$, $e = 1.6 \times 10^{-19}$ C, $K = 2.56 \times 10^{-12}$ J:

$$r_0 = \frac{(9 \times 10^9)(2)(79)(1.6 \times 10^{-19})^2}{2.56 \times 10^{-12}}$$

Numerator = $(9 \times 10^9)(2)(79)(2.56 \times 10^{-38}) = 3.64 \times 10^{-26}$

$$r_0 = \frac{3.64 \times 10^{-26}}{2.56 \times 10^{-12}}$$

$$r_0 = 1.42 \times 10^{-14}\ m$$

So for the α-particle, $\mathbf{r_0 \approx 1.42 \times 10^{-14}\ m}$ (about 14.2 fm).

(b) Repeating with a proton of the same energy. The distance of closest approach is $\mathbf{r_0 = \frac{k\, q_{proj}\, Z\, e}{K}}$, so for the same K and same target, $\mathbf{r_0 \propto q_{proj}}$, the charge of the projectile.

A proton carries charge e, which is half that of the α-particle (2e), so its distance of closest approach is halved:

$$r_0^{(p)} = \frac{r_0^{(\alpha)}}{2} = \frac{1.42 \times 10^{-14}}{2}$$

$$r_0^{(p)} = 7.1 \times 10^{-15}\ m$$

So for the proton the distance of closest approach is $\mathbf{7.1 \times 10^{-15}\ m}$ (about 7.1 fm).

Final answer. (a) r0 ≈ 1.42 x 10^-14 m for the α-particle; (b) r0 ≈ 7.1 x 10^-15 m for the proton (halved).

← *Back to Q130*

Q131

Draw a graph showing the variation of number of particles scattered (N) with the scattering angle θ in Geiger-Marsden experiment. Why only a small fraction of the particles are scattered at **θ > 90°***? Mention two limitations of Rutherford nuclear model of an atom.*

[2022 • Set 55-5-1]

SOLUTION

The graph of the number of scattered α-particles (N) against the scattering angle θ is a curve that starts very high near **θ = 0°** and falls steeply as θ

increases, becoming an almost flat, vanishingly small tail beyond **θ = 90°** and reaching its minimum near **θ = 180°**.

To describe the plot precisely: take θ on the x-axis (from 0° to 180°) and N on the y-axis. The curve is highest at small angles and drops rapidly, since N varies as $\frac{1}{\sin^4(\theta/2)}$. Mark the point near **θ = 90°** where N has already become very small.

Only a small fraction of particles scatter at **θ > 90°** because the nucleus is extremely tiny compared to the atom (nuclear size ~ 10^{-15} m against atomic size ~ 10^{-10} m). Most of the atom is empty space, so most α-particles pass through almost undeflected.

A large-angle deflection (**θ > 90°**) requires the α-particle to make a near head-on approach, i.e. a very small impact parameter. Since only a few particles happen to head almost straight at a nucleus, only a very small fraction is scattered through angles greater than 90°.

Two limitations of Rutherford's nuclear model are: (1) According to classical electromagnetic theory, an electron revolving in a circular orbit is accelerating and must continuously radiate energy. It would therefore lose energy and spiral into the nucleus, making the atom **unstable**, which contradicts the observed stability of atoms.

(2) During this continuous spiralling, the frequency of the emitted radiation would change continuously, giving a **continuous spectrum**. But atoms are actually observed to emit sharp, discrete line spectra, which Rutherford's model cannot explain.

Final answer. N is highest at small angles and falls as **$1/\sin^4(\theta/2)$**; few scatter beyond **90°** because the nucleus is tiny, and the model fails to explain atomic stability and discrete line spectra.

← *Back to Q131*

Q132

(a) Use Bohr's postulate to prove that the radius of nth *orbit in a hydrogen atom is proportional to* n^2.

(b) How will the energy of a hydrogen atom change if n increases from 1 to 2?

[2022 • Set 55-5-2]

SOLUTION

(a) In Bohr's model the electron revolves in a circular orbit, and the electrostatic (Coulomb) force of attraction between the nucleus (charge $+e$) and the electron (charge $-e$) provides the necessary centripetal force.

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n^2} = \frac{m\,v_n^2}{r_n}$$

Writing $\boldsymbol{k = \frac{1}{4\pi\varepsilon_0}}$ and simplifying:

$$m\,v_n^2\,r_n = k\,e^2 \quad ...(1)$$

By Bohr's quantisation postulate, the angular momentum is an integral multiple of $\frac{h}{2\pi}$:

$$m\,v_n\,r_n = \frac{n\,h}{2\pi} \quad ...(2)$$

From (2), $\boldsymbol{v_n = \frac{n\,h}{2\pi\,m\,r_n}}$. Substituting this value of v_n into equation (1):

$$m\left(\frac{n\,h}{2\pi\,m\,r_n}\right)^2 r_n = k\,e^2$$

$$\frac{n^2\,h^2}{4\pi^2\,m\,r_n} = k\,e^2$$

$$r_n = \frac{n^2\,h^2}{4\pi^2\,m\,k\,e^2}$$

Since h, m, k and e are all constants, $\boldsymbol{r_n \propto n^2}$. **Hence the radius of the nth orbit is proportional to n^2.** (For $n = 1$, $r_1 = a_0 = 0.53$ Å, the Bohr radius.)

(b) The energy of the nth orbit of a hydrogen atom is $\boldsymbol{E_n = \frac{-13.6}{n^2}}$ **eV.**

For $n = 1$: $\boldsymbol{E_1 = -13.6}$ **eV.**

For $n = 2$: $\boldsymbol{E_2 = \frac{-13.6}{4} = -3.4}$ **eV.**

Change in energy:

$$\Delta E = E_2 - E_1 = -3.4 - (-13.6) = +10.2 \text{ eV}$$

The positive sign shows the energy increases, so the atom **absorbs 10.2 eV** of energy in the transition from $n = 1$ to $n = 2$.

Final answer. Bohr's postulates give $r_n = \dfrac{n^2 h^2}{4\pi^2 m k e^2} \propto n^2$; and the energy increases by $\Delta E = +10.2$ **eV** (energy absorbed) from $n=1$ to $n=2$.

← *Back to Q132*

Q133

An electron in a hydrogen atom makes transitions from orbits of higher energies to orbits of lower energies.
(i) When will such transitions result in (a) Lyman (b) Balmer series?
(ii) Find the ratio of the longest wavelength in Lyman series to the shortest wavelength in Balmer series.

[2022 • Set 55-5-3]

SOLUTION

(i) The wavelength of the emitted radiation in a transition from a higher orbit n_i to a lower orbit n_f is given by Rydberg's formula:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

(a) **Lyman series** is produced when the electron jumps from any higher orbit to the first orbit, i.e. $n_f = 1$ and $n_i = 2, 3, 4, \ldots$. These lines lie in the ultraviolet region.

(b) **Balmer series** is produced when the electron jumps from any higher orbit to the second orbit, i.e. $n_f = 2$ and $n_i = 3, 4, 5, \ldots$. These lines lie in the visible region.

(ii) The longest wavelength in the Lyman series corresponds to the smallest energy jump, i.e. $n_i = 2 \rightarrow n_f = 1$:

$$\frac{1}{\lambda_L^{\max}} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R\left(1 - \frac{1}{4}\right) = \frac{3R}{4}$$

$$\lambda_L^{\max} = \frac{4}{3R}$$

The shortest wavelength in the Balmer series corresponds to the largest energy jump, i.e. $\mathbf{n_i = \infty \to n_f = 2}$:

$$\frac{1}{\lambda_B^{\min}} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R}{4}$$

$$\lambda_B^{\min} = \frac{4}{R}$$

Taking the ratio:

$$\frac{\lambda_L^{\max}}{\lambda_B^{\min}} = \frac{4/(3R)}{4/R} = \frac{4}{3R} \times \frac{R}{4} = \frac{1}{3}$$

Therefore $\lambda_L^{\max} : \lambda_B^{\min} = 1 : 3$.

Final answer. Lyman ends at $\mathbf{n_f=1}$, Balmer at $\mathbf{n_f=2}$; the ratio of longest Lyman to shortest Balmer wavelength is **1:3**.

← *Back to Q133*

Q134

In a Geiger-Marsden experiment, find the distance of closest approach to the gold nucleus (mass no. = 79) of a 7.7 MeV α*-particle before it comes momentarily to rest and reverses its direction. Why is it different from actual radius of gold nucleus? Plot a graph between number of scattered* α*-particles detected in gold foil experiment and angle of scattering. What is the main assumption in plotting this graph?*

[2021]

SOLUTION

At the distance of closest approach r_0, the α-particle is momentarily at rest, so its entire kinetic energy K has been converted into electrostatic potential energy between the α-particle (charge 2e) and the gold nucleus (charge Ze):

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0} \Rightarrow r_0 = \frac{2\,k\,Z\,e^2}{K}$$

Given: $Z = 79$, $k = 9 \times 10^9\ N\,m^2C^{-2}$, $e = 1.6 \times 10^{-19}\ C$, and

$$K = 7.7\ \text{MeV} = 7.7 \times 1.6 \times 10^{-13} = 1.232 \times 10^{-12}\ J$$

Substituting:

$$r_0 = \frac{(9 \times 10^9)(2)(79)(1.6 \times 10^{-19})^2}{1.232 \times 10^{-12}}$$

$$r_0 = \frac{3.64 \times 10^{-26}}{1.232 \times 10^{-12}}$$

$$r_0 \approx 2.96 \times 10^{-14}\ m \approx 30\ \text{fm}$$

The distance of closest approach is $r_0 \approx 2.96 \times 10^{-14}\ m$.

This value differs from the actual radius of the gold nucleus (about 7 fm) because the α-particle stops and reverses before it can actually touch the nucleus. It is halted purely by Coulomb repulsion at the point where its kinetic energy equals the potential energy. Thus r_0 is only an **upper limit** to the nuclear size, not the true nuclear radius.

The graph of the number of scattered α-particles N versus the scattering angle θ is a curve that is very high at small θ and falls steeply (as $1/\sin^4(\theta/2)$), becoming a small tail at large angles up to 180°. (Plot θ on the x-axis, N on the y-axis.)

The main assumption in plotting this graph is that the scattering is due to a single encounter with the point-like Coulomb field of the heavy nucleus (single scattering), the foil being thin enough that each α-particle is deflected by only one nucleus.

COMMON MISTAKE

*Always convert MeV to joules (**1 MeV = $1.6 \times 10^{-13}\ J$**) before substituting into **$r_0 = 2kZe^2/K$**, or the power of ten will be wrong.*

Final answer. $r_0 \approx 2.96 \times 10^{-14}\ m$ (≈ 30 fm); it exceeds the true nuclear radius as the α-particle stops before touching the nucleus, giving only an upper limit.

← *Back to Q134*

Q135

(a) Draw the graph of radius of orbit (r_n) in hydrogen atom as a function of orbit number (n).
(b) In a hydrogen atom, find the ratio of the time taken by the electron to complete one revolution in the first excited and in the second excited states.

[2020 • Set 55-2-3]

SOLUTION

(a) The radius of the nth orbit in a hydrogen atom is $\mathbf{r_n = n^2 a_0}$, where $\mathbf{a_0 = 0.53\ \text{Å}}$ is the Bohr radius. Since $\mathbf{r_n \propto n^2}$, the graph of r_n against n is an upward-opening **parabola** passing through the origin.

To draw it: take $\boldsymbol{n = 1, 2, 3,}$ on the x-axis and r_n on the y-axis. Plot the points $\mathbf{r_1 = a_0 = 0.53\ \text{Å}}$, $\mathbf{r_2 = 4a_0 = 2.12\ \text{Å}}$, $\mathbf{r_3 = 9a_0 = 4.77\ \text{Å}}$, and join them with a smooth curve that rises more and more steeply, showing that successive orbits get further apart.

(b) The time period of revolution of the electron in the nth orbit. We use $\mathbf{r_n \propto n^2}$ and $\mathbf{v_n \propto \frac{1}{n}}$.

$$T_n = \frac{2\pi r_n}{v_n} \propto \frac{n^2}{1/n} = n^3$$

So $\mathbf{T_n \propto n^3}$.

The first excited state corresponds to $\boldsymbol{n = 2}$ and the second excited state corresponds to $\boldsymbol{n = 3}$.

$$\frac{T_2}{T_3} = \frac{2^3}{3^3} = \frac{8}{27}$$

Therefore $T_2 : T_3 = 8 : 27$.

COMMON MISTAKE

*First excited state means **n=2** and second excited state means **n=3** (the ground state is **n=1**); mixing these up is the commonest slip here.*

Final answer. The r_n vs n graph is an upward parabola through the origin; the ratio of revolution times is $T_2 : T_3 = 8 : 27$.

← *Back to Q135*

Q136

Calculate the de-Broglie wavelength associated with the electron revolving in the first excited state of hydrogen atom. The ground state energy of the hydrogen atom is −13.6 eV.

[2020 • Set 55-5-1]

SOLUTION

The first excited state of hydrogen corresponds to $n = 2$. The total energy in this state is

$$E_2 = \frac{-13.6}{n^2} = \frac{-13.6}{4} = -3.4 \text{ eV}$$

In Bohr's model the kinetic energy equals the magnitude of the total energy, so

$$KE = +3.4 \text{ eV} = 3.4 \times 1.6 \times 10^{-19} = 5.44 \times 10^{-19} J$$

The momentum of the electron is

$$p = \sqrt{2\, m\, KE} = \sqrt{2 \times (9.1 \times 10^{-31}) \times (5.44 \times 10^{-19})}$$

$$p = \sqrt{9.90 \times 10^{-49}} = 9.96 \times 10^{-25} \text{ kg } m \text{ s}^{-1}$$

By de-Broglie's relation, the wavelength is

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{9.96 \times 10^{-25}}$$

$$\lambda \approx 6.66 \times 10^{-10}\, m = 6.66 \text{ Å}$$

The de-Broglie wavelength of the electron in the first excited state is about 6.6 Å.

This can be checked from the standing-wave condition $2\pi r_n = n\lambda$. For $n = 2$, $\lambda = \pi r_2 = \pi (4 a_0) = \pi \times 4 \times 0.53 \approx 6.66$ Å, which agrees with the value above.

Final answer. The de-Broglie wavelength of the electron in the first excited state (n=2) is $\lambda \approx 6.6$ Å.

← *Back to Q136*

Q137

Explain briefly how Rutherford scattering of α-particle by a target nucleus can provide information on the size of the nucleus.

[2019 • Set 55-1-3]

SOLUTION

When an α-particle of kinetic energy K is fired head-on towards a nucleus of charge +Ze, the strong Coulomb repulsion slows it down. At the point of **closest approach** the α-particle (charge +2e) comes momentarily to rest, and by conservation of energy all its kinetic energy is converted into electrostatic potential energy:

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{d}$$

Solving for the distance of closest approach d:

$$d = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K} = \frac{2\,k\,Z\,e^2}{K}$$

This distance d tells us how close the α-particle gets to the nucleus. As long as d is larger than the actual nuclear radius, the α-particle stays outside the nucleus, feels only the point-Coulomb field, and the scattering follows Rutherford's formula exactly.

Since $\boldsymbol{d \propto \frac{1}{K}}$, firing higher-energy α-particles makes d smaller, bringing the particle closer to the nucleus. When d becomes comparable to or smaller than the nuclear radius, the α-particle begins to feel short-range nuclear forces and even penetrates the nucleus. At this stage the observed scattering starts to **deviate** from Rutherford's prediction.

The value of d at which this deviation first appears gives an estimate of the **size (an upper limit on the radius) of the nucleus**. In this way Rutherford

scattering experimentally yields $R_{nuc} \sim 10^{-14}\ m$, which is about 10^4 times smaller than the atomic radius $\sim 10^{-10}\ m$.

Final answer. From $d = \frac{2kZe^2}{K}$, the smallest d at which Rutherford's formula still holds gives an upper limit on the nuclear radius ($\sim 10^{-14}\ m$).

← Back to Q137

Q138

Prove that the magnetic moment of the electron revolving around a nucleus in an orbit of radius r with orbital speed v is equal to evr/2. *Hence using Bohr's postulate of quantization of angular momentum, deduce the expression for the magnetic moment of hydrogen atom in the ground state.*

[2019 • Set 55-2-1]

SOLUTION

An electron revolving around the nucleus with speed v in an orbit of radius r behaves like a tiny current loop. The time for one revolution (period) is

$$T = \frac{2\pi r}{v}$$

The electron crossing any point once per period constitutes a current

$$I = \frac{e}{T} = \frac{e}{2\pi r / v} = \frac{e v}{2\pi r}$$

The magnetic moment of a current loop is $\mu = I A$, where $A = \pi r^2$ is the area of the orbit:

$$\mu = I A = \frac{e v}{2\pi r} \times \pi r^2 = \frac{e v r}{2}$$

Hence the magnetic moment is $\mu = \frac{e v r}{2}$. (Proved.)

Now, by Bohr's quantisation postulate, the angular momentum is

$$m_e v r = \frac{n h}{2\pi}$$

For the ground state $n = 1$:

$$v\,r = \frac{h}{2\pi\, m_e}$$

Substituting this into $\mu = \frac{e\,v\,r}{2}$:

$$\mu = \frac{e}{2} \times \frac{h}{2\pi\, m_e} = \frac{e\,h}{4\pi\, m_e}$$

This quantity is the **Bohr magneton** μ_B. Putting in numerical values:

$$\mu = \frac{(1.6 \times 10^{-19})(6.63 \times 10^{-34})}{4\pi\,(9.1 \times 10^{-31})} \approx 9.27 \times 10^{-24}\, J\,T^{-1}$$

Therefore the magnetic moment of the hydrogen atom in the ground state is $\mu = \frac{e\,h}{4\pi\, m_e} \approx 9.27 \times 10^{-24}\, J\,T^{-1}$.

Final answer. $\mu = \frac{evr}{2}$, and using Bohr's quantisation for n=1, $\mu = \frac{eh}{4\pi\, m_e} \approx 9.27 \times 10^{-24}\, J\,T^{-1}$ (the Bohr magneton).

← *Back to Q138*

Q139

Draw a plot of α-particle scattering by a thin foil of gold to show the variation of the number of the scattered particles with scattering angle. Describe briefly how the large angle scattering explains the existence of the nucleus inside the atom. Explain with the help of impact parameter picture, how Rutherford scattering serves a powerful way to determine an upper limit on the size of the nucleus.

[2019 • Set 55-2-1]

SOLUTION

The plot of the number of scattered α-particles N against the scattering angle θ is a curve that is extremely high at small θ and falls very steeply as θ increases, becoming vanishingly small as $\theta \to 180°$. It follows Rutherford's relation $N \propto$

$\frac{1}{\sin^4(\theta/2)}$. (Take θ from 0° to 180° on the x-axis and N on the y-axis; mark the steep fall away from $\theta = 0$.)

How large-angle scattering proves the existence of a nucleus: To deflect a fast, energetic α-particle (kinetic energy of the order of MeV) through an angle greater than 90°, an enormous repulsive Coulomb force is needed over a very short distance. Such a force can arise only if the entire **positive charge and almost the whole mass** of the atom are concentrated in a very small, dense central region, the nucleus. A diffuse, spread-out positive charge (as in Thomson's plum-pudding model) would exert only a weak force and could never produce these rare, large-angle deflections. The few α-particles that bounce back therefore confirm the existence of a tiny, massive nucleus.

Impact-parameter picture and upper limit on nuclear size: The impact parameter b is the perpendicular distance of the initial path of the α-particle from the centre of the nucleus. A large b gives a small deflection; a very small b (near head-on) gives a large deflection. For a head-on collision ($b = 0$), the particle reaches the distance of closest approach

$$d = \frac{2\,k\,Z\,e^2}{K}$$

As long as d is greater than the nuclear radius, the Coulomb-only analysis matches the experiment. By increasing the α-particle energy K, d shrinks (since $d \propto 1/K$) and approaches the nuclear radius. When d becomes comparable to the nuclear size, the observed scattering deviates from Rutherford's prediction, and the value of d at which this happens gives an **upper limit on the size of the nucleus** ($R_{nuc} \sim 10^{-14}\ m$).

Final answer. N falls steeply as $1/\sin^4(\theta/2)$; rare large-angle scattering proves a tiny massive nucleus, and the smallest $d = \frac{2kZe^2}{K}$ where Rutherford's formula holds gives an upper limit on its size.

← *Back to Q139*

Q140

(a) How is the stability of hydrogen atom in Bohr model explained by de-Broglie's hypothesis?
(b) A hydrogen atom initially in the ground state absorbs a photon which excites it to $n = 4$ *level. When it gets de-excited, find the maximum number of lines which are emitted by the atom. Identify the series to which these lines belong. Which of them has the shortest wavelength?*

[2019 • Set 55-2-2]

SOLUTION

(a) According to de-Broglie's hypothesis, the revolving electron is associated with a matter wave of wavelength $\lambda = \frac{h}{m v}$. An orbit is stable only if this wave forms a **standing wave** around the nucleus, i.e. the circumference of the orbit contains a whole number of wavelengths:

$$2\pi r_n = n\lambda = n\frac{h}{m v}$$

Rearranging gives the angular-momentum quantisation:

$$m v r_n = \frac{n h}{2\pi}$$

This is exactly Bohr's quantisation condition. Only these special orbits allow the matter wave to close on itself in phase; in any other orbit the wave would interfere destructively and cancel out. So these allowed orbits are **non-radiating and stable**, which explains the stability of the hydrogen atom.

(b) When the atom is excited to $n = 4$ and then de-excites, the maximum number of spectral lines emitted is

$$\text{Number of lines} = \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = \frac{12}{2} = 6$$

So 6 lines are emitted.

These lines belong to the following series:

Lyman series (transitions to $n = 1$): $4 \to 1$, $3 \to 1$, $2 \to 1$ (3 lines).

Balmer series (transitions to $n = 2$): $4 \to 2$, $3 \to 2$ (2 lines).

Paschen series (transition to $n = 3$): $4 \to 3$ (1 line).

The shortest wavelength corresponds to the largest energy difference. The biggest jump is $4 \to 1$ (a Lyman line), so the **$4 \to 1$ transition of the Lyman series has the shortest wavelength**.

Final answer. Standing matter waves ($2\pi r_n = n\lambda$) explain stable orbits; de-excitation from $n=4$ gives **6** lines (Lyman, Balmer, Paschen), and the **41** Lyman line has the shortest wavelength.

← *Back to Q140*

Q141

(a) Draw the energy level diagram for the line spectra representing Lyman series and Balmer series in the spectrum of hydrogen atom.
(b) Using the Rydberg formula for the spectrum of hydrogen atom, calculate the largest and shortest wavelengths of the emission lines of the Balmer series in the spectrum of hydrogen atom. (Use the value of Rydberg constant $R = 1.1 \times 10^7\ m^{-1}$)

[2019 • Set 55-2-3]

SOLUTION

Part (a): Energy level diagram.

Draw a set of horizontal lines, one for each level, labelled with the energy $E_n = -\frac{13.6}{n^2}$ eV.

The lowest line ($n=1$) sits at −13.6 eV, then $n=2$ at −3.4 eV, $n=3$ at −1.51 eV, $n=4$ at −0.85 eV, getting closer together as we go up, and $n=\infty$ at 0 eV (ionisation level) at the top.

From the higher levels $n=2,3,4,$ draw vertical down-arrows that all end on the $n=1$ line. This group of arrows is the **Lyman series** (lies in the ultraviolet region).

From the higher levels $n=3,4,5,$ draw vertical down-arrows that all end on the $n=2$ line. This group of arrows is the **Balmer series** (lies in the visible region).

Part (b): Largest and shortest wavelength of the Balmer series.

The Rydberg formula for the emission lines is

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For the Balmer series the electron falls to $\mathbf{n_f = 2}$, while $\mathbf{n_i = 3, 4, 5,}$

The largest (longest) wavelength comes from the smallest energy jump, which is $\mathbf{n_i = 3 \to n_f = 2}$:

$$\frac{1}{\lambda_{max}} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right) = R \times \frac{5}{36}$$

$$\lambda_{max} = \frac{36}{5R} = \frac{36}{5 \times 1.1 \times 10^7}$$

$$\lambda_{max} = 6.55 \times 10^{-7}\, m = 655 \text{ nm}$$

The shortest wavelength is the series limit, the largest jump $\mathbf{n_i = \infty \to n_f = 2}$, so $\mathbf{\frac{1}{n_i^2} = 0}$:

$$\frac{1}{\lambda_{min}} = R\left(\frac{1}{2^2} - 0\right) = \frac{R}{4}$$

$$\lambda_{min} = \frac{4}{R} = \frac{4}{1.1 \times 10^7}$$

$$\lambda_{min} = 3.64 \times 10^{-7}\, m = 364 \text{ nm}$$

So the Balmer series stretches from **655 nm (longest)** down to **364 nm (shortest, the series limit)**.

TIP

For any series, the longest wavelength is always the smallest jump (n=n_f+1 to n_f) and the shortest is the series limit (infinity to n_f).

Final answer. Balmer series: longest wavelength $\mathbf{\lambda_{max} = 6.55 10^{-7}\, m}$ (655 nm), shortest $\mathbf{\lambda_{min} = 3.64 10^{-7}\, m}$ (364 nm).

← *Back to Q141*

Q142

(a) Using Bohr's postulates, derive the expression for the orbital period of the electron moving in the nth *orbit of hydrogen atom.*
(b) Write Rydberg's formula for wavelengths of the spectral lines of hydrogen atom spectrum. Mention to which series in the emission spectrum of hydrogen, H_β *line belongs.*

[2019 • Set 55-4-1]

SOLUTION

Part (a): Orbital period of the electron in the nth orbit.

The time taken for one complete revolution is the circumference of the orbit divided by the speed:

$$T_n = \frac{2\pi r_n}{v_n}$$

From Bohr's postulates, the radius of the nth orbit is

$$r_n = \frac{\varepsilon_0 n^2 h^2}{\pi m e^2}$$

and the speed of the electron in the nth orbit is

$$v_n = \frac{e^2}{2\varepsilon_0 n h}$$

Substituting both into the expression for T_n:

$$T_n = \frac{2\pi}{v_n} \times r_n = \frac{2\pi \times \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}}{\dfrac{e^2}{2\varepsilon_0 n h}}$$

$$T_n = 2\pi \times \frac{\varepsilon_0 n^2 h^2}{\pi m e^2} \times \frac{2\varepsilon_0 n h}{e^2}$$

$$T_n = \frac{4\varepsilon_0^2 n^3 h^3}{m e^4}$$

So the **orbital period** is $\mathbf{T_n = \dfrac{4\varepsilon_0^2 n^3 h^3}{m e^4}}$, which shows $\mathbf{T_n \propto n^3}$ (higher orbits take much longer per revolution).

Part (b): Rydberg's formula and the H_β line.

The wavelengths of the spectral lines of hydrogen are given by

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

where $\boldsymbol{R = 1.097\times 10^7\ m^{-1}}$ is the Rydberg constant, n_f is the final (lower) level and n_i is the initial (higher) level.

The H_β line is the **second line of the Balmer series**, produced by the transition $\boldsymbol{n_i = 4 \to n_f = 2}$. It lies in the visible region (blue-green, $\boldsymbol{\lambda \approx 486\ nm}$).

Final answer. Orbital period $\boldsymbol{T_n = \frac{4\varepsilon_0^2 n^3 h^3}{me^4} \propto n^3}$; the $\boldsymbol{H_\beta}$ line (**42**) belongs to the Balmer series.

← Back to Q142

Q143

Using Bohr's postulates, derive the expression for the radius of the nth orbit in which the electron is revolving in hydrogen atom. How does de-Broglie's hypothesis explain the stability of hydrogen atom? Explain.

[2019 • Set 55-4-2]

SOLUTION

Bohr's postulates used here:

First postulate: an electron revolves in certain stable circular orbits around the nucleus without radiating energy. These are called **stationary orbits**.

Second postulate: the angular momentum of the electron in a stationary orbit is quantised, being an integral multiple of $\frac{h}{2\pi}$:

$$L = m\,v\,r = \frac{n\,h}{2\pi},\ n = 1, 2, 3,$$

Derivation of the radius r_n:

The electrostatic (Coulomb) attraction between the proton and the electron supplies the centripetal force needed for circular motion:

$$\frac{m\,v^2}{r} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r^2} \quad \ldots (1)$$

From the second postulate, the speed is

$$v = \frac{n\,h}{2\pi\,m\,r} \quad \ldots (2)$$

Substituting (2) into (1):

$$m\left(\frac{n\,h}{2\pi\,m\,r}\right)^2 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r}$$

$$\frac{n^2\,h^2}{4\pi^2\,m\,r^2} = \frac{e^2}{4\pi\varepsilon_0\,r}$$

Solving for r:

$$r_n = \frac{n^2\,h^2\varepsilon_0}{\pi\,m\,e^2} = n^2\,a_0$$

where $\mathbf{a_0 = \frac{h^2\varepsilon_0}{\pi\,m\,e^2} = 0.529}$ is the **Bohr radius** (the radius of the first orbit). Thus $\mathbf{r_n \propto n^2}$.

de Broglie's explanation of stability:

de Broglie proposed that a moving electron behaves like a wave of wavelength

$$\lambda = \frac{h}{m\,v}$$

For an orbit to be stable, the electron wave must form a **standing wave** around the orbit, otherwise the wave would interfere destructively with itself on successive turns and die out. This requires the circumference of the orbit to contain a whole number of de Broglie wavelengths:

$$2\pi\,r_n = n\lambda = \frac{n\,h}{m\,v}$$

Rearranging gives

$$m\,v\,r_n = \frac{n\,h}{2\pi}$$

This is exactly **Bohr's quantisation condition**. So only those orbits that contain an integral number of electron wavelengths are stable (non-radiating), which explains the stability of the hydrogen atom.

Final answer. $r_n = \dfrac{n^2 h^2 \varepsilon_0}{\pi m e^2} = n^2 a_0$; stable orbits hold a whole number of de Broglie waves: $2\pi r_n = n\lambda$, giving Bohr's quantisation.

← *Back to Q143*

Q144

Use Bohr's postulates to derive the expressions for the potential and kinetic energy of the electron moving in the nth *orbit of the hydrogen atom. How is the total energy of the electron expressed in terms of its kinetic and potential energies?*

[2019 • Set 55-4-3]

SOLUTION

Setting up the force balance (write $k = \dfrac{1}{4\pi\varepsilon_0}$):

The Coulomb attraction provides the centripetal force for the electron in the nth orbit:

$$\frac{m v_n^2}{r_n} = \frac{k e^2}{r_n^2}$$

$$\Rightarrow m v_n^2 = \frac{k e^2}{r_n}$$

Kinetic energy of the electron:

$$KE_n = \frac{1}{2} m v_n^2 = \frac{1}{2} \cdot \frac{k e^2}{r_n} = \frac{k e^2}{2 r_n}$$

Potential energy of the electron:

The electrostatic potential energy of the electron ($-e$) in the field of the proton ($+e$), taking infinity as the zero of energy, is

$$U_n = \frac{k(+e)(-e)}{r_n} = -\frac{k e^2}{r_n}$$

Notice that $\mathbf{U_n = -2\,KE_n}$, the standard result for an inverse-square (Coulomb) force.

Total energy:

$$E_n = KE_n + U_n = \frac{k\,e^2}{2\,r_n} - \frac{k\,e^2}{r_n} = -\frac{k\,e^2}{2\,r_n}$$

Substituting $\mathbf{r_n = n^2\,a_0}$:

$$E_n = -\frac{k\,e^2}{2\,a_0\,n^2} = -\frac{13.6}{n^2}\ \text{eV}$$

Relation between the energies:

$$E_n = -KE_n = \frac{1}{2}U_n$$

So the **total energy equals the negative of the kinetic energy and half the potential energy**. The total energy is negative, which means the electron is bound to the nucleus.

Final answer. $\mathbf{KE_n = \frac{ke^2}{2r_n}}$, $\mathbf{U_n = -\frac{ke^2}{r_n}}$, $\mathbf{E_n = -\frac{ke^2}{2r_n} = -KE_n = \frac{1}{2}U_n = -\frac{13.6}{n^2}\ eV}$.

← *Back to Q144*

Q145

Obtain the first Bohr's radius and the ground state energy of a muonic hydrogen atom i.e. an atom where the electron is replaced by a negatively charged muon (μ^-) *of mass about* 207 m_e *that orbits around a proton.*
(Given for hydrogen atom, radius of first orbit and ground state energy are 0.53×10^{-10} *m and* −13.6 *eV respectively)*

[2019 • Set 55-5-1]

SOLUTION

For a Bohr-type atom, the first Bohr radius and the ground-state energy depend on the mass m of the orbiting particle as

$$r_1 \propto \frac{1}{m} \text{ and } E_1 \propto m$$

(the charge Z of the nucleus and all other constants stay the same). In the muonic atom the muon has mass $\mathbf{m_\mu = 207\ m_e}$.

First Bohr radius of the muonic atom:

$$r_\mu = a_0 \times \frac{m_e}{m_\mu} = \frac{0.53 \times 10^{-10}}{207}$$

$$r_\mu = 2.56 \times 10^{-13}\ m \approx 256\ \text{fm}$$

So the muon orbits 207 times closer to the proton than the electron does.

Ground-state energy of the muonic atom:

$$E_\mu = E_1 \times \frac{m_\mu}{m_e} = (-13.6\ \text{eV}) \times 207$$

$$E_\mu = -2815.2\ \text{eV} \approx -2.82\ \text{keV}$$

The muonic atom is therefore much more tightly bound (about 207 times) than ordinary hydrogen.

CONCEPT

Remember the scaling: smaller orbit (r down by 207) but deeper energy (E up by 207). Same factor, opposite direction.

Final answer. Muonic hydrogen: first Bohr radius $\mathbf{r_\mu 2.5610^{-13}\ m}$ and ground-state energy $\mathbf{E_\mu \approx -2.82\ keV}$.

← *Back to Q145*

Q146

A hydrogen atom in its excited state emits radiations of wavelengths 1218 *A and* 974.3 *A when it finally comes to the ground state. Identify the energy levels from where transitions occur. Given Rydberg constant* $\boldsymbol{R = 1.1 \times 10^7\ m^{-1}}$. *Also specify the spectral series to which these lines belong.*

[2019 • Set 55-5-2]

SOLUTION

Since the atom finally reaches the ground state, both emitted lines end at $\mathbf{n_f = 1}$, so they belong to the **Lyman series**. We use the Rydberg formula to find the starting level n_i for each line:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right),\ n_f = 1$$

First line, $\boldsymbol{\lambda_1 = 1218 = 1218\times 10^{-10}\ m}$. Try $\mathbf{n_i = 2}$:

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R\times\frac{3}{4} = \frac{3}{4}\times 1.1\times 10^7 = 8.25\times 10^6\ \mathrm{m}^{-1}$$

$$\lambda = \frac{1}{8.25\times 10^6} = 1.212\times 10^{-7}\ m = 1212$$

This matches 1218 , so the first line is the transition $\mathbf{n_i = 2 \to n_f = 1}$.

Second line, $\boldsymbol{\lambda_2 = 974.3 = 974.3\times 10^{-10}\ m}$:

$$\frac{1}{\lambda_2} = \frac{1}{974.3\times 10^{-10}} = 1.026\times 10^7\ \mathrm{m}^{-1}$$

$$1.026\times 10^7 = 1.1\times 10^7\left(1 - \frac{1}{n_i^2}\right)$$

$$1 - \frac{1}{n_i^2} = \frac{1.026\times 10^7}{1.1\times 10^7} = 0.933$$

$$\frac{1}{n_i^2} = 1 - 0.933 = 0.067 \Rightarrow n_i^2 \approx 15 \Rightarrow n_i \approx 4$$

So the second line is the transition $\mathbf{n_i = 4 \to n_f = 1}$.

Hence the transitions occur from $\boldsymbol{n = 2}$ and from $\boldsymbol{n = 4}$ down to $\boldsymbol{n = 1}$, and **both lines belong to the Lyman series** (ultraviolet).

Final answer. Transitions are n=21 (1218 \AA) and n=41 (974.3 \AA); both belong to the Lyman series.

← *Back to Q146*

Q147

Hydrogen atom in third excited state de-excites to the first excited state. Obtain the expressions for the frequency of radiation emitted in this process.
Also determine the ratio of the wavelengths of the emitted radiations when the atom de-excites from the third excited state to the second excited state and from the third excited state to the first excited state.

[2019 • Set 55-5-3]

SOLUTION

Identify the levels: the third excited state is $\mathbf{n_i = 4}$ and the first excited state is $\mathbf{n_f = 2}$ (ground state is n=1, first excited n=2, second excited n=3, third excited n=4).

Frequency of radiation for the **4 → 2** transition:

The emitted photon energy equals the difference between the two levels, $\mathbf{E_n = -\frac{13.6}{n^2}}$ **eV**:

$$\Delta E = E_4 - E_2 = 13.6\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 13.6\left(\frac{1}{4} - \frac{1}{16}\right)$$

$$\Delta E = 13.6\times\frac{3}{16} = 2.55 \text{ eV} = 2.55\times 1.6\times 10^{-19} J = 4.08\times 10^{-19} J$$

The frequency is

$$\nu = \frac{\Delta E}{h} = \frac{4.08\times 10^{-19}}{6.626\times 10^{-34}}$$

$$\nu = 6.16\times 10^{14} \text{ Hz}$$

Ratio of wavelengths, **4→ 3** (third to second excited) and **4→ 2** (third to first excited):

Using $\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$, so $\lambda \propto \frac{1}{\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)}$.

$$\frac{1}{\lambda_{4\to 3}} \propto \frac{1}{3^2} - \frac{1}{4^2} = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$$

$$\frac{1}{\lambda_{4\to 2}} \propto \frac{1}{2^2} - \frac{1}{4^2} = \frac{1}{4} - \frac{1}{16} = \frac{3}{16} = \frac{27}{144}$$

Taking the ratio:

$$\frac{\lambda_{4\to 3}}{\lambda_{4\to 2}} = \frac{27/144}{7/144} = \frac{27}{7}$$

So $\lambda_{4\to 3} : \lambda_{4\to 2} = 27 : 7$ and the frequency of the $4\to 2$ line is about 6.16×10^{14} Hz.

Final answer. $\nu_{42} 6.1610^{14}$ **Hz** and $\lambda_{43}:\lambda_{42}=27:7$.

← *Back to Q147*

Q148

(a) State Bohr's postulate to define stable orbits in hydrogen atom. How does de Broglie's hypothesis explain the stability of these orbits ?
(b) A hydrogen atom initially in the ground state absorbs a photon which excites it to the $n = 4$ *level. Estimate the frequency of the photon.*

[2018]

SOLUTION

Part (a): Bohr's stable-orbit postulate.

An electron can revolve only in those orbits for which its angular momentum is an integral multiple of $\frac{h}{2\pi}$:

$$L = m\,v\,r = \frac{n\,h}{2\pi},\ n = 1, 2, 3,$$

While moving in these allowed (stationary) orbits, the electron does not radiate energy, so the orbits are stable.

de Broglie's explanation:

de Broglie assigned a wavelength to the orbiting electron,

$$\lambda = \frac{h}{m\,v}$$

A stable orbit is one in which the electron wave forms a **standing wave**, so the circumference must hold a whole number of wavelengths:

$$2\pi\, r_n = n\lambda = \frac{n\,h}{m\,v}$$

Rearranging gives

$$m\,v\,r_n = \frac{n\,h}{2\pi}$$

which is exactly Bohr's quantisation condition. Orbits that do not satisfy this would produce destructive interference of the electron wave and so cannot exist. This is how de Broglie's hypothesis explains the stability of the allowed orbits.

Part (b): Frequency of the absorbed photon (**1 → 4**).

The photon energy equals the difference between the ***n*=4** and ***n*=1** levels, with $\mathbf{E_n = -\frac{13.6}{n^2}\ eV}$:

$$\Delta E = E_4 - E_1 = 13.6\left(\frac{1}{1^2} - \frac{1}{4^2}\right) = 13.6\left(1 - \frac{1}{16}\right)$$

$$\Delta E = 13.6\times\frac{15}{16} = 12.75\ \text{eV} = 12.75\times 1.6\times 10^{-19} = 2.04\times 10^{-18}\ J$$

The frequency of the photon is

$$\nu = \frac{\Delta E}{h} = \frac{2.04\times 10^{-18}}{6.626\times 10^{-34}}$$

$$\nu = 3.08\times 10^{15}\ \text{Hz}$$

This frequency lies in the ultraviolet region.

Final answer. Stable orbits hold a whole number of de Broglie waves ($\mathbf{2\pi\, r_n = n\lambda}$); the absorbed photon has frequency $\mathbf{\nu 3.08 10^{15}\ Hz}$.

← *Back to Q148*

Q149

Using Bohr's postulates, derive the expression for the orbital period of the electron moving in the nth *orbit of hydrogen atom.*

[2017]

SOLUTION

The orbital period is the time for one full revolution, equal to the circumference divided by the orbital speed:

$$T_n = \frac{2\pi r_n}{v_n}$$

From Bohr's postulates, the radius of the nth orbit is

$$r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m e^2}$$

and the speed of the electron in the nth orbit is

$$v_n = \frac{e^2}{2\varepsilon_0 h n}$$

Substituting both into T_n:

$$T_n = \frac{2\pi \times \dfrac{n^2 h^2 \varepsilon_0}{\pi m e^2}}{\dfrac{e^2}{2\varepsilon_0 h n}}$$

$$T_n = 2\pi \times \frac{n^2 h^2 \varepsilon_0}{\pi m e^2} \times \frac{2\varepsilon_0 h n}{e^2}$$

$$T_n = \frac{4\varepsilon_0^{\,2} h^3 n^3}{m e^4}$$

So the **orbital period** is $\mathbf{T_n = \dfrac{4\varepsilon_0^{\,2} h^3 n^3}{m e^4}}$, giving $\mathbf{T_n \propto n^3}$. For hydrogen the period of the first orbit works out to $\mathbf{T_1 \approx 1.52 \times 10^{-16}\ s}$.

Final answer. Orbital period $\mathbf{T_n = \dfrac{4\varepsilon_0^{\,2} h^3 n^3}{m e^4} \propto n^3}$.

← *Back to Q149*

Q150

In the study of Geiger-Marsden experiment on scattering of α particles by a thin foil of gold, draw the trajectory of α-particles in the coulomb field of target nucleus. Explain briefly how one gets the information on the size of the nucleus from this study.
From the relation $\boldsymbol{R = R_0 A^{1/3}}$*, where* R_0 *is constant and A is the mass number of the nucleus, show that nuclear matter density is independent of A.*

[2015]

SOLUTION

Trajectory of the α-particles:

Draw the target nucleus as a small positive charge. Show several α-particles coming in along parallel straight lines from the left, each with a different impact parameter *b* (the perpendicular distance of the initial path from the nucleus). Each particle follows a **hyperbolic path**: it approaches the nucleus, is repelled by the Coulomb force and deflects through a scattering angle θ, then recedes along a straight asymptote.

A particle with a large impact parameter *b* is deflected only slightly (small θ). A particle aimed almost straight at the nucleus (small *b*) is deflected through a large angle, and one heading dead-on ($\boldsymbol{b = 0}$) is turned right back ($\boldsymbol{\theta = 180°}$).

Estimating the size of the nucleus:

For a head-on collision the α-particle slows down, stops momentarily at a distance r_0 from the nucleus (the distance of closest approach), and then returns. At that point all of its kinetic energy has been converted into electrostatic potential energy:

$$\frac{1}{2} m_\alpha v^2 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(2e)(Ze)}{r_0}$$

Solving for r_0:

$$r_0 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2 Z e^2}{K}, K = \frac{1}{2} m_\alpha v^2$$

Since the α-particle is never seen to penetrate the nucleus, r_0 gives an **upper limit on the size of the nucleus** (the actual radius is smaller than r_0). This is

how the experiment shows that the nucleus is extremely small, of the order of $10^{-15}\ m$.

Nuclear density is independent of A:

With nuclear radius $\boldsymbol{R = R_0\, A^{1/3}}$, the volume of the nucleus is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0{}^3 A$$

The mass of a nucleus of mass number A is approximately $\boldsymbol{M = A\, m_n}$, where m_n is the mass of one nucleon. Hence the density is

$$\rho = \frac{M}{V} = \frac{A\, m_n}{\frac{4}{3}\pi R_0{}^3 A} = \frac{3\, m_n}{4\pi R_0{}^3}$$

The mass number A cancels out, so ρ does **not depend on A**. Numerically, $\boldsymbol{\rho \approx 2.3\times 10^{17}\ kg/m^3}$, the same for all nuclei.

CONCEPT

Because density is independent of A, nuclear matter is incompressible and all nuclei pack nucleons equally tightly.

Final answer. Closest approach gives an upper bound on nuclear size; density $\boldsymbol{\rho = \frac{3m_n}{4\pi R_0{}^3} 2.3 10^{17}\ kg/m^3}$, independent of $\boldsymbol{A}$.

← *Back to Q150*

Q151

A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. Upto which energy level the hydrogen atoms would be excited ?
Calculate the wavelengths of the first member of Lyman and first member of Balmer series.

[2014]

SOLUTION

The energy of an electron in the nth orbit of hydrogen is $E_n = -\frac{13.6}{n^2}$ eV, with the ground state at $E_1 = -13.6$ eV.

The energy needed to raise the atom from the ground state ($n=1$) to a level n is $\Delta E = E_n - E_1 = 13.6\left(1 - \frac{1}{n^2}\right)$ eV.

For $n=3$: $\Delta E = 13.6\left(1 - \frac{1}{9}\right) = 12.09$ eV.

For $n=4$: $\Delta E = 13.6\left(1 - \frac{1}{16}\right) = 12.75$ eV.

The beam supplies only 12.5 eV, which is more than 12.09 eV but less than 12.75 eV. So the atom can be excited up to $n=3$ but not to $n=4$. Hence the **hydrogen atoms are excited up to the $n=3$ level**.

First member of the Lyman series is the transition $n=2 \to n=1$.

$$\Delta E = 13.6\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 13.6 \times \frac{3}{4} = 10.2 \text{ eV}.$$

$$\lambda = \frac{1240}{\Delta E(\text{in eV})} \text{ nm} = \frac{1240}{10.2} \text{ nm}.$$

$\lambda_{\text{Lyman}} \approx 121.6$ nm (ultraviolet region).

First member of the Balmer series is the transition $n=3 \to n=2$.

$$\Delta E = 13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = 13.6 \times \frac{5}{36} = 1.889 \text{ eV}.$$

$\lambda = \frac{1240}{1.889}$ nm ≈ 656.3 nm (the red H_α line, visible region).

Thus the atoms reach $n=3$, with $\lambda_{\text{Lyman}} \approx 121.6$ nm and $\lambda_{\text{Balmer}} \approx 656.3$ nm.

> **TIP**
>
> Use the handy shortcut $\lambda(\text{nm}) = \frac{1240}{\Delta E(\text{eV})}$; it comes from $hc = 1240$ eV·nm and saves time in the exam.

Final answer. Atoms are excited up to n = 3; first Lyman line ≈ 121.6 nm and first Balmer line ≈ 656.3 nm.

← *Back to Q151*

Q152

A 12.9 eV beam of electrons is used to bombard gaseous hydrogen at room temperature. Upto which energy level the hydrogen atoms would be excited ? Calculate the wavelength of the first member of Paschen series and first member of Balmer series.

[2014]

SOLUTION

The energy of the electron in the nth orbit of hydrogen is $E_n = -\frac{13.6}{n^2}$ eV, with $E_1 = -13.6$ eV for the ground state.

The energy required to excite the atom from the ground state to level n is $\Delta E = E_n - E_1 = 13.6\left(1 - \frac{1}{n^2}\right)$ eV.

For $n=3$: $\Delta E = 13.6\left(1 - \frac{1}{9}\right) = 12.09$ eV.

For $n=4$: $\Delta E = 13.6\left(1 - \frac{1}{16}\right) = 12.75$ eV.

For $n=5$: $\Delta E = 13.6\left(1 - \frac{1}{25}\right) = 13.06$ eV.

The beam supplies 12.9 eV, which exceeds 12.75 eV (needed for $n=4$) but is less than 13.06 eV (needed for $n=5$). So the **hydrogen atoms are excited up to the $n=4$ level**.

First member of the Paschen series is the transition $n=4 \to n=3$.

$$\Delta E = 13.6\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 13.6\left(\frac{1}{9} - \frac{1}{16}\right) = 13.6 \times \frac{7}{144} = 0.661 \text{ eV}.$$

$\lambda = \frac{1240}{0.661}$ nm ≈ 1876 nm ≈ 1.876 µm (infrared region).

First member of the Balmer series is the transition $n=3 \to n=2$.

$$\Delta E = 13.6\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 13.6 \times \frac{5}{36} = 1.889 \text{ eV}.$$

$\lambda = \frac{1240}{1.889}$ nm ≈ 656.3 nm (the red H_α line, visible region).

Hence the atoms reach $n=4$, with $\lambda_{Paschen} \approx 1.876$ µm and $\lambda_{Balmer} \approx 656.3$ nm.

Final answer. Atoms are excited up to n = 4; first Paschen line ≈ 1.876 μm and first Balmer line ≈ 656.3 nm.

← *Back to Q152*

Q153

Using Bohr's postulates, obtain the expression for the total energy of the electron in the stationary states of the hydrogen atom. Hence draw the energy level diagram showing how the line spectra corresponding to Balmer series occur due to transition between energy levels.

[2013]

SOLUTION

By Bohr's postulates, the electron of charge $-e$ moves in a circular orbit of radius r_n around the nucleus of charge $+e$. The electrostatic (Coulomb) attraction supplies the centripetal force.

$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n^2}=\frac{mv^2}{r_n}$, where $k=\frac{1}{4\pi\varepsilon_0}$.

From this, $mv^2=\frac{ke^2}{r_n}$, so the kinetic energy of the electron is

$K=\frac{1}{2}mv^2=\frac{ke^2}{2r_n}$.

The potential energy of the electron in the field of the nucleus is

$U=-\frac{ke^2}{r_n}$.

The total energy is the sum of the two.

$E_n=K+U=\frac{ke^2}{2r_n}-\frac{ke^2}{r_n}=-\frac{ke^2}{2r_n}$.

Using Bohr's quantisation of angular momentum, the radius of the nth orbit works out to $r_n=\frac{\varepsilon_0 h^2 n^2}{\pi m e^2}$. Substituting this for r_n gives

$E_n=-\frac{me^4}{8\varepsilon_0^2 h^2}\cdot\frac{1}{n^2}$.

Putting in the constants, this equals $\mathbf{E_n = -\frac{13.6}{n^2}\ eV}$. The negative sign shows the electron is bound to the nucleus.

Energy level diagram: Draw a set of horizontal lines, one for each level, with energy increasing upward. The lowest line is the ground state $\mathbf{E_1 = -13.6\ eV}$ at the bottom. Above it, more closely spaced, draw $\mathbf{E_2 = -3.4\ eV}$, $\mathbf{E_3 = -1.51\ eV}$, $\mathbf{E_4 = -0.85\ eV}$, $\mathbf{E_5 = -0.54\ eV}$, and the topmost line $\mathbf{E_\infty = 0\ eV}$ (ionisation limit). Label each line with its n value and energy on the right.

Balmer series: Draw downward vertical arrows ending on the $\boldsymbol{n=2}$ line, starting from $\boldsymbol{n=3,4,5,\ldots}$ Label these arrows H_α (**32**, red), H_β (**42**, blue-green), H_γ (**52**, violet), and so on. Each arrow represents the emission of a visible photon of energy

$\mathbf{h\nu = E_n - E_2}$, with $\boldsymbol{n=3,4,5,\ldots}$

These transitions produce the **Balmer series** lines lying in the visible region of the hydrogen spectrum.

Final answer. Total energy E_n = −13.6/n^2 eV, and the Balmer series arises from transitions n = 3, 4, 5, … → n = 2.

← *Back to Q153*

Q154

In a Geiger-Marsden experiment, calculate the distance of closest approach to the nucleus of $\boldsymbol{Z = 80}$*, when an* α*-particle of* 8 MeV *energy impinges on it before it comes momentarily to rest and reverses its direction.*
How will the distance of closest approach be affected when the kinetic energy of the α*-particle is doubled?*
OR
The ground state energy of hydrogen atom is −13.6 eV*. If an electron makes a transition from an energy level* −0.85 eV *to* −3.4 eV*, calculate the wavelength of the spectral line emitted. To which series of hydrogen spectrum does this wavelength belong?*

[2012]

SOLUTION

At the distance of closest approach r_0, the α-particle momentarily stops, so its entire initial kinetic energy K has been converted into electrostatic potential energy between the α-particle (charge 2e) and the nucleus (charge Ze).

$$K=\frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0}=\frac{2kZe^2}{r_0}.$$

Rearranging, $r_0=\frac{2kZe^2}{K}$.

Here $Z=80$, $k=910^9\ N\cdot m^2C^{-2}$, $e=1.610^{-19}\ C$, and $K=8\ \text{MeV}=81.610^{-13}=1.2810^{-12}\ J$.

$$r_0=\frac{(910^9)(2)(80)(1.610^{-19})^2}{1.2810^{-12}}.$$

$$r_0=\frac{(910^9)(2)(80)(2.5610^{-38})}{1.2810^{-12}}=\frac{3.68610^{-26}}{1.2810^{-12}}.$$

$$r_0\approx 2.8810^{-14}\ m\approx 28.8\ \text{fm}.$$

Since $r_0\propto\frac{1}{K}$, doubling the kinetic energy halves the distance of closest approach.

$$r_0'=\frac{r_0}{2}\approx 1.4410^{-14}\ m\approx 14.4\ \text{fm}.$$

Hence $r_0\approx 2.8810^{-14}\ m$, and on doubling the energy it becomes **about 14.4 fm (halved)**.

OR

First find the principal quantum numbers of the two levels using $E_n=-\frac{13.6}{n^2}\ \text{eV}$.

For $E=-0.85\ \text{eV}$: $n^2=\frac{13.6}{0.85}=16\Rightarrow n=4$.

For $E=-3.4\ \text{eV}$: $n^2=\frac{13.6}{3.4}=4\Rightarrow n=2$.

So the transition is from $n=4$ to $n=2$.

Energy of the emitted photon: $\Delta E=E_4-E_2=(-0.85)-(-3.4)=2.55\ \text{eV}$.

$$\lambda=\frac{1240}{\Delta E(\text{eV})}\ \text{nm}=\frac{1240}{2.55}\ \text{nm}\approx 486\ \text{nm}.$$

Since the transition ends on $n=2$, the line belongs to the **Balmer series** (the H_β line).

Hence $\lambda \approx$ **486 nm, belonging to the Balmer series**.

Final answer. $r_0 \approx 2.88 \times 10^{-14}$ m, halving to ≈ 14.4 fm when energy is doubled; OR $\lambda \approx 486$ nm in the Balmer series.

← *Back to Q154*

Q155

(a) Using Bohr's second postulate of quantization of orbital angular momentum show that the circumference of the electron in the nth *orbital state in hydrogen atom is n times the de Broglie wavelength associated with it.*
(b) The electron in hydrogen atom is initially in the third excited state. What is the maximum number of spectral lines which can be emitted when it finally moves to the ground state?

[2012]

SOLUTION

(a) By Bohr's second postulate, the angular momentum of the electron in the nth orbit is quantised.

$L = mvr_n = \frac{nh}{2\pi}$, where $n = 1, 2, 3, \ldots$

According to de Broglie, the wavelength of the matter wave associated with the electron is

$\lambda = \frac{h}{mv}$, so $mv = \frac{h}{\lambda}$.

Substitute $mv = \frac{h}{\lambda}$ into the quantisation condition.

$\frac{h}{\lambda} r_n = \frac{nh}{2\pi}$.

Cancel h and rearrange.

$2\pi r_n = n\lambda$.

The left side $2\pi r_n$ is the circumference of the nth orbit. Hence **the circumference of the** n^{th} **orbit equals** n **times the de Broglie wavelength**. Physically, the orbit must contain a whole number of electron wavelengths so

that the matter wave forms a stable standing wave (it interferes constructively with itself after each revolution).

(b) The third excited state corresponds to $n=4$ (ground state $n=1$, first excited $n=2$, second excited $n=3$, third excited $n=4$).

The maximum number of spectral lines emitted when the electron de-excites from level n to the ground state is

$$N=\frac{n(n-1)}{2}.$$

$$N=\frac{4(4-1)}{2}=\frac{43}{2}=6.$$

Hence a maximum of **6 spectral lines** can be emitted.

Final answer. Circumference $2\pi r_n = n\lambda$; and a maximum of 6 spectral lines are emitted from n = 4 to the ground state.

← *Back to Q155*

Q156

Using Bohr's postulates for hydrogen atom, show that the total energy (E) of the electron in the stationary states can be expressed as the sum of kinetic energy (K) and potential energy (U), where $\mathbf{K = -2U}$. *Hence deduce the expression for the total energy in the* n^{th} *energy level of hydrogen atom.*

[2012]

SOLUTION

By Bohr's postulates, the Coulomb attraction between the nucleus (charge +Ze) and the electron (charge $-e$) provides the centripetal force for the circular orbit of radius r.

$$\frac{kZe^2}{r^2}=\frac{mv^2}{r}, \text{ where } k=\frac{1}{4\pi\varepsilon_0}.$$

This gives $mv^2=\frac{kZe^2}{r}$, so the kinetic energy is

$$K=\frac{1}{2}mv^2=\frac{kZe^2}{2r}.$$

The potential energy of the electron in the field of the nucleus is

$U = -\dfrac{kZe^2}{r}$.

Comparing the two expressions, we see that

$U = -\dfrac{kZe^2}{r} = -2\left(\dfrac{kZe^2}{2r}\right) = -2K.$

So $U = -2K$, which is the same as the required relation $K = -\dfrac{U}{2}$.

The total energy is the sum of kinetic and potential energy.

$E = K + U = K + (-2K) = -K.$

Hence the **total energy** $E = -K = \dfrac{U}{2}$, showing the electron is bound (negative total energy).

Now using the quantised radius $r_n = \dfrac{\varepsilon_0 h^2 n^2}{\pi m Z e^2}$ and $E_n = -\dfrac{kZe^2}{2r_n}$, we get

$E_n = -\dfrac{mZ^2e^4}{8\varepsilon_0^2 h^2} \cdot \dfrac{1}{n^2}.$

In electron-volts this becomes $E_n = -\dfrac{13.6\,Z^2}{n^2}$ eV. For hydrogen ($Z = 1$), $E_n = -\dfrac{13.6}{n^2}$ eV.

CONCEPT

A clean memory hook: $E = -K = \dfrac{1}{2}U$. If you know any one of E, K, U in a Bohr orbit, the other two follow instantly.

Final answer. K = −U/2 (so U = −2K) and E = −K, giving E_n = −13.6 Z^2/n^2 eV.

← *Back to Q156*

Q157

(a) Using de Broglie's hypothesis, explain with the help of a suitable diagram, Bohr's second postulate of quantization of energy levels in a hydrogen atom.

(b) The ground state energy of hydrogen atom is −13.6 eV. *What are the kinetic and potential energies of the electron in this state?*

[2011]

SOLUTION

(a) According to de Broglie, a moving electron has an associated matter wave of wavelength

$$\lambda=\frac{h}{mv}.$$

For the orbit to be a stable, non-radiating (stationary) orbit, this matter wave must close on itself and form a standing wave. This happens only when the circumference of the orbit contains a whole number of wavelengths.

$2\pi r_n=n\lambda$, with $n=1,2,3,\ldots$

Substitute $\lambda=\frac{h}{mv}$.

$$2\pi r_n=n\,\frac{h}{mv}.$$

Rearranging gives Bohr's second postulate (quantisation of angular momentum).

$$mvr_n=\frac{nh}{2\pi}.$$

Diagram: Draw a circle representing the electron's orbit and superimpose a wavy line (the matter wave) running all the way around it so that the crests and troughs join smoothly end to end, completing a whole number of wavelengths (for example $n=4$ shows four full wavelengths fitting exactly around the circle). Label the radius r_n and mark one full wavelength λ. If the circumference were not an integer multiple of λ, the wave would not join up and would destroy itself by destructive interference, so no stable orbit would exist. Only the discrete values $n=1,2,3,\ldots$ give allowed stationary orbits.

(b) In Bohr's model the total energy is $E_n=-K_n=\frac{U_n}{2}$, i.e. $K_n=-E_n$ and $U_n=2E_n$.

For the ground state, $E_1=-13.6$ eV.

Kinetic energy: $K_1=-E_1=-(-13.6)=+13.6$ eV.

Potential energy: $U_1=2E_1=2(-13.6)=-27.2$ eV.

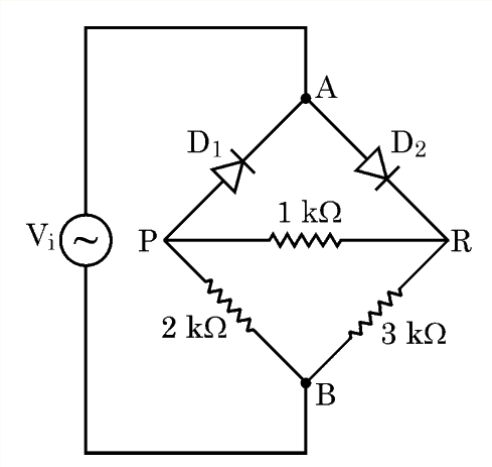
Hence in the ground state, **kinetic energy =+13.6 eV and potential energy =−27.2 eV** (their sum gives back $\mathbf{E_1}$**=−13.6 eV**).

Final answer. Standing-wave condition $2\pi r_n = n\lambda$ gives $mvr_n = nh/2\pi$; for the ground state K = +13.6 eV and U = −27.2 eV.

← *Back to Q157*

Q158

The energy levels of a hypothetical atom are shown below. Which of the shown transitions will result in the emission of a photon of wavelength 275 *nm?*
Which of these transitions correspond to emission of radiation of (i) maximum and (ii) minimum wavelength?



[2011 • Set 55-1-1]

SOLUTION

When an electron jumps from a higher to a lower energy level, the emitted photon carries energy equal to the gap between the levels, $\Delta E = hc/\lambda$. Using the shortcut $E(\text{eV}) = \dfrac{1240}{\lambda(\text{nm})}$, the photon energy for $\lambda = 275$ **nm** is

$$E = \frac{1240}{275} \approx 4.5 \text{ eV}.$$

The given levels are at 0, −2, −4.5 and −10 eV. Compute the gap for each possible downward transition.

0→−2: $\Delta E = 2$ **eV.**

0→−4.5: Δ E=4.5 eV.

0→−10: Δ E=10 eV.

−2→−4.5: Δ E=2.5 eV.

−2→−10: Δ E=8 eV.

−4.5→−10: Δ E=5.5 eV.

The only gap matching 4.5 eV is the **0 eV→−4.5 eV** transition. So this transition emits the **275 nm photon**.

Since $\lambda = \frac{1240}{\Delta E}$, wavelength is largest when Δ E is smallest and smallest when Δ E is largest.

(i) Maximum wavelength: smallest gap is **Δ E=2 eV** for the **0→−2** transition, giving $\lambda = \frac{1240}{2} = 620$ **nm**.

(ii) Minimum wavelength: largest gap is **Δ E=10 eV** for the **0→−10** transition, giving $\lambda = \frac{1240}{10} = 124$ **nm**.

Hence **275 nm comes from 0→−4.5 eV; maximum λ from 0→−2 eV and minimum λ from 0→−10 eV**.

TIP

Wavelength and energy gap are inverses: smallest jump = longest wavelength, biggest jump = shortest wavelength. Compare Δ E values, no need to compute every λ.

Final answer. 275 nm corresponds to the 0 → −4.5 eV transition; maximum λ from 0 → −2 eV and minimum λ from 0 → −10 eV.

← *Back to Q158*

Q159

Using the postulates of Bohr's model of hydrogen atom, obtain an expression for the frequency of radiation emitted when the atom makes a transition from the higher

energy state with quantum number n_i *to the lower energy state with quantum number* n_f ($n_f < n_i$).

[2011 • Set 55-2-1]

SOLUTION

By Bohr's first and second postulates, the energy of the electron in a stationary state with quantum number n is

$$E_n = -\frac{me^4}{8\varepsilon_0^2 h^2}\cdot\frac{1}{n^2}.$$

By Bohr's third (frequency) postulate, when the atom makes a transition from a higher state n_i to a lower state n_f (with $n_f < i$), it emits a single photon whose energy equals the difference of the two level energies.

$$h\nu = E_{n_i} - E_{n_f}.$$

Substitute the expression for the energy of each level.

$$h\nu = \left(-\frac{me^4}{8\varepsilon_0^2 h^2 n_i^2}\right) - \left(-\frac{me^4}{8\varepsilon_0^2 h^2 n_f^2}\right).$$

$$h\nu = \frac{me^4}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right).$$

Divide both sides by h to get the frequency of the emitted radiation.

$$\nu = \frac{me^4}{8\varepsilon_0^2 h^3}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right).$$

Since $n_f < i$, the bracket is positive, so $\nu > 0$, as it must be for emission.

This may be written in terms of the **Rydberg constant** $R = \frac{me^4}{8\varepsilon_0^2 h^3 c} = 1.09710^7$ m^{-1}.

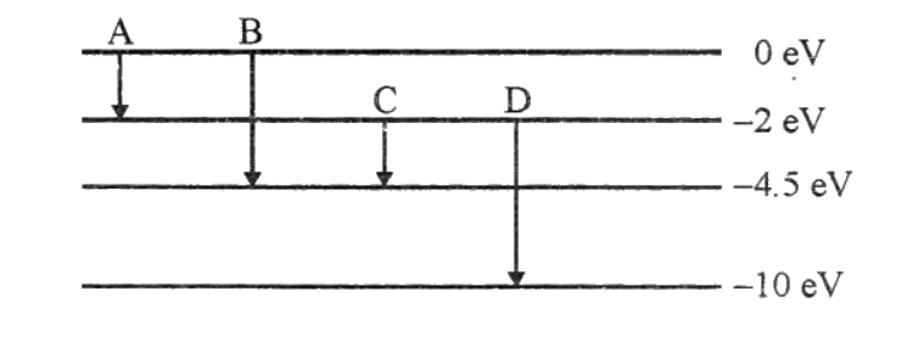
$\nu = Rc\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$, which is the **frequency of the emitted radiation**.

Final answer. $\nu = (me^4/8\varepsilon_0^2 h^3)(1/n_f^2 - 1/n_i^2) = Rc(1/n_f^2 - 1/n_i^2)$.

← *Back to Q159*

Q160

(a) The energy levels of an atom are as shown below. Which of them will result in the transition of a photon of wavelength 275 nm*?*



(b) Which transition corresponds to emission of radiation of maximum wavelength?

[2009]

SOLUTION

When an electron falls from a higher to a lower level, the emitted photon's energy equals the gap between the levels: $\Delta E = \frac{hc}{\lambda}$. Using $E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$, the photon energy for $\lambda = 275$ **nm** is

$E = \frac{1240}{275} \approx 4.5$ **eV.**

The levels shown are at 0, −2, −4.5 and −10 eV. Find the energy gap for each downward transition.

0→−2: $\Delta E = 2$ eV.

0→−4.5: $\Delta E = 4.5$ eV.

0→−10: $\Delta E = 10$ eV.

−2→−4.5: $\Delta E = 2.5$ eV.

−2→−10: $\Delta E = 8$ eV.

−4.5→−10: $\Delta E = 5.5$ eV.

(a) The gap of 4.5 eV matches the **0 eV→−4.5 eV** transition, so **this transition produces the λ=275 nm photon**.

(b) Wavelength is maximum when the energy gap is minimum, since $\lambda = \frac{1240}{\Delta E}$.

The smallest gap is $\Delta E = 2$ **eV** for the **0→−2 eV** transition.

$\lambda_{max} = \frac{1240}{2} = 620$ **nm.**

Hence the **0→−2 eV transition gives the maximum wavelength of 620 nm**.

Final answer. 275 nm comes from the 0 → −4.5 eV transition; the 0 → −2 eV transition gives the maximum wavelength (620 nm).

← *Back to Q160*

Q161

Draw a schematic arrangement of the Geiger-Marsden experiment. How did the scattering of α-particles by a thin foil of gold provide an important way to determine an upper limit on the size of the nucleus? Explain briefly.

[2009]

SOLUTION

Schematic arrangement of the Geiger-Marsden (alpha-scattering) experiment: A radioactive source such as $^{214}_{83}$Bi emits a beam of α-particles. This beam is passed through a lead collimator (a small slit) so that a fine, narrow, well-directed beam of α-particles is obtained.

The collimated beam is made to fall on a very thin gold foil of thickness about $2.1 \times 10^{-7}\ m$. The foil is kept thin so that the α-particles mostly undergo single scattering.

A rotatable detector, consisting of a zinc-sulphide (ZnS) fluorescent screen backed by a microscope, is placed on the other side of the foil. It can be moved to any angle θ about the foil. Wherever an α-particle strikes the screen, a tiny flash of light (scintillation) is produced, and these flashes are counted at different scattering angles θ.

(Diagram in words: source → lead collimator slit → narrow α-beam → thin gold foil at the centre; the ZnS screen + microscope detector is mounted on an arm

that sweeps in a circle around the foil to read the count at each angle θ, measured from the original beam direction.)

How it gives an upper limit on nuclear size: An incoming α-particle of charge +2e is repelled by the positive nucleus of charge +Ze (for gold $Z = 79$). The α-particle slows down as it approaches and momentarily stops at the distance of closest approach r_0, where all its kinetic energy has been converted into electrostatic potential energy.

$$K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r_0}$$

$$\Rightarrow r_0 = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}$$

Here K is the kinetic energy of the α-particle. So long as the α-particle does not actually touch or enter the nucleus, only the Coulomb (electrostatic) force acts, and the observed scattering matches Rutherford's Coulomb-scattering formula.

For typical α-particles of energy about 5.5 MeV incident on gold, this gives $\mathbf{r_0 \approx 4 \times 10^{-14}\ m}$, and the scattering still obeys the Coulomb law. This means the actual nucleus is even smaller than r_0. Therefore r_0 provides an **upper limit on the size of the nucleus**, showing that the entire positive charge and nearly all the mass of the atom are concentrated in an extremely tiny central region (the nucleus) of size of the order of $10^{-14}\ m$, which is far smaller than the atomic size of about $10^{-10}\ m$.

CONCEPT

Closest approach r_0 is an UPPER limit only: the real nucleus is smaller, because the Coulomb formula is still obeyed at r_0, meaning the α-particle never reached the nuclear surface.

Final answer. The distance of closest approach $\mathbf{r_0 = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}}$ (about $\mathbf{410^{-14}\ m}$ for **5.5 MeV α**-particles on gold) fixes the upper limit on the nuclear size.

← *Back to Q161*

Q162

The energy level diagram of an element is given below. Identify, by doing necessary calculations, which transition corresponds to emission of a spectral line of wavelength 102.7 nm.

[2008]

SOLUTION

The energy of the emitted photon is related to its wavelength by the relation

$$E = \frac{hc}{\lambda}$$

Using the convenient form **hc = 1240 eV nm** and **λ = 102.7 nm**,

$$E = \frac{1240 \text{ eV nm}}{102.7 \text{ nm}}$$

$$E = 12.07 \text{ eV}$$

So the transition responsible for this line must have an energy difference of about 12.07 eV between the two levels involved.

The energies of the hydrogen-like levels are

$$E_1 = -13.6 \text{ eV}, E_2 = -3.4 \text{ eV}, E_3 = -1.51 \text{ eV}, E_4 = -0.85 \text{ eV}$$

Now I check the energy released in transitions ending at the ground state (***n*=1**), since the energy gap needed is large:

$$E_2 - E_1 = -3.4 - (-13.6) = 10.2 \text{ eV}$$

$$E_3 - E_1 = -1.51 - (-13.6) = 12.09 \text{ eV}$$

$$E_4 - E_1 = -0.85 - (-13.6) = 12.75 \text{ eV}$$

The value 12.09 eV matches the required photon energy of 12.07 eV almost exactly.

Hence the spectral line of wavelength 102.7 nm is emitted when the electron jumps from the third level to the first level, i.e. the ***n* = 3 → *n* = 1 transition**.

Since it ends on the ground state (n=1), it belongs to the **Lyman series** (it is the second member, Lyman-β).

Final answer. The line **102.7 nm** corresponds to the $n = 3 \to n = 1$ transition (Lyman series).

← *Back to Q162*

Q163

Show that Bohr's second postulate, 'the electron revolves around the nucleus only in certain fixed orbits without radiating energy' can be explained on the basis of de Broglie hypothesis of wave nature of electron.

[2008]

SOLUTION

According to the de Broglie hypothesis, every moving particle behaves like a wave. An electron of mass m moving with speed v has a matter wave of wavelength

$$\lambda = \frac{h}{mv}$$

where h is Planck's constant.

Consider the electron revolving in a circular orbit of radius r around the nucleus. For this orbit to be a stable, non-radiating (stationary) orbit, the matter wave associated with the electron must form a standing wave around the circumference. This is possible only if the wave joins onto itself smoothly after each revolution, i.e. the electron wave is in phase with itself.

A standing wave around the orbit forms only when the total length of the orbit (its circumference) contains a whole number of de Broglie wavelengths:

$$2\pi r = n\lambda,\ n = 1, 2, 3,$$

(Picture in words: the circular orbit drawn as a closed loop with a continuous wavy line running along it; for an allowed orbit the crests and troughs exactly meet head-to-tail after one full loop, giving n complete wavelengths around the

circle. If the circumference were not a whole multiple of λ, the wave would not join smoothly.)

Substituting $\lambda = \frac{h}{mv}$,

$$2\pi r = \frac{nh}{mv}$$

Rearranging,

$$mvr = \frac{nh}{2\pi}$$

This is exactly **Bohr's second postulate**, the quantisation of angular momentum, namely $L = mvr = \frac{nh}{2\pi}$.

If $2\pi r$ were not an integral multiple of λ, the wave on successive revolutions would arrive out of phase and undergo destructive interference, so no stable orbit could exist. Only those discrete orbits for which $2\pi r = n\lambda$ allow the electron wave to be self-sustaining without destructive interference. These permitted orbits are the **stationary (non-radiating) orbits**. Thus the de Broglie standing-wave picture gives a physical explanation for Bohr's postulate that the electron revolves only in certain fixed orbits without radiating energy.

Final answer. The standing-wave condition $2\pi r = n\lambda$ gives $mvr = \frac{nh}{2\pi}$, which is Bohr's quantisation postulate, so only discrete non-radiating orbits are allowed.

← *Back to Q163*